MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)
Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi
Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution



PROJECT REPORT

23AISE409 PROJECT PHASE – II

"PROBABILISTIC CASCADING FAILURE AND AGENT ASSISTED ROOT CAUSE ANALYSIS FOR MICROSERVICE SYSTEMS"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

by

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
2026–2027

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CERTIFICATE

This is to certify that the 23AISE409 PROJECT PHASE – II work entitled "**Probabilistic Cascading Failure and Agent Assisted Root Cause Analysis for Microservice Systems**" is a bonafide work carried out by **ANISH C (4MT23AI008), ROYSON SALIS (4MT23AI046), DHANUSH K MURTHY (4MT23AI401), K BHARATH PARMAR (4MT24AI402)** in partial fulfillment for the award of degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** of **Mangalore Institute of Technology & Engineering (Autonomous)** during the year 2026–2027. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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DECLARATION

We declare that this 23AISE409 PROJECT PHASE – II report titled "**Probabilistic Cascading Failure and Agent Assisted Root Cause Analysis for Microservice Systems**" submitted in partial fulfillment of the degree of B.E. in Artificial Intelligence and Machine Learning is a record of original work carried out by us under the supervision of Dr. Manjunath Prasad H R, and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

When a microservice fails, the failure does not stay where it starts. A slow datastore drags down the services that query it, their callers time out, retries amplify the load, and within a minute the operator faces dozens of alarms in which the originating fault cannot be told from its victims by severity alone. Recent diagnosis methods localise the fault with graph neural networks over fused metrics, traces and logs; they need labelled failures, GPU inference, and give an output the on-call engineer cannot audit.

This project builds the alternative as a complete, unsupervised, single-core pipeline that observes Docker containers from outside the application, over SSH, and assists the operator with a cause, a prediction of what fails next, a fault type and a runbook. A platform of about 36 000 lines of TypeScript and 329 automated tests recovers the call graph from container socket tables, shows it in 2D and 3D, and runs the diagnosis. Its core is a probabilistic cascading-failure model whose per-edge propagation probabilities are learned online as Beta–Binomial posteriors; a hybrid detector in which a per-service Isolation Forest and a robust rule layer jointly raise the alarm, with container facts (stopped, crash-looping) treated as evidence rather than degree; a ranker that separates a service's own deviation from what its neighbours explain; and call directions learned from telemetry. A Detection engines page runs the Rules, ML and Hybrid engines in parallel and shows what each detected, computed and predicted.

The evaluation uses 62 faults injected blind into two real containerised systems over 217 minutes each, and 178 held-out synthetic faults over 16 topologies, with bootstrap intervals, paired tests, an ablation and an amplification study. The hybrid names the true cause first for 74% of real faults against 65% for the rule layer (paired difference $+9.7$ points, 95% CI $[+1.6, +17.7]$); learned propagation discriminates better than fixed edge probabilities (AUROC 0.847 against 0.784) on 80 observed cascades; learned directions stay correct when service names carry no information (28 of 31 pairs against none); and change-point alarms roughly halve the median detection delay (10.5 s to 4.9 s). A decision window costs about 6 to 7 ms. The report also gives what did not work, a measurement bug that had hidden half of the real faults, and operational findings from the benchmark hosts.

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ABBREVIATIONS AND NOTATIONS

Abbreviations

AIOps	Artificial Intelligence for IT Operations
API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic curve
AP	Average Precision
CI	Confidence Interval
C+N	Caught and named (detected, and the true cause ranked first)
cgroup	Linux control group (resource limits for a container)
ECE	Expected Calibration Error
EC2	Amazon Elastic Compute Cloud
FA/h	False-alarm episodes per fault-free hour
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
MAD	Median Absolute Deviation
ML	Machine Learning
PEAS	Performance, Environment, Actuators, Sensors
RCA	Root Cause Analysis
REST	Representational State Transfer
SSH	Secure Shell
TCP	Transmission Control Protocol
USN	University Seat Number
WS	WebSocket

Notations

S, N	the set of services and its size
$x_s(t)$	the reading of service s at time t (eight metrics, missing values kept as $\varnothing$)
Δ	the collection period, 5 seconds
$w(e)$, $c(e)$, $d(e)$	weight, evidence confidence and dependency share of edge e
level(x)	forest level of a reading: 0 typical, 1 on the service's alarm line
λ_s, ρ_s, f_s	forest level, rule score and fact flag of service s
τ_λ, τ_ρ	alarm thresholds on the forest level (2) and the rule score (1.25)
$o(s)$, $\sigma(s)$	own level and symptom level of service s
π_s	random-walk share of service s
α	restart probability of the walk (0.15)
$\hat{p}(c \leftarrow x)$	learned probability that trouble in x reaches its caller c
$P(c)$	risk that service c is reached from a suspected root
A	amplification factor of the synthetic generator

CHAPTER 1

INTRODUCTION

1.1 Background

Most large applications today are built as many small services that talk to each other over the network. A shopping site, for example, has separate services for the catalogue, the cart, payment, shipping and orders, each with its own database or cache [1]. The style lets teams deploy and scale each piece on its own. It also creates a kind of failure that a single program does not have: a fault does not stay where it starts.

Take a slow database. The service that queries it starts waiting, so it answers slowly. The services that call *that* service begin to time out, retry, and add load on top of load. Within a minute a dozen services are raising alarms together, and the operator on call is looking at a wall of red where the first thing that broke is no louder than the things it broke. Practitioners call this an alert storm. The real cost is not the number of alerts but the time spent deciding which one deserves action [2, 3].

Two facts make the problem manageable. First, trouble travels along the call graph: from a callee to the services that wait on it, not in random directions. Second, it travels *probabilistically*. A caller that has a timeout, a cache or a circuit breaker may live through a failing dependency; a caller without those protections will not. A tool that knows the call graph, and how likely trouble is to cross each edge, can answer two questions that a ranked list of alarms cannot: which service is the origin, and which healthy services are about to be pulled in.

Recent research answers the first question with heavy machinery: models that fuse metrics, traces and logs into learned representations and localise the fault with graph neural networks [4, 5, 6]. They work, but they need labelled failure data that most teams do not have, they need accelerated hardware at inference time, and their output (a cluster number, an embedding ranking) cannot be checked by the engineer against the numbers on the dashboard. Classic observability products, on the other hand, usually ask for changes inside the application, such as client libraries, agents or a sidecar in every pod, before they show anything.

This project takes a different route. It watches the containers *from outside*, over SSH and the Docker socket, so that nothing in the monitored system is modified. From those observations it builds a live dependency graph, and on top of that graph it runs an unsupervised, single-core diagnosis pipeline that proposes a root cause, a prediction of what fails next, a fault type and a short runbook, each backed by numbers the operator can inspect.

1.2 Problem Statement

The project addresses four linked problems that a team running containerised microservices meets in practice:

- **Invisible dependencies.** The real call graph of a running system is rarely documented, and it drifts. Compose files list what was *declared*, not what is actually talking. Without instrumentation, nobody can say which service calls which, or in which direction.
- **Alert storms.** When one service fails, its callers fail too. Ranking alarms by severity points at the loudest service, which is often a victim and not the cause.
- **No view of what comes next.** Operators learn that a service is affected only after it alarms. A model of how failure spreads on this particular system could warn earlier.
- **Black-box diagnosis.** Methods that need training data and return an opaque score are hard to trust at three in the morning. An engineer needs to see the evidence and how it was combined.

The problem is therefore to *observe a containerised system without touching it, recover its call graph, detect trouble per service against that service's own normal, rank the likely origin of an incident, predict how it will spread, and show every step in a form an engineer can verify*, with no labelled data and no specialised hardware.

1.3 Motivation and Need for the Project

Small teams and student projects run exactly the systems for which heavyweight AIOps products are a poor fit: a handful of Docker hosts on a cloud account, no budget for per-host licences, and applications that nobody wants to re-instrument. Yet these systems fail in the same cascading way as large ones. We saw it ourselves while running the benchmark applications for this project: killing one datastore container produced alarms on the services that call it within seconds.

Three observations shaped the work. First, much of what an operator needs is already visible from the host: container state, CPU and memory against the container's own limit, network counters, and the table of open TCP sockets inside each container. Second, a per-service model of "normal" removes the need for global thresholds, which fail because services differ by orders of magnitude in every metric. Third, a result that cannot be reproduced or audited is of little use, so the evaluation had to be as careful as the tool: ground truth that the detector never sees, confidence intervals, paired tests, and an honest account of what did not work.

1.4 Objectives

1. To build a platform that connects to Docker hosts over SSH, discovers the service topology, and shows live health, metrics and dependencies in a 2D and a 3D view, without changing the monitored applications.
2. To recover call *directions* from observed sockets and, where names carry no information, from the services' own behaviour.
3. To detect trouble per service against its own recent history with a hybrid of a learned detector (Isolation Forest) and a robust statistical rule layer, treating container facts such as "stopped" or "crash-looping" as evidence rather than degree.
4. To rank the root cause of an incident by separating a service's own deviation from what its neighbours explain, and by running a random walk over the call graph.
5. To model cascading failure with learned per-edge propagation probabilities, predict which services are next, and warn early about gradual faults.
6. To classify the fault type and attach a short runbook, so the output is something an engineer can act on.
7. To make the three engines (Rules, ML, Hybrid) switchable and to show, side by side and on the same readings, what each one detects, computes and predicts.
8. To evaluate all of it on real and synthetic fault campaigns with ground truth the detector never sees, reporting confidence intervals, paired tests, an ablation and the negative results.

1.5 Scope of the Project

The system targets Docker-based microservice stacks on Linux hosts that can be reached over SSH. It reads container state, statistics and socket tables; it does not read application payloads and it does not require code changes. It was developed and measured on the following applications:

- **ShopFlow**, a 13-service retail checkout system and **LedgerLine**, a 12-service retail-banking transfer system. Both were built for this study as test benches, each on its own EC2 host with its own Fault Lab and a Prometheus–Grafana–Loki monitoring stack (Section 5.9); both were run under open-loop traffic for 217 minutes with 62 faults injected;
- **Sock Shop**, the open microservices demo, used for development;
- a social-network application from the DeathStarBench suite [7] (27 containers) and the OpenTelemetry demo, used to check that discovery works on third-party systems;

- a synthetic generator that writes runs in the same recorded format over 16 random topologies, used to enlarge the evaluation.

The mapper is not a distributed tracing tool: there are no request identifiers or spans, and the "Traces" page shows sampled TCP connection snapshots. It does not replace an application performance monitoring suite. No language model takes part in the diagnosis; the "assistant" is a transparent composition of statistical models, and Chapter 7 discusses where a language-model agent could be added.

1.6 Significance and Expected Contribution

The project shows that useful root-cause analysis does not need instrumentation, labels or a GPU, and it measures how far that goes. Its contributions are:

- an **instrumentation-free observation layer** that recovers topology and call direction from container sockets and Docker state;
- a **probabilistic cascading-failure model** whose per-edge propagation probabilities are Beta–Binomial posteriors learned online from the system's own incidents, with risk flowing over the graph by noisy-OR;
- a **hybrid detector** in which a per-service Isolation Forest and a robust rule layer jointly raise the alarm and container facts count as evidence, with an operating contract that reports alarm *time* as well as false-alarm episodes;
- a **fact-aware, decomposition-based ranker** for the root cause, and **call directions learned from telemetry**;
- a **Detection engines page** that runs Rules, ML and Hybrid in parallel and shows what each detected, computed and predicted;
- a **reproducible evaluation** with a blind fault injector, a held-out synthetic benchmark, bootstrap intervals, paired tests, an ablation and an amplification study, together with the results that did not help and a measurement bug that had hidden half of the real faults;
- a working **platform** of about 36 000 lines of TypeScript with 329 automated tests, a traffic generator and a host-check tool, used throughout the experiments.

1.7 Organization of the Report

Chapter 2 reviews related work on anomaly detection, root-cause localisation, cascade modelling and observability tooling, and states the gap. Chapter 3 formulates the problem and presents the proposed system, its architecture and its two agents. Chapter 4 describes the methodology: the models, the evaluation measures, the blind fault injector and the statis-

tics. Chapter 5 describes the implementation: modules, interfaces, core code, algorithms, configuration, the prototype and testing. Chapter 6 reports the results of the experiments, the comparison with existing approaches, and the operational findings from the hosts. Chapter 7 concludes and lists limitations and future work. The appendices give the configuration constants, the fault classes, the test suites and a short user guide.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This project draws on five bodies of work: anomaly detection on time series, root-cause localisation in microservice systems, models of how failure spreads, evaluation practice for anomaly and diagnosis methods, and the observability tools that supply the data. Each statement about a cited work below restates what that work documents; where we compare, we say what we compared.

2.2 Review of Existing Work

2.2.1 Anomaly detection

Isolation Forest [8, 9] isolates an outlier with few random cuts and needs no density estimate, which suits per-service streams of mixed and skewed metrics. Distance and density methods [10, 11], the histogram-based score HBOS [12] and the multivariate control chart of Hotelling [13, 14] with a shrinkage covariance estimate [15] are natural alternatives, and we compare all of them in Chapter 6. Robust location and scale from the median and the median absolute deviation [16, 17] underpin the rule layer of the proposed system. Surveys of streaming and learned anomaly detection [18, 19, 20] and deployed detectors for monitoring data [21, 22] describe the setting we work in: unlabelled streams, drifting baselines and a high cost of false alarms.

2.2.2 Evaluation of anomaly detectors

Several studies criticise the common practice of scoring anomaly detectors point by point with an adjustment that credits a whole event if any point inside it is flagged [23, 24], and argue for range-based precision and recall that respect the duration of an event [25]. This is why the evaluation in Chapter 4 is event-level: a fault counts as detected only when an alarm names a truly involved service while the fault is active, and false alarms are counted as episodes outside every fault window.

2.2.3 Root-cause localisation in microservices

MonitorRank [26] introduced a personalised random walk over the call graph, biased by how anomalous each service looks, and ranks the services by the walk's stationary share.

MicroRCA [27], CloudRanger [28], Microscope [29], AutoMAP [30] and MicroRank [31] extend the idea with anomaly-weighted transitions, causal-graph construction and trace spectra. Causal approaches recover structure with the PC algorithm or hierarchical discovery [32, 33, 34]. Learned methods localise with graph autoencoders and multi-source fusion of metrics, traces and logs [5, 4, 6, 35]. A survey of AIOps approaches to failure management [3] and one of anomaly and root-cause work specific to microservices [36] place these methods in context. Most of them need traces, labelled failures or training, and some need accelerated hardware; the random-walk family needs only the call graph and an anomaly score, which is the family this project builds on.

2.2.4 Cascading failure and change detection

The independent-cascade model of influence spread [37] is the basis of our propagation model, with the edge probabilities estimated online instead of assumed. Probabilistic forecasts are judged with proper scoring rules, reliability diagrams and expected calibration error [38, 39, 40]. Detection latency can be attacked with classical change-point detection [41, 42]. PageRank [43] and reciprocal-rank fusion [44] inform ranking; Granger causality [45] and transfer entropy [46] inform the study of call direction; random forests and CART [47, 48] back the fault-type classifier.

2.2.5 Benchmarks and chaos engineering

Open benchmark applications such as DeathStarBench [7] and Train-Ticket [49] provide realistic microservice systems to test on, and chaos-engineering practice [50] provides the method for injecting faults on purpose. Our fault injector follows that practice but, importantly, keeps its ledger private: nothing inside the system under test is labelled.

2.2.6 Observability tooling

Production observability usually combines three kinds of tool. *Metric systems* scrape numbers from endpoints that each service exposes. *Distributed tracing* follows a request across services and needs the application, or a proxy in front of it, to propagate trace identifiers. *Application performance monitoring* products add agents inside the runtime. All three give an accurate picture once the instrumentation exists; none works on a stack that nobody is willing to modify. Service meshes can supply topology without code changes, but they require a sidecar in every pod. The site-reliability literature [2] is clear that the hard part is not collecting signals but deciding what to look at first, which is the gap that root-cause ranking targets.

Table 2.1: Review of existing work (what each documents, what it needs, and where it stops).

Work	Approach	Needs	Limitation for our setting
Isolation Forest [8]	Isolate outliers with random trees; score by path length	Unlabelled history	Gives a score, not a cause
MonitorRank [26]	Personalised random walk over the call graph	Call graph, anomaly score	Direction and edge weights are inputs
MicroRCA [27], CloudRanger [28], Microscope [29], AutoMAP [30]	Extend the walk with anomaly-weighted transitions and graph construction	A graph and monitoring data (sources differ by method)	The graph is built from data the host alone does not give
MicroRank [31]	Trace spectrum analysis with PageRank	Request traces	Needs tracing in every service
RCD [32], CIRCA [33]	Causal discovery or structural causal graph for localisation	Metrics, structure	Assumes a faithful causal model
Eadro [6], DeepHunt [5], DiagFusion [4]	Learned fusion of metrics, traces, logs; graph neural or autoencoder models	Labelled or curated failures; GPU	Not auditable; training data needed
Independent cascade [37]	Probabilistic spread over a graph	Edge probabilities	Probabilities must be estimated
Change-point tests [41, 42]	Detect a shift in a stream	A stream	Alone, no cause or graph

2.3 Comparative Analysis

Table 2.2 compares the approaches on the properties that decide whether a method can be used on an unmodified Docker stack. The proposed system is deliberately not the most accurate method one could imagine; it is the one that can be switched on in minutes, explains itself, and has been measured honestly.

Table 2.2: Approaches compared on practical properties (✓ yes, – no, ∼ partly).

Property	**Metric tools**	**Tracing**	**Learned fusion**	**Walk RCA**	**This project**
No change to applications	–	–	–	∼	✓
Works without labelled failures	✓	✓	–	✓	✓
Recovers the call graph itself	–	✓	∼	–	✓
Learns call direction	–	✓	–	–	✓
Per-service learned baseline	∼	–	✓	∼	✓
Ranks a root cause	–	∼	✓	✓	✓
Predicts what fails next	–	–	–	–	✓
Needs a GPU	–	–	✓	–	–
Every score can be audited	✓	✓	–	∼	✓

2.4 Research and Design Gap

Four gaps motivate the design.

1. **Observation without instrumentation.** The graph that random-walk and causal methods assume is usually given. Recovering it, with direction, from the host alone is rarely addressed, and the observation noise that results (a socket is seen from both ends) changes what the downstream methods must tolerate.
2. **Facts are not degrees.** A stopped container is not "very anomalous"; it is a fact. Methods that fold container state into one score lose the most reliable signal there is, and our ablation shows how much that costs.
3. **Propagation is assumed, not learned.** Cascade prediction normally uses fixed edge probabilities. Learning them from the system's own incidents, with a prior that makes rarely used edges behave sensibly, is cheap and measurable.
4. **Evaluation is optimistic.** Point-adjusted scoring, tuned thresholds and a single data set make results look better than they are. We use event-level scoring, a held-out split, paired tests, and we publish what failed.

2.5 Summary

Existing root-cause methods are strong on accuracy but assume instrumentation, labels or hardware that small teams do not have, and they are usually evaluated optimistically. The proposed system occupies a different point in the design space: instrumentation-free, unsupervised, auditable and honest about its limits. The next chapter formulates the problem and describes the system.

CHAPTER 3

PROPOSED SYSTEM AND SCOPE

3.1 Problem Formulation

A system is a set S of N services with a directed call graph. An edge $u \to v$ means that u calls v, so a failure travels *against* the arrow, from v to the callers that wait on it. Every $\Delta = 5$ s the collector produces, for each service s, a reading

$$x_s(t) \in (\mathbb{R} \cup \{\varnothing\})^8,$$

holding CPU, memory, CPU throttling, network in and out, request rate, error share and 95th-percentile latency. The symbol $\varnothing$ marks a metric that was not reported. It is never replaced by zero, because a stopped container must be distinguishable from an idle one. Each reading comes with a container state (running, paused, restarting, exited, detached).

At time t the pipeline must decide four things, using only readings up to t:

(i) whether an incident is in progress;

(ii) which service is its root cause r;

(iii) which currently healthy services will be affected before the incident ends;

(iv) what kind of fault it is.

The pipeline must do this without labelled failures, without changing any application, and fast enough that a decision window finishes well inside the 5 s collection period.

3.2 Proposed Solution

The proposed system is an *agent-assisted* pipeline in a concrete sense. A **collector agent** observes the containers from outside the application. A **diagnosis assistant** turns those observations into a cause, a forecast of what fails next, a fault type, a runbook and an explanation the operator can check. Between the two sits the evidence-weighted dependency graph. Figure 3.1 shows the whole system and Figure 3.2 one 5-second cycle.

The solution has seven steps, each described in Chapter 4:

1. **Observe.** Read container state, statistics and socket tables over SSH; install nothing on the host.

2. **Build the graph.** Merge declared dependencies (from the Compose file) with observed ones; orient each observed socket by the listening port; give each edge a confidence and a dependency share.
3. **Learn directions.** Where the heuristic cannot tell who calls whom, vote over dynamic and structural signals.
4. **Detect.** Judge every service against its own recent past with two Isolation Forests and a robust rule layer; treat container facts as evidence.
5. **Rank.** Split each service's deviation into what its callees explain and what they do not; run a personalised random walk over the call graph; put facts first.
6. **Predict.** Spread risk over the graph by noisy-OR with edge probabilities learned from the system's own incidents; warn early about gradual faults.
7. **Diagnose and explain.** Classify the fault type, attach a short runbook and show the evidence behind every number.

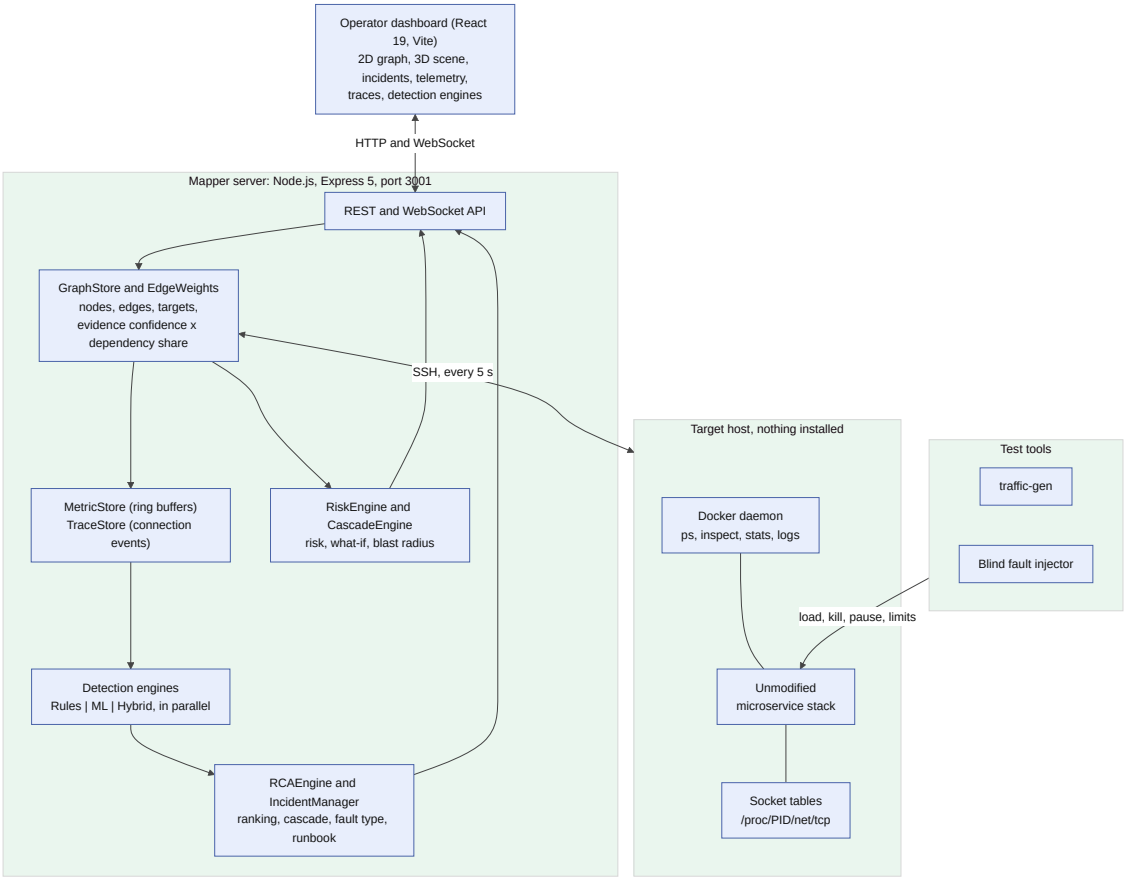


Figure 3.1: System architecture. The server reaches the target host over SSH and reads Docker and the container socket tables; nothing is installed on the host. The three detection engines share the same readings and graph; the dashboard receives updates over a WebSocket.

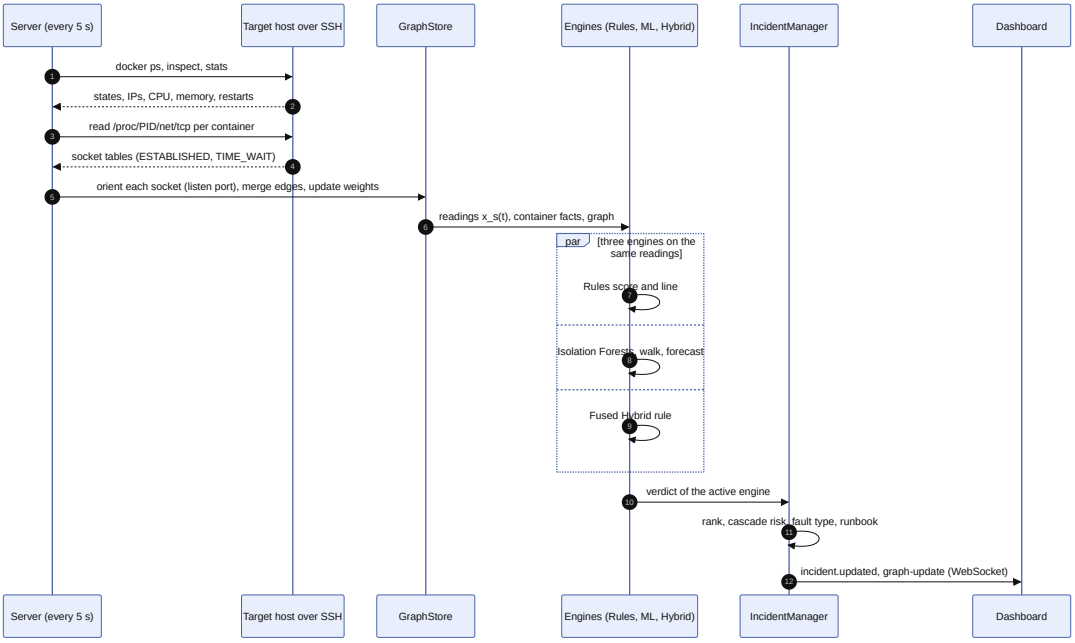


Figure 3.2: One 5-second cycle: observation, graph update, the three engines in parallel, ranking and the push to the dashboard.

3.3 Requirements

3.3.1 Functional requirements

Table 3.1: Functional requirements and where they are met.

ID	Requirement	Component
FR1	Connect to a Docker host over SSH with a key pair and discover its containers and Compose dependencies	Connection manager, discovery engine
FR2	Collect CPU, memory, throttling, network, restarts and request statistics every 5 s	Metric collector, MetricStore
FR3	Recover observed dependencies and their direction from container socket tables	Collector, GraphStore, EdgeWeights
FR4	Show the live topology in 2D and in 3D, with health, link activity and layouts to choose from	Dashboard (graph canvas, 3D scene)
FR5	Detect abnormal behaviour per service against its own baseline	Rules, ML and Hybrid engines
FR6	Raise an incident and rank the root cause with an evidence breakdown	RCAEngine, IncidentManager
FR7	Predict which healthy services are likely next and give an early warning for gradual faults	CascadeEngine, ML engine
FR8	Classify the fault type and give a runbook	Diagnosis layer, remediation guide
FR9	Let the operator switch the engine and compare all three on the same readings	Engine switch, Detection engines page
FR10	Generate load and run experiments against a target	TrafficController, traffic-gen
FR11	Record runs and score the engines against a private fault ledger	Evaluation harness (`run/cli`)
FR12	Report host and mapper problems separately	`check-host`, `mapper-edges`

3.3.2 Non-functional requirements

Table 3.2: Non-functional requirements.

Quality	Requirement
Non-intrusive	No code, agent or sidecar is added to the monitored application; the host needs only Docker and SSH access
Unsupervised	No labelled failures and no training phase beyond the first minutes of normal readings
Auditable	Every score shown to the operator can be traced to a measurement and a stated rule
Fast	A decision window finishes in a small fraction of the 5-second collection period on one CPU core
Honest	Container state is shown as a fact, missing metrics as missing, and measured limits are stated rather than hidden
Safe to connect	Remote commands come from a fixed catalogue whose arguments are sanitised; keys stay outside the repository
Reproducible	Every reported number is regenerated by a script in the repository
Portable	Runs on macOS, Linux and Windows with Node.js; a Python sidecar is optional

3.4 System Architecture

The architecture has five layers (Figure 3.1).

Dashboard. A React 19 application built with Vite. It holds no polling logic: the server pushes graph and incident updates over a WebSocket, and REST calls fetch history and configuration. The views are described in Section 5.7.

Server. A Node.js process (Express 5, TypeScript) that owns all state and is the only component that writes to disk. The `GraphStore` keeps nodes, edges and targets; the `MetricStore` keeps a ring buffer of readings per service; the `TraceStore` keeps connection events; the `IncidentManager` drives the engines and turns their verdicts into incidents.

Connection and collection. A connection manager keeps one SSH session per target and runs commands from a fixed catalogue (`docker ps`, `inspect`, `stats`, `logs`, and reads of `/proc/<pid>/net/tcp`). Reading the socket table from the host through the container's process id works for every image, including images with no shell. An optional standalone collector can run on a host and push to the same ingest endpoint.

Target workloads. Unmodified Docker Compose applications.

Test tools. A traffic generator with per-application workflows and profiles, and a fault injector that works on containers (kill, pause, resource limits, network detach) and writes its ledger to a file the detector never reads.

Table 3.3 lists the network endpoints.

Table 3.3: Network endpoints used by the system.

Port	Component	Protocol	Direction
3001	Mapper server	HTTP and WebSocket	Dashboard to server
5173	Dashboard (Vite)	HTTP	Browser to dashboard
4400	Traffic generator	HTTP and WebSocket	Server to traffic generator
22	Target host	SSH	Server to host
8080 (typical)	Target application front end	HTTP	Traffic generator to host

3.5 Agent Design

3.5.1 The two agents

Table 3.4 describes the two agents with the PEAS scheme (performance measure, environment, actuators, sensors).

Table 3.4: PEAS description of the two agents.

	Collector agent	Diagnosis assistant
Performance	Complete, timely and truthful readings; correct edges and directions	Caught-and-named rate; alarm time; calibration of predictions
Environment	A Docker host reached over SSH, running an unknown microservice stack	The evidence-weighted graph and the readings it holds
Actuators	SSH commands from a fixed catalogue	Incidents, rankings, predictions, runbooks, explanations in the dashboard
Sensors	Docker state and statistics, socket tables, logs, interface counters	Readings, container facts, graph, learned directions

3.5.2 Decision logic of the diagnosis assistant

The assistant runs one decision window per cycle. Its alarm rule is shown in Figure 3.3; its ranking is shown in Figure 3.4. An incident moves through the states in Figure 3.5.

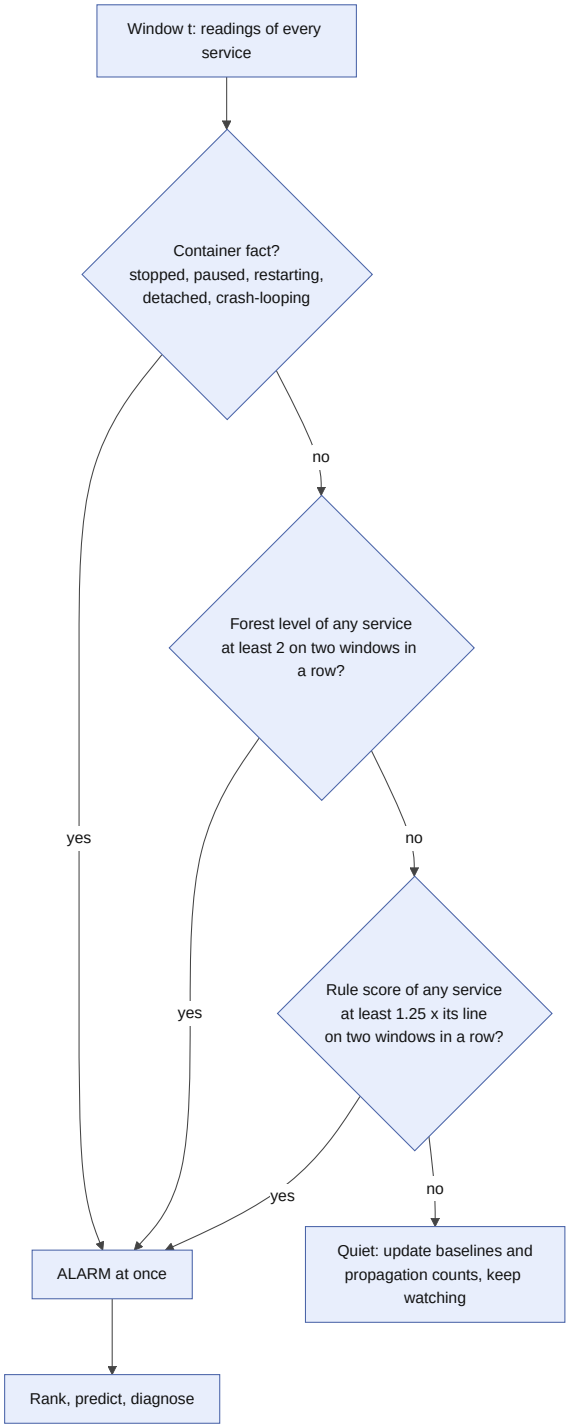


Figure 3.3: The hybrid alarm rule. A container fact raises the alarm at once; the forest and rule conditions must hold for two windows in a row.

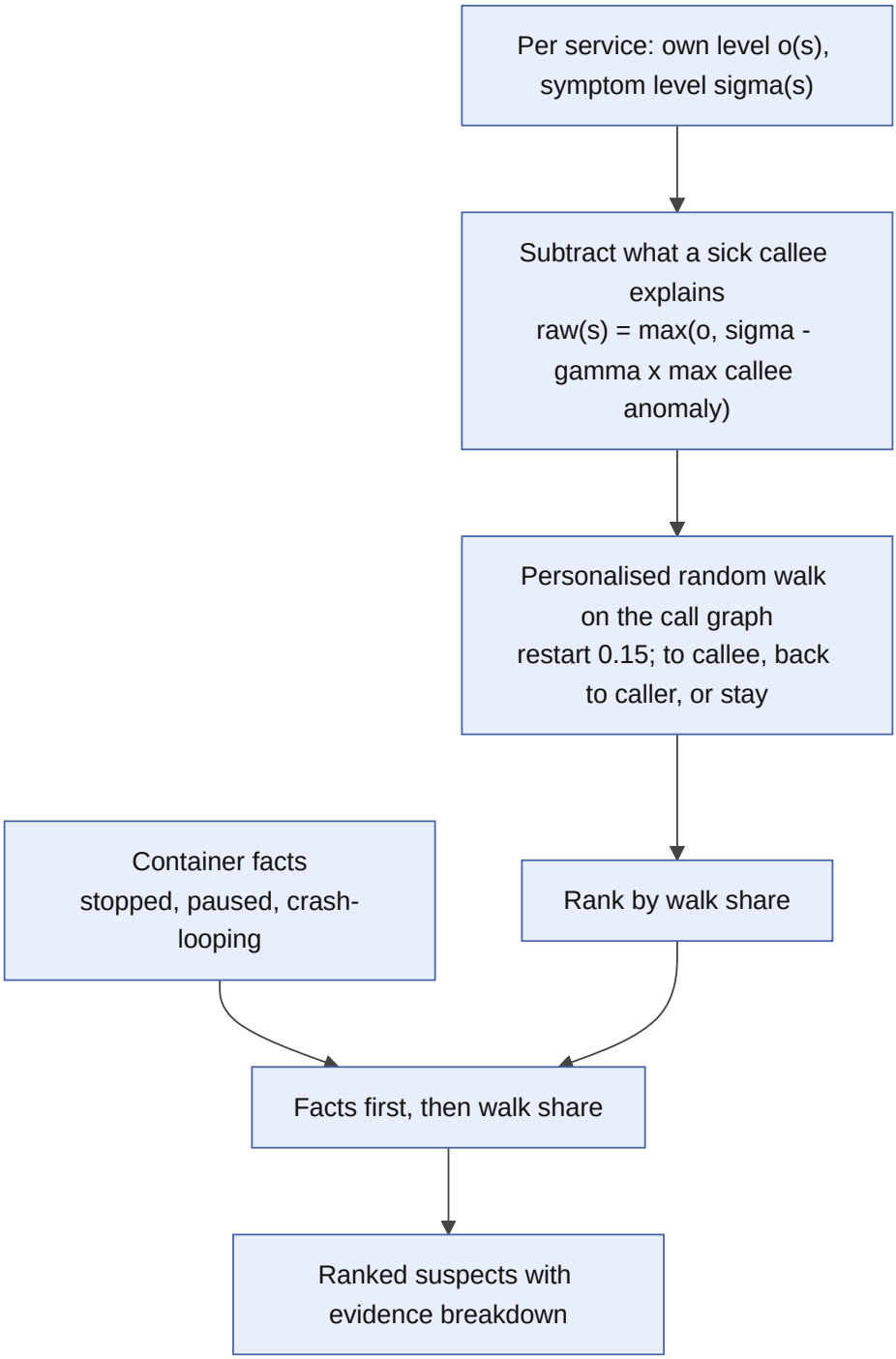


Figure 3.4: Root-cause ranking: split each deviation into own and explained parts, run the random walk, put facts first.

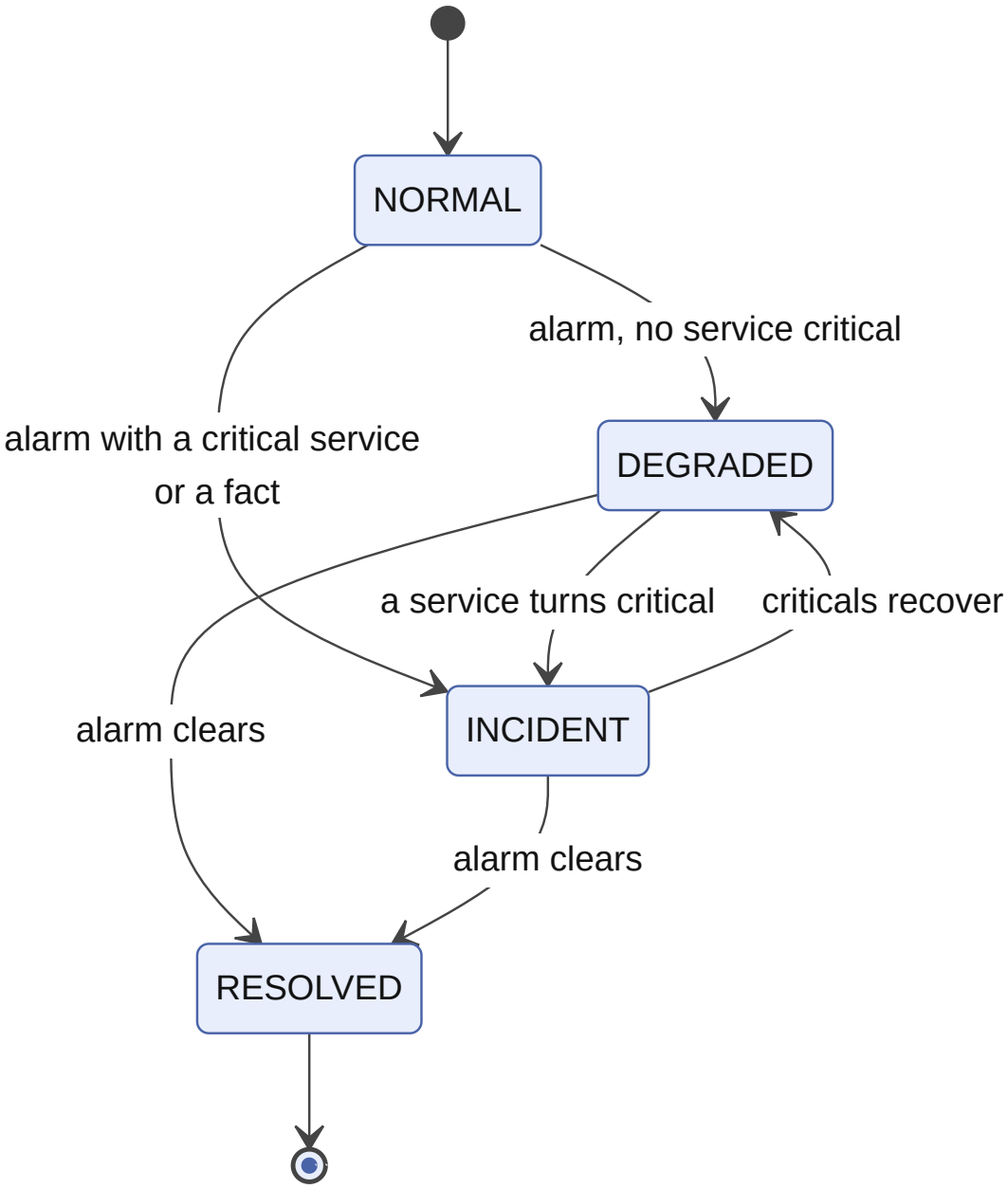


Figure 3.5: Incident lifecycle as implemented in the incident manager.

3.5.3 Three engines on the same readings

The dashboard has a switch with three positions: Rules, Hybrid and ML. Only the active engine decides the incident, but all three are computed on every cycle from the same readings and graph (Figure 3.6). This makes two things possible: switching engines does not start the learned detector from nothing, and the Detection engines page can show what each engine detected, computed and predicted at the same moment.

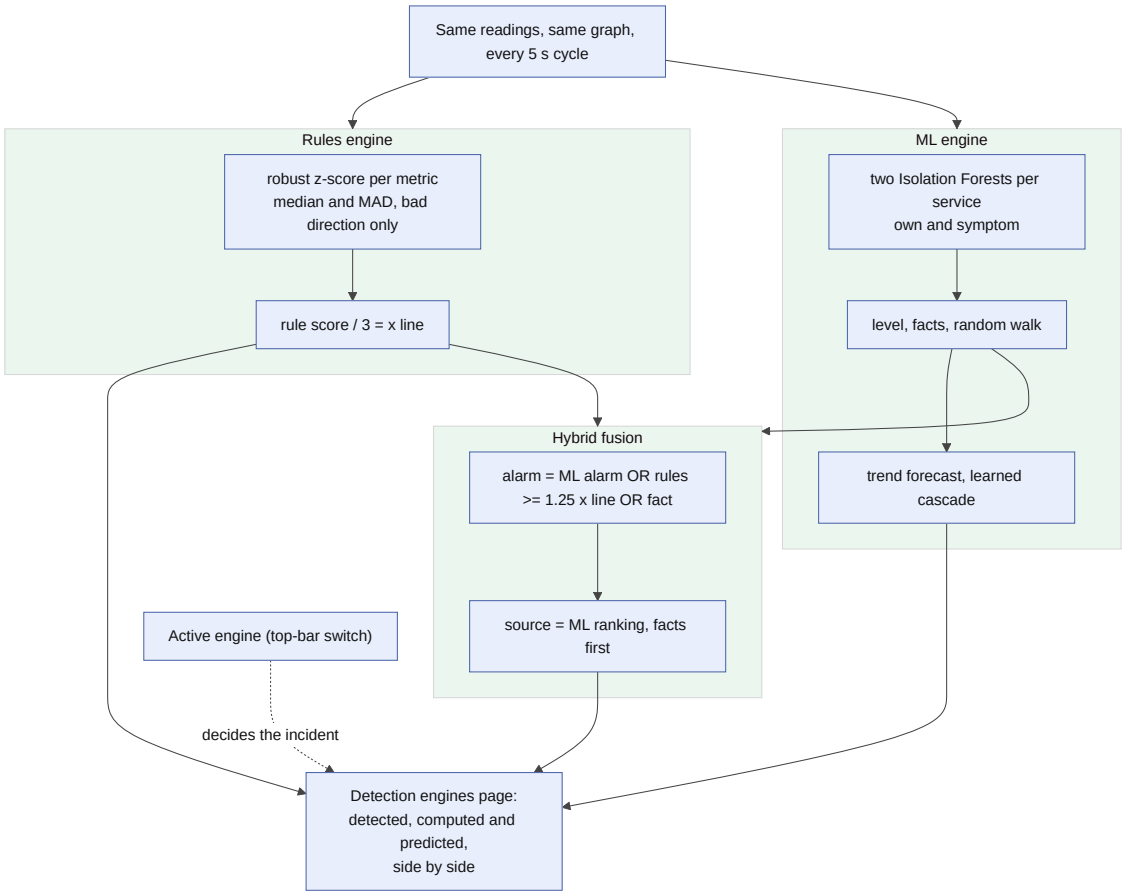


Figure 3.6: The three engines in parallel. The rules score, the ML engine and the fused Hybrid rule are computed on the same inputs; the switch only chooses which one decides the incident.

3.5.4 Messages and data exchanged

Table 3.5: The main messages exchanged between components.

Message	From → to	Content
Reading	Collector → MetricStore	Eight metrics per service with missing values kept as missing
Connection event	Collector → TraceStore	Caller, callee, port, state, connection key
Graph update	Server → dashboard	Nodes, edges, weights, targets (WebSocket)
Incident update	IncidentManager → dashboard	Root cause, ranking, evidence, cascade, runbook
Engine info	Server → dashboard	Mode, per-engine verdicts, configuration, offline scores
Ingest payload	Standalone collector → server	Nodes, edges, metrics, connection events

3.6 Functional Components

Table 3.6 lists the main components with their responsibility and location in the repository.

Table 3.6: Main components.

Component	Responsibility	Location
ConnectionManager	One SSH session per target; fixed command catalogue; timeouts sized for a loaded host	`server/telemetry-platform`
MetricCollector	Docker statistics, socket tables, interface counters; orients each socket	`server/telemetry-platform`
EndpointDiscoveryEngine	Topology from Compose and running containers; routes and endpoint health	`server/discovery`
GraphStore	Nodes, edges, targets; merges declared and observed edges; prunes stale ones	`server/graph`
EdgeWeights	Evidence confidence and dependency share per edge	`server/graph`
MetricStore, TraceStore	Ring buffers of readings; connection events with a 24-hour window	`server/telemetry`, `server/traces`
Detection algorithms	Rule scores, container facts, features, thresholds	`server/algorithms/detection`
ML engine	Isolation Forests, random walk, forecast, learned cascade, hybrid fusion	`server/algorithms/ml`
RCAEngine, IncidentManager	Ranking, explanation, incident lifecycle	`server/algorithms/rca`, `server/rca`
RiskEngine, CascadeEngine	Per-service risk, what-if failure, blast radius	`server/graph`
TrafficController	Starts and stops load; resolves the target URL; any project is accepted	`server/traffic`
traffic-gen	Workflows, profiles, virtual users	`traffic-gen`
Evaluation harness	Records runs, scores engines against a ledger	`run/cli`, `v2`
Dashboard	Views, 3D scene, engine switch	`client/src`

3.7 Scope and Constraints

In scope. Docker-based services on Linux hosts reachable over SSH; container-level metrics, socket tables and logs; unsupervised detection; ranking; cascade prediction; fault typing; a dashboard; a reproducible evaluation.

Out of scope. Distributed tracing (the "Traces" page shows sampled TCP connection snapshots, not requests); application payloads; Kubernetes clusters (not tested); a language-model agent (discussed as future work); training on labelled data.

Constraints that shape the design.

- *Coarse signals.* Observation from outside gives eight metrics per service every 5 s. A brownout on an almost idle service moves none of them enough to alarm; Chapter 6 reports this as an open problem.
- *Each socket is seen from both ends.* Direction has to be recovered, or every dependency would appear as a cycle.
- *Loaded hosts answer slowly.* Command timeouts are sized for a host under heavy load. A host that cannot answer `docker ps` at all shows nothing, and Section 6.1.12 gives a measured example.
- *One write.* The collection path only reads. The one write the mapper makes on a host is restarting a container that exited with a non-zero code, limited to a few attempts per container and switched off with `MAPPER_AUTO_RESTART=off`. Load generators are never restarted by it.

CHAPTER 4

METHODOLOGY

4.1 Objective of the Study

The study asks how far an instrumentation-free, unsupervised pipeline can go in answering four operational questions about a running microservice system: is something wrong, where did it start, who is next, and what kind of fault is it. It also asks, with equal weight, how reliable the answers are. On the synthetic track the settings were chosen on a tuning split kept apart from the data used for the reported numbers; on the real track they were not tuned on a held-out system, and the report says so where it matters.

4.2 Method and Approach

The approach is a pipeline of small, separately testable models rather than one large model. Each stage has a stated input, a stated output and a measured contribution (Section 6.1.4). The pipeline is built in this order:

1. an **evidence-weighted graph** from observed sockets and declared dependencies;
2. **learned call directions** for pairs the heuristic cannot orient;
3. a **hybrid detector** of per-service Isolation Forests and a robust rule layer, with container facts;
4. a **ranker** built on a deviation decomposition and a personalised random walk;
5. a **cascade model** with Beta–Binomial edge probabilities and noisy-OR risk, plus trend-based early warning and change-point alarms;
6. a **fault-type classifier** and runbook.

The evaluation then replays recorded runs through the engines and scores them against a ledger the engines never see.

4.3 Measurement and Evaluation Parameters

Table 4.1 defines the measures. They are event-level on purpose: point-wise scoring with a window adjustment rewards an engine that flags one point of a long fault, which is not how an operator experiences an alarm [23, 24, 25].

Table 4.1: Measures used in the evaluation.

Measure	Definition
Detected (Det.)	Share of faults for which an alarm during the fault (or within a 60 s grace period) places a truly involved service in the top three
Caught and named (C+N)	Share of all faults that are detected and for which the faulted service is first at the last alarming window before the fault heals
A@k	Share of detected faults whose true cause is in the top k
False alarms per hour	Alarm episodes outside every fault window, per fault-free hour
Alarm time	Fraction of fault-free time spent in alarm
Detection delay	Seconds from fault onset to the first qualifying alarm
AUROC, average precision	Service-level discrimination: positives are the faulted service during a fault; negatives are any service when no fault is active; container facts are excluded
Brier score, ECE	Accuracy and calibration of who-is-next probabilities
Direction accuracy	Share of observed true-call pairs oriented correctly, wrongly or left unresolved
Cost	Milliseconds per decision window on one core

Two operating contracts. An operator can budget false-alarm *episodes* per fault-free hour, but one unbroken alarm counts as a single episode, so an engine can meet an episode budget while alarming almost continuously. We therefore also report *alarm time*. An earlier version of our own detector had a low episode rate and an alarm time of 86% on the real systems; the refinements of Section 4.6.3 bring it to 5%.

4.4 Tools, Equipment and Resources

Table 4.2: Tools, equipment and resources.

Item	Detail
Operator machine	Apple Mac mini running the server and the dashboard; Node.js 22
Target hosts	AWS EC2 `c7i-flex.large` instances (2 vCPUs, 4 GiB) in `ap-south-1`, Ubuntu, Docker, Docker Compose
Server stack	TypeScript, Node.js, Express 5, `ws` (WebSocket), `ssh2`
Dashboard stack	React 19, Vite 8, TypeScript, `@xyflow/react` (2D graph), Three.js with react-three-fiber (3D), Recharts
Optional ML sidecar	Python 3 with scikit-learn (Isolation Forest, LOF, one-class SVM) and a z-score baseline
Applications under test	ShopFlow (13 services), LedgerLine (12 services), Sock Shop, a DeathStarBench social network (27 containers), the OpenTelemetry demo
Load and faults	`traffic-gen` (workflows and profiles), a blind fault injector (kill, pause, cgroup limits, network detach)
Analysis and typesetting	Python with matplotlib for figures; LaTeX and Mermaid for this report
Test tooling	Node's built-in test runner; 329 tests in 19 suites

4.5 Experimental and Implementation Procedure

1. **Prepare the applications.** Start ShopFlow and LedgerLine with Docker Compose on the lab hosts.
2. **Connect the mapper.** Add each host (address, user, key) in the dashboard; the server discovers containers and Compose dependencies and starts the 5-second cycle.
3. **Drive traffic.** Run open-loop traffic with a daily wave, bursts and flash crowds for 217 minutes per system.
4. **Inject faults blind.** The injector draws faults from a Poisson process whose rate follows the load, applies them physically to containers, and writes the ledger to a private file.
5. **Record.** The collector's readings and container facts are recorded in a trace format that the engines can replay offline.
6. **Replay and score.** Each engine runs on the recorded trace; scores are computed against the ledger after the run.

7. **Extend synthetically.** A generator writes runs in the same format over 16 further topologies, tuned on one split and scored once on a disjoint split.
8. **Report.** Intervals, paired tests, an ablation and an amplification study are generated by scripts in the repository.

4.6 The Models

4.6.1 The evidence-weighted graph

Each edge has a *confidence* $c(e)$, the strongest evidence behind it: 1.0 for a call seen in request logs or oriented by a listening port, 0.85 when oriented only by the ephemeral-port range, 0.70 when observed with no direction evidence, 0.50 when the direction is unresolved, and 0.40 when declared in the Compose file but never seen carrying traffic. The *dependency share* $d(e)$ of an edge leaving a caller is its slow-baseline traffic normalised over the caller's edges. A slow baseline is used because a link goes quiet exactly when its callee dies; weighting on the instantaneous rate would make a dead dependency look less and less important. The weight is

$$w(e) = \mathrm{clip}\big(c(e)\, d(e)\, \phi,\, 0,\, 1\big),$$

with ϕ discounting fire-and-forget telemetry edges to a tenth. *The offline benchmarks replay recordings that do not keep edge evidence, so they use unit weights; the effect of the evidence weighting is therefore not measured in this report.*

4.6.2 Learned call directions

The collector sees most pairs from both ends (in the recorded runs 15 of 32 pairs in one system and 18 of 36 in the other), so direction must be recovered before a walk can follow it. The shipped heuristic reads names (an entry point calls a service, a service calls a datastore) and fails when names carry no role. We instead learn the direction of each observed pair online from the services' own histories, with no labels and no names. Four *dynamic* signals use what services do during trouble: upward symptoms, downward traffic drops, implication ($P(T_a \mid T_b) - P(T_b \mid T_a)$, with T meaning "in trouble"), and a Granger-style F-test on the symptom series. Two *structural* signals use the observed graph: the depth of each service from the entry point, and whether a service reports request metrics of its own (a datastore does not). The learned direction is the sign of $\sum_m \omega_m s_m(a,b)$ with weights $\omega_m \in \{0.5, 1, 2\}$ chosen by grid search on the tuning split. A pair is left undecided if the magnitude is below 0.05.

4.6.3 Per-service Isolation Forests

A service is judged against its own recent past, never against a global threshold, because services differ by orders of magnitude in every metric. The eight features are split into two groups, each with its own forest of 100 trees trained on a sliding window of the last $W = 120$ readings judged normal: *own* (CPU, memory, throttling), which a dependency cannot cause, and *symptom* (traffic, errors, latency, network), which can be inherited. A reading is scored by the forest built *before* it arrived. With $\mathbb{E}[h(x)]$ the mean path length and $c(\psi)$ the usual normaliser, the raw score $s(x) = 2^{-\mathbb{E}[h(x)]/c(\psi)}$ becomes a level

$$\text{level}(x) = \frac{s(x) - \tilde{s}}{\ell - \tilde{s}}, \qquad \ell = \tilde{s} + \kappa \cdot 1.4826 \cdot \text{MAD}(s_{\text{train}}), \tag{4.1}$$

where $\tilde{s}$ is the median training score and $\kappa = 3$. Level 1 is the service's own alarm line. Only readings with level below one join the training window, so a fault never becomes the baseline. Four refinements were each found by testing and are ablated in Section 6.1.4:

- **training jitter** at each feature's floor, so that a feature that never moved can still register a later jump;
- a **range guard** counting floors beyond the training range, because a forest scores an out-of-range value no higher than its most extreme training point;
- **one-sided features**: less CPU, memory, throttling, error or latency than usual is never a fault of the service itself (it happens after every restart);
- **two-speed drift**: a small steady shift (level < 3) is accepted as the new normal within two minutes, a large one is quarantined for thirty.

4.6.4 Rules and container facts

A robust one-sided z-score per metric (median and MAD, bad direction only) gives a rule score ρ_s normalised so that 1 is its three-sigma line. A container that is exited, paused, restarting or network-detached is a *fact*: evidence, not degree. Two facts that raw container state misses were added after inspecting the real runs. A `docker stop` exits with code zero, which looks like a finished job, so a service that had been running and then exits is *stopped*. A crash loop looks healthy between kills, so two outages within three minutes mark it.

4.6.5 The hybrid alarm

Let λ_s be a service's forest level and $f_s \in \{0, 1\}$ its fact flag. The hybrid alarms at window t if

$$\max_s f_s = 1 \quad \text{or} \quad \max_s \lambda_s \geq \tau_\lambda \quad \text{or} \quad \max_s \rho_s \geq \tau_\rho, \tag{4.2}$$

with $\tau_\lambda = 2$ and $\tau_\rho = 1.25$: the rules alone are trusted only well past their own line. In the live system a fact raises the alarm at once and the other two conditions must hold for two consecutive windows; the offline evaluation requires two consecutive windows for all three, which is the stricter reading.

4.6.6 Deviation decomposition and the random walk

With own level $o(s)$, symptom level $\sigma(s)$ and callees $C(s)$,

$$\begin{aligned} x(s) &= \gamma \max_{c \in C(s)} a(c), \\ \mathrm{raw}(s) &= \max\big(o(s), \max(0, \sigma(s) - x(s))\big), \end{aligned} \tag{4.3}$$

so a service whose symptoms are fully explained by a sick callee drops out, while one that is abnormal in its own resources does not. The ranker is a personalised-PageRank walk in the MonitorRank form [26]. A walker on the directed graph steps to a callee v with weight $w(u \to v)\, a(v)$, back to a caller with weight $\rho_{\mathrm{b}}\, a(u)$, and stays with weight $\max(0, a(u) - \max_c a(c), o(u))$, restarting on the anomalous set with probability $\alpha = 0.15$. The stationary share π_s is the score and facts rank first. The stay term lets the source, whose trouble nothing downstream explains, collect the most mass.

4.6.7 The cascade model

Each call edge $c \to x$ carries an unknown probability $p(c \leftarrow x)$ that trouble in x reaches its caller c. Whenever a service x turns abnormal, every healthy caller c is one trial; it succeeds if c turns abnormal within one minute. With k successes in n trials and a Beta prior of mean π_0 and strength m the posterior mean is

$$\hat{p}(c \leftarrow x) = \frac{k + m\,\pi_0}{n + m}, \tag{4.4}$$

so a rarely exercised edge starts at its prior and moves with observation. No labels are needed: the trials are the system's own history. All experiments use an uninformative prior ($\pi_0 = 0.5$, $m = 2$). Given a suspected root r, risk flows up the call graph by noisy-OR, the independent-cascade model [37]:

$$P(r) = 1, \qquad P(c) = 1 - \prod_{x \in C(c)} \big(1 - P(x)\, \hat{p}(c \leftarrow x)\big). \tag{4.5}$$

Services above a risk cut are reported as likely next, and $\sum_c P(c)$ estimates the blast radius. Calibration is assessed with the Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (p_i - y_i)^2$ [38], the expected calibration error $\mathrm{ECE} = \sum_b \frac{n_b}{n} |\bar{p}_b - \bar{y}_b|$ over equal-width bins [40], and reliability diagrams.

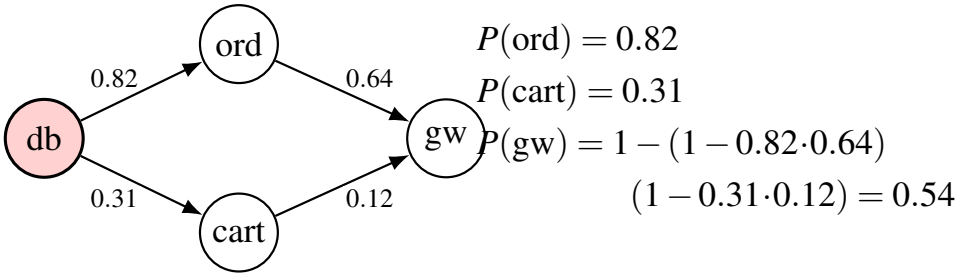


Figure 4.1: The cascade model on a four-service example. Edge labels are learned propagation probabilities $\hat{p}(c \leftarrow x)$; risk at the gateway combines its two routes by noisy-OR.

4.6.8 Early warning, change points and fault typing

For each rising-only feature a least-squares line is fitted over the last two minutes and projected two minutes ahead through the range guard; a service not yet at the alarm level whose trend will reach it is flagged. Separately, online Page–Hinkley change-point detection [41] on the raw metrics can raise the alarm before the forest line is crossed. At the alarm, 35 features of the top suspect's telemetry (container-state history, exit code, number of separate outages, memory slope, CPU and throttling change, latency and error ratios against a pre-fault baseline, network drop, number of anomalous callers) go to a random forest [47] that names the fault type (crash, stop, pause, crash-loop, flap, memory, brownout, latency, partition). A short runbook of two to four concrete checks per type is attached. This layer changes neither detection nor ranking.

4.7 Error Sources and Mitigation

Table 4.3: Error sources and how each is handled.

Source of error	Mitigation
A socket is seen from both ends	Orient by the side that holds the listening port; fall back to the ephemeral-port range; mark unresolved pairs with lower confidence
Short-lived connections between samples	Count recently closed sockets (TIME_WAIT) as well as established ones
A stopped container exits with code 0	Treat “was running, now exited” as stopped
A crash loop looks healthy between kills	Two outages within three minutes mark a crash loop
Services differ by orders of magnitude	Per-service baselines; robust median and MAD
A fault becoming the baseline	Train only on readings below the alarm line; quarantine large shifts for thirty minutes
Restarts lower memory and CPU	One-sided features: only increases count as the service’s own fault
A loaded host answers slowly	Command timeouts sized for a loaded host; the check tool reports a host that cannot answer
Optimistic evaluation	Event-level scoring; held-out synthetic split scored once; bootstrap and paired tests; negative results reported
Ledger parsing	A regression test after a bug that hid half of the real faults (Section 4.9.5)

4.8 Data Processing and Analysis

Readings keep missing values as missing. Each recorded trace stores, per window, the readings, container facts and the engine inputs, so the engines can be replayed deterministically. Warm-up (the first five minutes of each run) is excluded from false-alarm and ROC counts. Intervals are 95% percentile bootstrap intervals over faults [51]; comparisons between engines are *paired* bootstrap differences over the same faults, with two-sided p values; seed-averaged results are means over seeds. Overlapping faults within a run are not independent, so intervals are, if anything, optimistic. With 62 real faults a difference below about 8 points cannot be resolved, and no claim is made below that.

4.9 Evaluation Design

4.9.1 Systems and traffic

Two containerised applications were built for the study. *ShopFlow* is a 13-service retail checkout system (HTTP fan-out, two Postgres instances, two Redis caches, a NATS consumer). *LedgerLine* is a 12-service retail-banking transfer system (a saga orchestrator, a double-entry ledger, a third-party FX provider, a batch worker). Both run for 217 minutes under open-loop traffic with a daily wave, bursts and flash crowds. Each has its own EC2 host, Fault Lab and Grafana (Section 5.9).

4.9.2 Blind fault injection

Faults arrive by a non-homogeneous Poisson process whose rate follows the load, each with a class, a role-weighted target, a heavy-tailed duration and a severity; with some probability a follower is spawned in the target's one-hop neighbourhood, so incidents overlap. Faults are physical (`SIGKILL`, pause, cgroup CPU and memory limits, network detach) and nothing is labelled inside the system under test. The injector writes a private ledger with, per fault, the true root cause and the services structurally able to be affected, read only after the run. The two systems contribute 62 faults over seven classes (Figure 4.2).

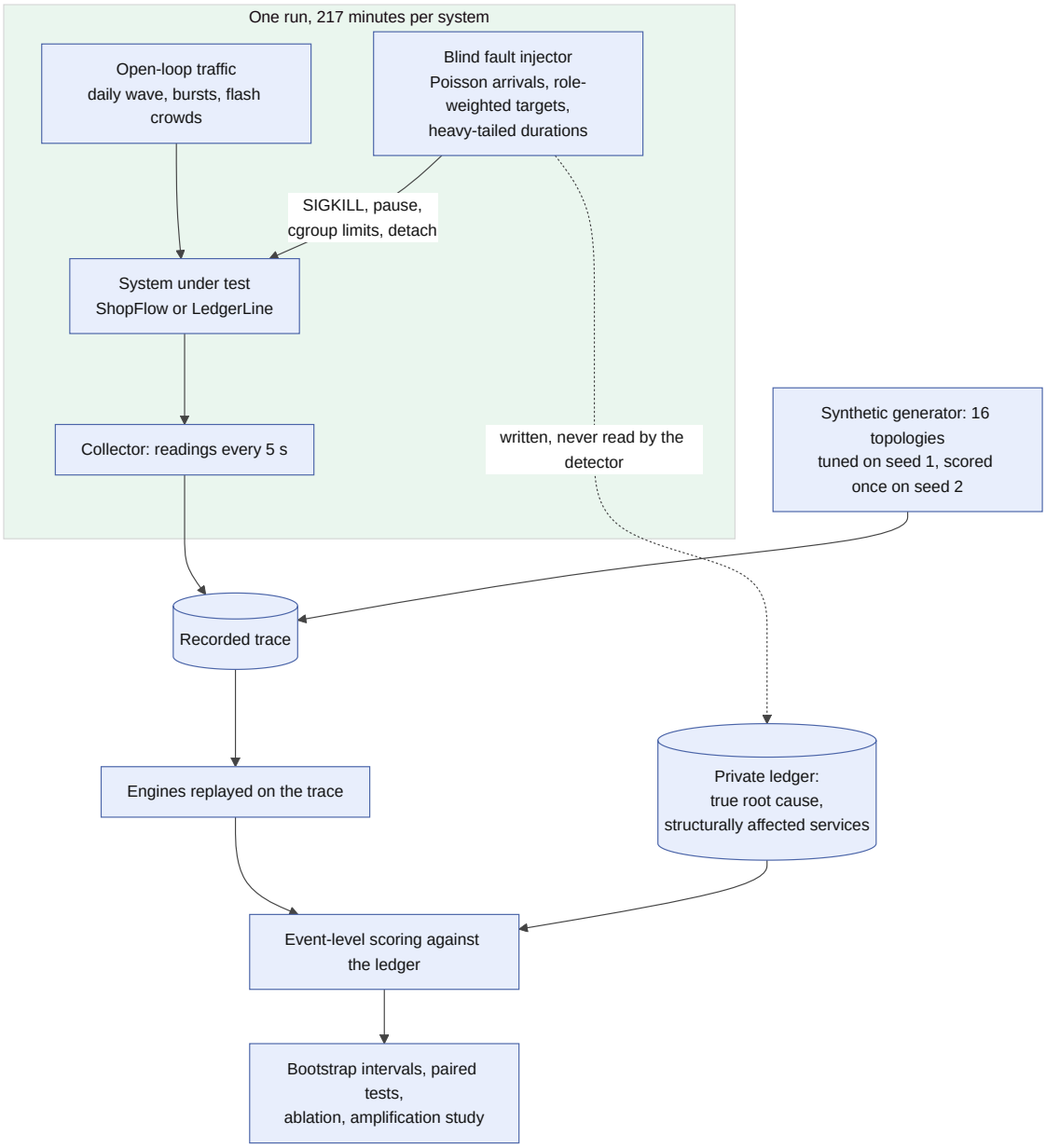


Figure 4.2: Evaluation method. The ledger is written by the injector and read only when scoring.

4.9.3 Synthetic benchmark

To enlarge the sample a generator writes runs in the identical recorded format: the two topologies with their measured per-service baselines plus random topologies, nine fault classes that propagate up and down the graph, overlapping faults and unlabelled noise (noisy neighbours, deploys, missing readings). Settings were chosen on a 16-campaign tuning split (170 faults, seed 1), and every reported synthetic number comes from a disjoint 16-campaign split (178 faults, seed 2) scored once. The generator also takes an *amplification factor A* that scales how strongly callers feel a fault (retry storms), and a class filter; it underlies the amplification study.

4.9.4 Questions the evaluation answers

Q1. Does the hybrid name the true cause more often than the rule layer, and how does it compare with ranking by anomaly alone and with the ML engine?

Q2. Which detector separates faulted from healthy services best, and is the Isolation Forest worth keeping?

Q3. How much does each mechanism contribute (ablation), and does graph-aware ranking help when victims are louder than the cause (amplification)?

Q4. Do learned propagation probabilities predict cascades better than fixed ones, and are they calibrated?

Q5. Do learned directions stay correct when names carry no information, and do change-point alarms cut detection delay?

4.9.5 A measurement bug

Each recorded run contains two fault campaigns that both number their faults from one. The ledger reader kept only the second, hiding 14 of 31 faults per system for every engine and counting their outages as false alarms. We fixed the reader, added a regression test, and every number in this report uses all 62 faults.

CHAPTER 5

IMPLEMENTATION AND EXPERIMENTAL WORK

5.1 Implementation Environment

Table 5.1: Implementation environment.

Part	Technology
Language and runtime	TypeScript on Node.js 22 (server run with `tsx`); a small amount of JavaScript in the traffic generator; optional Python 3
Server	Express 5, `ws` for WebSocket, `ssh2` for SSH (no `ssh` binary needed)
Dashboard	React 19, Vite 8, `@xyflow/react` for the 2D graph, Three.js with react-three-fiber for the 3D scene, Recharts, Framer Motion
Persistence	JSON files under `server/data` (graph state, remote configuration, per-target connection events); metric ring buffers in memory
Source size	About 18 400 lines of TypeScript in the server and 17 200 in the dashboard; 5 400 in the evaluation tools; 2 500 in the `v2` studies
Tests	329 automated tests in 19 suites, run with Node's built-in test runner
Version control	Git, with GitHub as the remote

5.2 Module-wise Implementation

5.2.1 Connection and collection

The connection manager keeps one SSH session per target and runs commands from a fixed catalogue. The catalogue is a list of command builders; every argument that reaches a shell is sanitised (container identifiers must match the shapes Docker itself produces, timestamps and tail lengths are validated). Timeouts are sized for a loaded host: on a swap-thrashing 2 GB machine with eighteen JVM services, `docker ps` was measured at 45 seconds, and a timeout shorter than that does not degrade gracefully, it makes the whole target disappear from the graph.

Each cycle the collector asks for the container list, inspects containers it has not seen, reads statistics, and reads each container's socket table. Reading the table through the container's process id on the host (Listing 5.1) works for every image, including images with no shell,

which had silently hidden the front end of one benchmark for weeks before it was changed. The same command returns the interface byte counters, which are exact, where `docker stats` prints values rounded to three significant figures.

```
  dockerContainerTcp: (containerId: string): RemoteCommand<string>
    => ({
    name: "docker.container.tcp",
    // `if` rather than `a && b && c` on purpose: a stopped
   container reports
    // PID 0, and that should read as "no connections" (exit 0,
   empty output),
    // which is true, not as a scan failure the caller warns about
   .
    //
    // The same network namespace's /proc/<pid>/net/dev rides
   along after a
    // marker line: exact byte counters per interface, read in the
    same exec.
    // docker stats prints NET I/O rounded to three significant
   figures
    // ("527MB"), so a rate taken between two of its readings
   moved in steps
    // of ~200 KB/s at that size; these counters are exact.
    command: `PID=$(docker inspect ${sanitizeContainerId(
   containerId)} --format '{{.State.Pid}}' 2>/dev/null); if [ -n "
   $PID" ] && [ "$PID" != "0" ]; then cat /proc/$PID/net/tcp /proc
   /$PID/net/tcp6 2>&1; echo '${NETDEV_MARKER}'; cat /proc/$PID/
   net/dev 2>/dev/null; fi`,
    timeoutMs: 10_000,
    maxOutputBytes: 65_536,
  }),
```

Listing 5.1: Reading a container's socket table and interface counters from the host (RemoteCommand.ts).

5.2.2 Edge direction, weights and the graph

Every connection appears in the socket tables of both ends. Recording "scanned container calls peer" on both sides produced a reverse duplicate for every real link, so the graph was effectively undirected: it put a cycle on every pair, which made "downstream of X" reach almost the whole graph and made blast radius and criticality saturate. Direction is now recovered from listening ports: whichever end owns the listening port is the callee (Listing 5.2). The ephemeral port range is a fallback when a container's listen table could not be read, and when neither settles it the edge is marked unresolved instead of being given a coin flip dressed as a measurement.

```ts
export function orientConnection(
  localPort: number,
  remotePort: number,
  scannedListenPorts: Set<number>,
  peerListenPorts: Set<number>,
): { scannedIsCaller: boolean; evidence: 'direction:listen-port' | 'direction:ephemeral-port' | 'direction:unresolved' } {
  const scannedIsServer = scannedListenPorts.has(localPort);
  const peerIsServer = peerListenPorts.has(remotePort);

  if (peerIsServer && !scannedIsServer) return { scannedIsCaller: true, evidence: 'direction:listen-port' };
  if (scannedIsServer && !peerIsServer) return { scannedIsCaller: false, evidence: 'direction:listen-port' };

  const localEphemeral = localPort >= EPHEMERAL_MIN && localPort <= EPHEMERAL_MAX;
  const remoteEphemeral = remotePort >= EPHEMERAL_MIN && remotePort <= EPHEMERAL_MAX;
  if (localEphemeral && !remoteEphemeral) return { scannedIsCaller: true, evidence: 'direction:ephemeral-port' };
  if (remoteEphemeral && !localEphemeral) return { scannedIsCaller: false, evidence: 'direction:ephemeral-port' };

  return { scannedIsCaller: true, evidence: 'direction:unresolved' };
}
```

Listing 5.2: Which end of a TCP connection made the call (MetricCollector.ts).

Observed and declared edges are merged, and an observed edge expires ten minutes after it was last seen. A connection that is seen only as a recently closed socket (TIME_WAIT, which lingers about a minute) is counted once by its connection key, so a burst of short calls does not look like many links. A call made through the host's address to a port that one service publishes is attributed to that service (a connection to a private or bridge-gateway address on a published port).

5.2.3 Container facts

Listing 5.3 turns the container runtime's state into one of a few facts. A load generator and a container that finished cleanly are *benign*: traffic is started and stopped by people, not by faults.

```ts
export function containerFact(n: ServiceNode): ContainerFact {
  const m = (n.metadata ?? {}) as any;
```

```
  const state = String(m.state ?? '').toLowerCase();
  const base = state.split(/[\s(]/)[0];
  if (m.networkDetached && base === 'running') return 'detached';
  if (base === 'paused') return 'paused';
  if (base === 'restarting') return 'restarting';
  if (base === 'created') return 'benign';
  if (base === 'exited' || base === 'dead' || base === 'removing'
    || state === 'not found' || m.vanished) {
    if (LOAD_GENERATOR.test(n.name ?? '')) return 'benign';
    if (base === 'exited' && exitCodeOf(n) === 0) return 'benign';
    return 'down';
  }
  // No runtime state at all, but the source marked it critical (
    for example: listed by the compose file, absent from docker ps)
    .
  if (!base && n.status === 'critical') return 'down';
  return null;
}
```

Listing 5.3: Container state as a fact (containerFacts.ts).

5.2.4 The Isolation Forest level

Listing 5.4 is the judgement of one reading by a fitted forest: the one-sided guard, the level relative to the line, and the range guard.

```
  judge(v: (number | null)[]): { score: number; level: number } {
    if (!this.models.length) return { score: 0, level: 0 };
    // a one-sided feature below its usual value is judged as if
    it were usual
    const x = this.project(v).map((val, k) => (this.p.oneSided &&
    UPPER_ONLY.has(this.cols[k]) ? Math.max(val, this.fill[k]) :
    val));
    const levels = this.models.map((m) => { const sc = m.det.score
    (x); return { sc, lv: Math.max(0, (sc - m.typical) / Math.max(1
    e-6, m.line - m.typical)) }; });
    const modelLevel = levels.reduce((a, b) => a + b.lv, 0) /
    levels.length;
    // range guard: a value far outside anything seen in training
    would be cut off at the root of every tree, but the
    // forest scores it no higher than its most extreme training
    point. Count how many floors beyond the range it is.
    let rangeLevel = 0;
    if (this.p.rangeK > 0) x.forEach((val, k) => {
      const excess = Math.max(0, val - this.hi[k], this.lo[k] -
    val);
      rangeLevel = Math.max(rangeLevel, excess / (this.p.floors[
```

```
    this.cols[k]] * this.p.rangeK));
    });
    return { score: levels[0].sc, level: Math.max(modelLevel,
    rangeLevel) };
  }
```

Listing 5.4: Judging a reading (serviceBaseline.ts).

5.2.5 Ranking by random walk

Listing 5.5 is the personalised random walk. The “stay” weight is what lets the true source keep the walker: a service whose trouble nothing downstream explains collects the most time.

```
export function rootCauseWalk(ids: string[], a: Map<string, number
    >, edges: WalkEdge[], p: Partial<WalkParams> = {}, own?: Map<
    string, number>): Map<string, number> {
  const { alpha, rho, iterations } = { ...WALK_DEFAULTS, ...p };
  const n = ids.length, index = new Map(ids.map((id, i) => [id, i
    ]));
  const an = ids.map((id) => Math.max(0, Math.min(1, a.get(id) ??
    0)));
  const ao = ids.map((id) => Math.max(0, Math.min(1, own?.get(id)
    ?? 0)));
  const total = an.reduce((s, x) => s + x, 0);
  const out = new Map<string, number>(ids.map((id) => [id, 0]));
  if (!n || total <= 0) return out;

  const callees: { j: number; w: number; mutual: boolean }[][] =
    ids.map(() => []);
  const callers: number[][] = ids.map(() => []);
  for (const e of edges) {
    const i = index.get(e.from), j = index.get(e.to);
    if (i === undefined || j === undefined || i === j) continue;
    callees[i].push({ j, w: Math.max(0, e.weight), mutual: !!e.
    mutual });
    callers[j].push(i);
  }
  // transition rows
  const rows: { j: number; p: number }[][] = ids.map((_, i) => {
    const moves: { j: number; p: number }[] = [];
    let worstCallee = 0;
    // a link whose direction is unknown lets the walker move, but
    cannot explain this service away
    for (const { j, w, mutual } of callees[i]) { moves.push({ j, p
    : w * an[j] }); if (!mutual) worstCallee = Math.max(worstCallee
```

```
    , an[j]); }
24      for (const k of callers[i]) moves.push({ j: k, p: rho * an[k]
        });
25      // stay: trouble not explained by anything it calls, and its
        own (resource) trouble, which nothing downstream can cause
26      moves.push({ j: i, p: Math.max(0, an[i] - worstCallee, ao[i])
        });
27      const sum = moves.reduce((s, m) => s + m.p, 0);
28      return sum > 0 ? moves.map((m) => ({ j: m.j, p: m.p / sum }))
        : [{ j: i, p: 1 }];
29    });
30    const restart = an.map((x) => x / total);
31    let pi = [...restart];
32    for (let it = 0; it < iterations; it++) {
33      const next = restart.map((r) => alpha * r);
34      for (let i = 0; i < n; i++) for (const m of rows[i]) next[m.j]
        += (1 - alpha) * pi[i] * m.p;
35      pi = next;
36    }
37    const s = pi.reduce((x, y) => x + y, 0) || 1;
38    ids.forEach((id, i) => out.set(id, pi[i] / s));
39    return out;
40  }
```

Listing 5.5: Root cause by random walk (randomWalk.ts).

5.2.6 Learned propagation

Listings 5.6 and 5.7 are the two halves of the cascade model: counting trials and successes per edge, and spreading risk up the graph by noisy-OR.

```
1   update(step: number, onsets: string[], abnormal: Set<string>,
      callersOf: (id: string) => string[]): void {
2       // settle trials: success as soon as the caller is abnormal,
        failure when the window runs out
3       const still: Trial[] = [];
4       for (const t of this.open) {
5         const c = this.counts.get(this.key(t.caller, t.callee)) ?? {
        trials: 0, successes: 0 };
6         if (abnormal.has(t.caller)) { c.trials++; c.successes++;
        this.counts.set(this.key(t.caller, t.callee), c); }
7         else if (step >= t.until) { c.trials++; this.counts.set(this
        .key(t.caller, t.callee), c); }
8         else still.push(t);
9       }
10      this.open = still;
11      for (const x of onsets) for (const caller of callersOf(x)) {
```

```
      if (abnormal.has(caller) || this.open.some((t) => t.caller === caller && t.callee === x)) continue;
      this.open.push({ caller, callee: x, until: step + this.p.windowSteps });
    }
  }
```

Listing 5.6: Settling and opening propagation trials (propagation.ts).

```
  risk(root: string, abnormal: Set<string>, callersOf: (id: string) => string[], calleesOf: (id: string) => string[]): Map<string, number> {
    const risk = new Map<string, number>([[root, 1]]);
    // breadth-first up the callers; a few sweeps settle the noisy-OR on small graphs
    for (let sweep = 0; sweep < 4; sweep++) {
      const frontier = [...risk.keys()];
      const seen = new Set(frontier);
      while (frontier.length) {
        const x = frontier.shift()!;
        for (const c of callersOf(x)) {
          if (c === root) continue;
          let keep = 1;
          for (const y of calleesOf(c)) { const ry = risk.get(y); if (ry) keep *= 1 - ry * this.prob(c, y); }
          risk.set(c, 1 - keep);
          if (!seen.has(c)) { seen.add(c); frontier.push(c); }
        }
      }
    }
    risk.delete(root);
    for (const id of abnormal) risk.delete(id);
    return risk;
  }
```

Listing 5.7: Risk by noisy-OR from a suspected root (propagation.ts).

5.2.7 Hybrid fusion and the three engines in parallel

Listing 5.8 is the fusion of rules and ML. The default detection mode is `or-strict`: the alarm value is the larger of the ML level and the rule score divided by 1.25. Listing 5.9 is the part of the incident manager that records, on every cycle, the verdict of each engine whichever one is active.

```
export function fuseMoment(x: MomentInputs, p: FusionParams): Fused {
```

```
  const { detection, ranking, rulesAlone, mlWeight } = { ...HYBRID_DEFAULTS, ...p };
  const ids = x.services;
  const r = (id: string) => Math.max(0, x.rules.get(id) ?? 0), l = (id: string) => x.level.get(id) ?? 0;
  const maxR = Math.max(0, ...ids.map(r)), maxL = Math.max(0, ...ids.map(l));
  const fact = x.hard.size > 0 ? 10 : 0;

  let max: number;
  switch (detection) {
    case 'rules': max = maxR; break;
    case 'ml': max = maxL; break;
    case 'or': max = Math.max(maxR, maxL); break;
    case 'and': max = Math.max(Math.min(maxR, maxL), fact); break;
    case 'or-strict': max = Math.max(maxL, maxR / rulesAlone); break;
    case 'agree': max = Math.max(fact, ...ids.map((id) => Math.min(r(id), l(id)))); break;
  }

  const rankIn = (order: string[]) => new Map(order.map((id, i) => [id, i + 1]));
  const rulesOrder = [...ids].sort((a, b) => r(b) - r(a) || a.localeCompare(b));
  const topW = Math.max(1e-9, ...ids.map((id) => x.walk.get(id) ?? 0)), topR = Math.max(1e-9, maxR);
  const blend = new Map(ids.map((id) => [id, mlWeight * ((x.walk.get(id) ?? 0) / topW) + (1 - mlWeight) * (r(id) / topR)]));
  let score: (id: string) => number;
  if (ranking === 'rules') score = (id) => r(id);
  else if (ranking === 'ml') { const k = rankIn(x.mlRanked); score = (id) => -(k.get(id) ?? 1e6); }
  else if (ranking === 'rrf') { const a = rankIn(x.mlRanked), b = rankIn(rulesOrder); score = (id) => 1 / (60 + (a.get(id) ?? ids.length)) + 1 / (60 + (b.get(id) ?? ids.length)); }
  else score = (id) => blend.get(id)!;
  const ranked = [...ids].sort((a, b) => (Number(x.hard.has(b)) - Number(x.hard.has(a))) || (score(b) - score(a)) || a.localeCompare(b));
  return { max, ranked, scores: ids.map((id) => blend.get(id)!), blend };
}
```

Listing 5.8: Fusing rules and ML (hybrid.ts).

```
      const rulesHotNow = rulesTop >= 1, mlHotNow = r.hot,
```

```
    hybridHotNow = fused.max >= 1;
2       sh.moments++;
3       if (rulesHotNow) sh.rulesHot++;
4       if (mlHotNow) sh.mlHot++;
5       if (hybridHotNow) sh.hybridHot++;
6       const round = (x: number) => +x.toFixed(2);
7       const fact = (id: string) => r.services.get(id)?.fact;
8       sh.now = {
9         rulesTop: round(rulesTop), mlTop: round(mlTop), hybridMax:
     round(Math.min(fused.max, 10)),
10        rulesRanked: [...rulesLine].sort((a, b) => b[1] - a[1]).
    filter(([, v]) => v > 0).slice(0, 3).map(([id]) => id),
11        mlRanked: r.ranked.slice(0, 3),
12        hybridRanked: fused.ranked.slice(0, 3),
13      };
14      sh.engines = {
15        rules: { alarm: rulesHotNow && prev.rules, peak: round(
    rulesTop), suspects: [...rulesLine].sort((a, b) => b[1] - a[1])
    .filter(([, v]) => v > 0).slice(0, 5).map(([id, v]) => ({ id,
    score: round(v), fact: fact(id) })) },
16        ml: { alarm: mlHotNow && prev.ml, peak: round(mlTop),
    suspects: r.ranked.slice(0, 5).map((id) => ({ id, score: +(r.
    rank.get(id) ?? 0).toFixed(3), fact: fact(id) })) },
17        hybrid: { alarm: hybridHotNow && prev.hybrid, peak: round(
    Math.min(fused.max, 10)), suspects: fused.ranked.slice(0, 5).
    map((id) => ({ id, score: round(fused.blend.get(id) ?? 0), fact
    : fact(id) })) },
18      };
19      sh.prev = { rules: rulesHotNow, ml: mlHotNow, hybrid:
    hybridHotNow };
```

Listing 5.9: Recording each engine's verdict on the same cycle (IncidentManager.ts).

5.2.8 Incidents

The incident manager, once per target per cycle, runs the three engines' shared step, builds the evidence for the active engine, calls the root-cause engine, and either updates the open incident or opens a new one. A new incident takes the status `DEGRADED` or `INCIDENT` according to whether any service is critical (Figure 3.5); it is resolved when the alarm clears. Severity is raised when more than a configured number of services are critical, and lowered for a service that nothing depends on and that depends on nothing, which has no blast radius.

5.2.9 Traffic generation

The traffic controller starts and stops load through the traffic generator's HTTP API. Each application has a workflow file (a set of weighted user journeys with think times) and a set of profiles (baseline, moderate, heavy, stress, ramp) that fix the number of virtual users and their pace. An application that has no file of its own gets a generic front-page workflow (Listing 5.10); before that change the server refused every project outside five named ones with "no workflow defined", which made benchmark hosts such as `bench1` undrivable.

```
/** A project with no workflow of its own gets the generic one,
    rather than a refusal. */
const moduleFor = (id) => TARGETS[id] || generic;
```

Listing 5.10: Fallback to a generic workflow (traffic-gen/server.js).

5.3 Algorithms

Algorithm 5.1 is one decision window. Algorithms 5.2, 5.3 and 5.4 give the observation, learning and parallel-engine steps.

Algorithm 5.1 One decision window of the diagnosis assistant

Input: readings $x_s(t)$, container states, graph G with learned directions

Output: alarm, ranked causes, likely-next set, fault type, runbook

1: **facts**: mark each service stopped, paused, restarting, detached or crash-looping
2: **score**: forest levels λ_s (own and symptom), rule scores ρ_s, anomalies a_s
3: **alarm**: apply the hybrid rule; if no alarm, update the propagation counts and stop
4: **rank**: services with a fact first, then by the walk share π_s over G
5: **explain**: report own versus symptom evidence and what the neighbours explain
6: **predict**: $P(c)$ by Eq. (4.5) from the top suspect; early-warning flags
7: **diagnose**: classify the fault type of the top suspect; attach its runbook
8: update $\hat{p}(c \leftarrow x)$ from services that newly turned abnormal

Algorithm 5.2 Orienting the sockets of one container

Input: socket table of container a, listening ports of every container, IP-to-service map

Output: directed edges with direction evidence

for each socket row in state ESTABLISHED or TIME_WAIT **do**
    find peer b from the remote address; if unknown, test the published-port rule
    if the peer owns the remote port as a listening port and a does not own the local port **then**
        edge $a \to b$, evidence *listen-port*
    else if a owns the local port and the peer does not own the remote port **then**
        edge $b \to a$, evidence *listen-port*
    else if exactly one port lies in the ephemeral range **then**
        the side with the ephemeral port is the caller, evidence *ephemeral-port*
    else
        edge $a \to b$, evidence *unresolved*
    end if
    count the connection once by its key (caller ip:port $>$ callee ip:port)
end for

Algorithm 5.3 Learning propagation and predicting the next services

Input: set of services newly abnormal O, set of abnormal services A, call graph

Output: edge probabilities $\hat{p}$; risk $P(c)$ for healthy services

for each open trial (c,x) **do**
    if $c \in A$ **then** record a success
    else if the window of 12 steps ran out **then** record a failure
    end if
end for
for each $x \in O$ and each healthy caller c of x **do** open the trial (c,x)
end for
$\hat{p}(c \leftarrow x) \leftarrow (k + m\pi_0)/(n+m)$ for every edge
for a suspected root r: $P(r) = 1$; sweep the callers upwards, $P(c) = 1 - \prod_{x \in C(c)} (1 - P(x)\hat{p}(c \leftarrow x))$
drop services that are already abnormal; report the rest above the risk cut

Algorithm 5.4 Computing the three engines in parallel on one cycle

Input: nodes, edges, new readings since the last cycle

Output: a verdict for Rules, ML and Hybrid; the incident for the active engine

1: step the ML engine with the new readings (forests, facts, walk, forecast, cascade)
2: score the same moment with the rule layer and divide by its line
3: fuse: Hybrid maximum $= \max(\text{ML level}, \text{rules}/1.25)$, facts $\rightarrow 10$; ranking by the ML order, facts first
4: for each engine record: alarm (over its line now **and** on the previous cycle), peak against its line, top five suspects
5: append the three peaks to a rolling history of 90 cycles
6: build the incident from the *active* engine only

5.4 Application Programming Interfaces

The server exposes 64 REST endpoints and one WebSocket. Table 5.2 groups them.

Table 5.2: The REST API by area (the full list is in Appendix B).

Area	Example endpoints	Purpose
Graph	`GET /api/graph`, `/api/nodes/:id`, `/api/nodes/:id/metrics`	Topology, node detail, metric history
Targets	`GET /api/targets`, `/api/targets/:id/diagnostics`, `/discovery`, `/services`	Registered hosts, discovery summary
Reliability	`GET /api/reliability/cascade`, `/whatif/:id`, `/blast-radius/:id`	Risk, what-if failure, blast radius
Incidents	`GET /api/incidents`, `/:id/evidence`, `/:id/propagation`, `/:id/timeline`	Incident list and detail
Engines	`GET /api/engine`, `/api/engine/info`, `POST /api/engine`	Active engine, per-engine verdicts, switch
Traffic	`POST /api/traffic/start`, `/stop`, `GET /status`, `/entrypoints`	Start and stop load, entry points
Experiments	`GET /api/experiments`, `POST /start`	Experiment history
ML sidecar	`GET /api/ml/status`, `/features`, `POST /retrain`	Optional Python pipeline
Configuration	`GET/POST /api/config/remote`, `POST /api/data/clear`	Hosts, keys, reset
Ingest	`POST /api/ingest`	Push from the optional standalone collector

The WebSocket carries `graph-update` (nodes, edges, targets, every 5 seconds), `incident.updated`, `trace-events` (the live connection feed) and the embedded terminal's data. The dashboard does no polling for graph state.

5.5 Configuration and Deployment

Table 5.3: Environment variables.

Variable	Default	Meaning
`PORT`	3001	Server port
`POLLING_INTERVAL_MS`	5000	Collection period
`METRIC_HISTORY_SIZE`	720	Readings kept per service (one hour at 5 s)
`INGEST_TOKEN`	built-in token	Token for `POST /api/ingest`; change it when the server is reachable by others
`MAPPER_ENGINE`	saved choice, else Hybrid	Engine at start (overrides the saved choice)
`MAPPER_BASELINE_MIN`	30	Minutes of baseline a project needs before its learned rule model is used (until then only container facts raise incidents)
`MAPPER_AUTO_RESTART`	on	Set to `off` to stop the mapper restarting crashed containers
`TRAFFIC_GEN_PORT`	4400	Traffic generator port
`MAPPER_PYTHON`, `PYTHON`	`python3`	Interpreter for the optional ML sidecar
`COMPOSE_PROJECT`	all	Restrict the standalone collector to one Compose project

A host is added in the dashboard with its address, SSH user and key path. The configuration is saved to `server/data/remote_config.json`; private keys are never copied into the repository, and `*.pem` is git-ignored as a backstop. `npm start` installs what is missing and starts the server and the dashboard; `npm run doctor` checks the machine; `npm run hosts` lists hosts and tests port 22; `npm run check-host` and `npm run mapper-edges` diagnose a host and what the mapper holds for it (Section 5.10).

Table 5.4: Engine parameters (defaults).

Parameter	Value	Meaning
Trees per forest / sample size	100 / 64	Isolation Forest size
Training window	120 readings	About ten minutes of normal behaviour
Readings before judging	30	First fit
Refit every	10 readings	Sliding-window refit
Alarm line	median + 3 robust spreads	Level 1
Alarm level / persistence	2 / 2 windows	Forest condition of the hybrid alarm
Rules alone trusted at	1.25 × line	Rule condition of the hybrid alarm
Drift	24 or 360 readings	Mild shift accepted in two minutes; large one quarantined for thirty
Walk restart / iterations	0.15 / 60	Random walk
Propagation trial window	12 windows	One minute
Propagation prior	mean 0.5, weight 2	Uninformative
Forecast window / horizon / R^2	24 / 24 / 0.6	Two minutes fitted, two minutes ahead

5.6 Experimental Setup and Test Cases

Table 5.5: Experimental setup.

Item	Setting
Real systems	ShopFlow (13 services) and LedgerLine (12 services), 217 minutes each
Traffic	Open-loop, daily wave, bursts, flash crowds
Faults on real systems	62 over seven classes, Poisson arrivals, role-weighted targets, heavy-tailed durations
Synthetic tuning split	16 campaigns, 170 faults, seed 1
Synthetic test split	16 disjoint campaigns, 178 faults, seed 2, scored once
Collection period	5 seconds
Statistics	95% percentile bootstrap; paired bootstrap for engine contrasts

5.7 Prototype and System Implementation

The figures below are screenshots of the running dashboard. They were taken on a simulated five-service-plus-datastore shop that the ground-truth simulator drives (10 services, 9 edges), not on the benchmark hosts, so that the same data could be shown with and without a fault. The scenario in the fault screenshots is the loss of the `orders-db` container.

5.7.1 Architecture view

The 2D view draws the dependency graph with a choice of arrangements (clusters, hierarchy, tiers, grid, circle, polygon, radial, spiral) and filters by service type. Nodes show CPU and memory against the container's own allowance; edges show the dependency and, when observed, its activity.

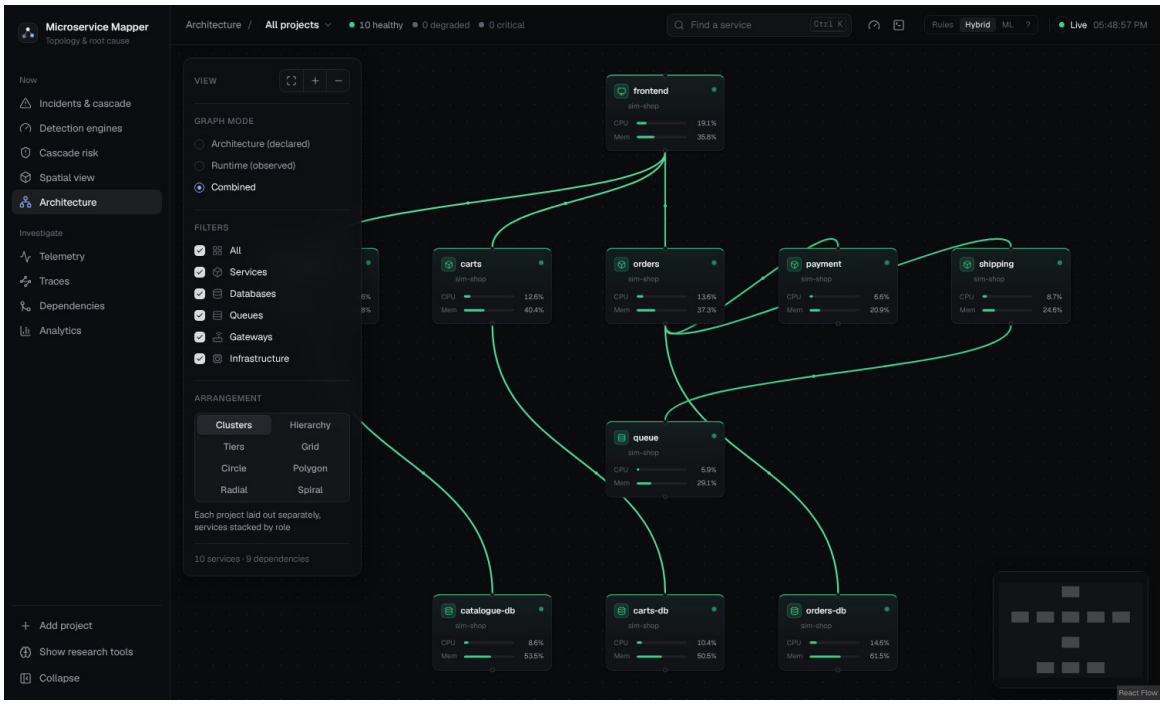


Figure 5.1: Architecture view: the dependency graph of the simulated shop, with arrangement and filter controls.

5.7.2 Incidents and cascade

When the alarm is raised the page shows the most likely root cause, the confidence, the severity and the number of affected services. Below it, the cascade panel shows the source, the services already affected (observed) and those next in line (predicted); the ranking lists each service with the strongest signal behind its score; the "Why" panel splits a score into its own-forest, symptom-forest and walk components and lists the evidence that raised it; and the next steps give concrete commands for the host.

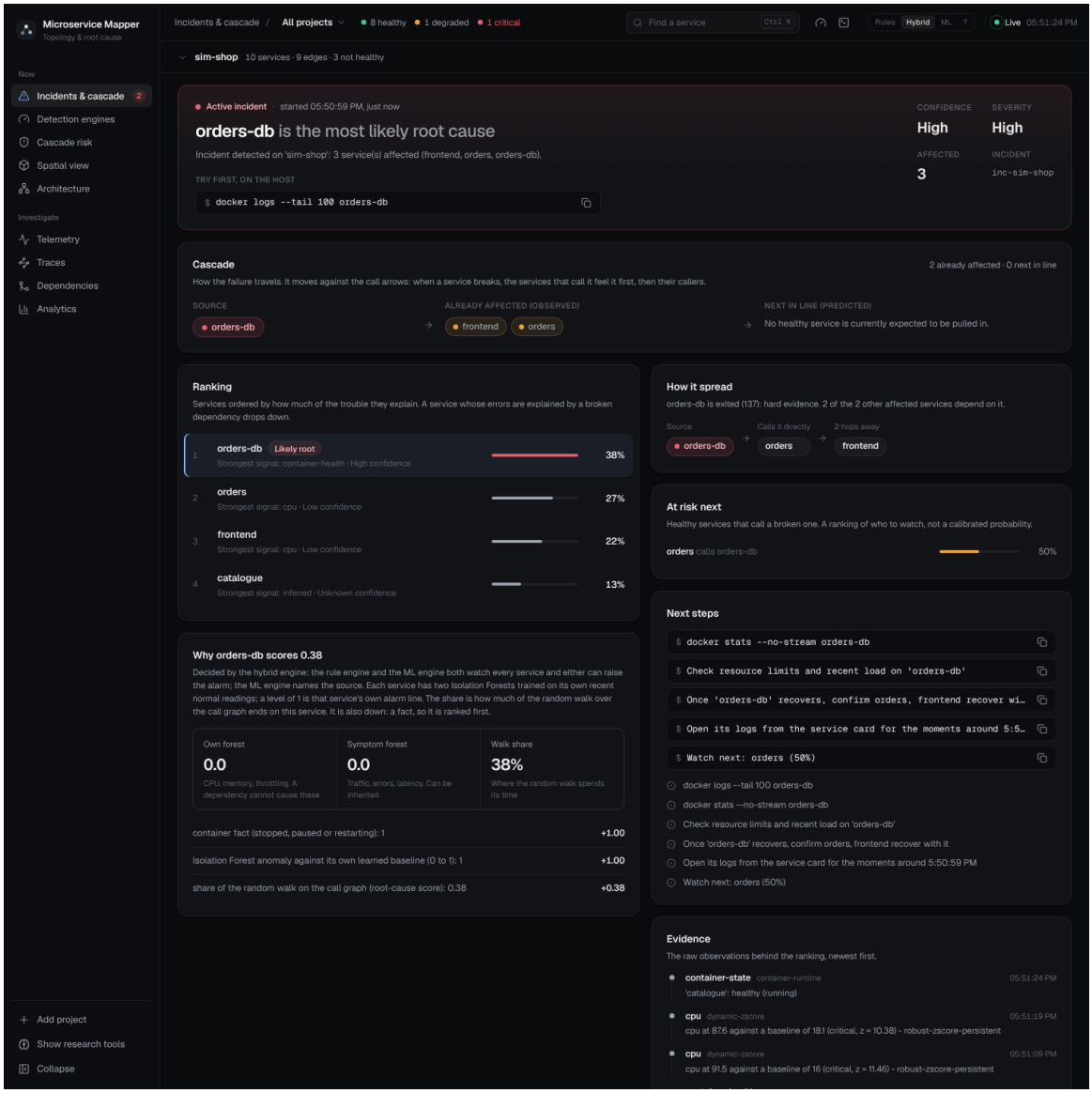


Figure 5.2: An active incident after `orders-db` was killed: root cause, cascade, ranking, evidence breakdown and next steps.

5.7.3 Detection engines

The Detection engines page (Figure 5.3) is the place to see that the engine switch does something. It explains the three engines, defines Hybrid, shows the three engines side by side on the same readings, and then lists the configuration of the rules and of the ML engine and the offline scores. When the system is quiet, all three columns read “quiet”; during the fault in Figure 5.4 all three alarm, and the table of every service shows each engine’s number for each service.

5.7.4 Cascade risk

The cascade-risk page shows what is failing now and on its own measurements, what is exposed, and what would happen if a chosen service failed (what-if), with the model’s name

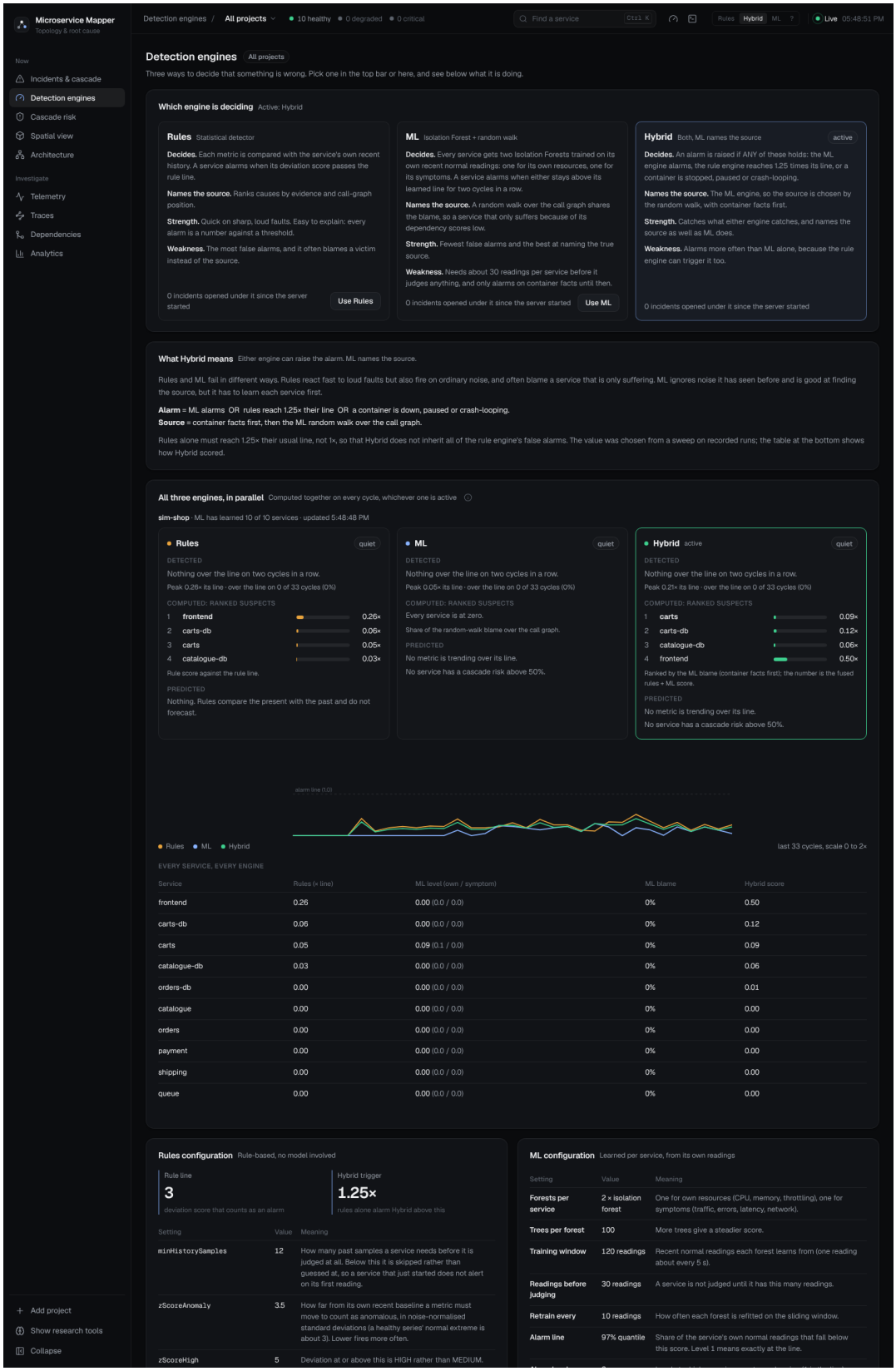


Figure 5.3: The Detection engines page while the system is healthy: the three engines, the Hybrid definition, the parallel view, and the configuration.

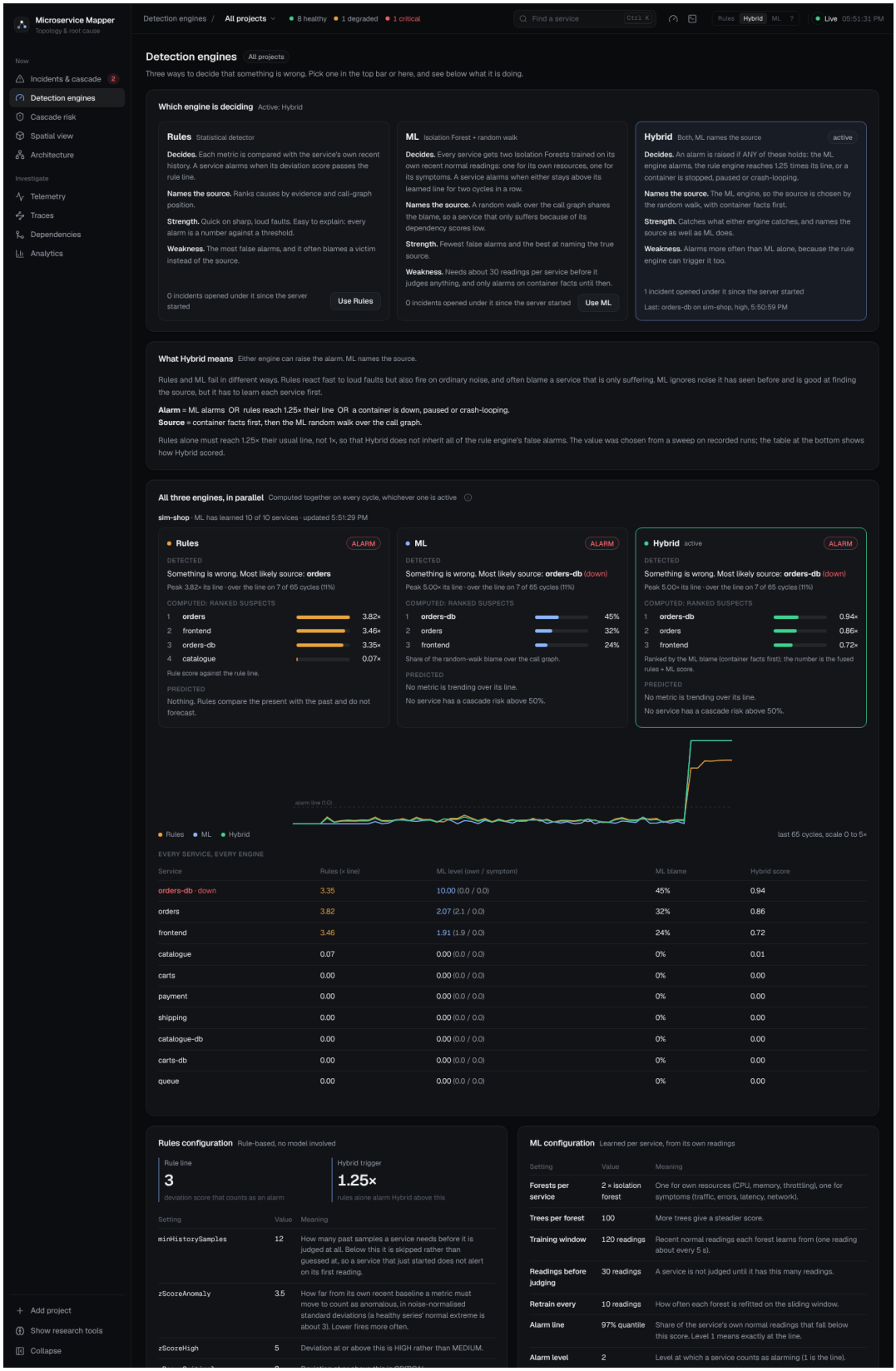


Figure 5.4: The same page during the `orders-db` fault: all three engines alarm; the chart shows each engine's peak against its own alarm line; the table lists every service for every engine.

and parameter beside every prediction rather than a bare number.

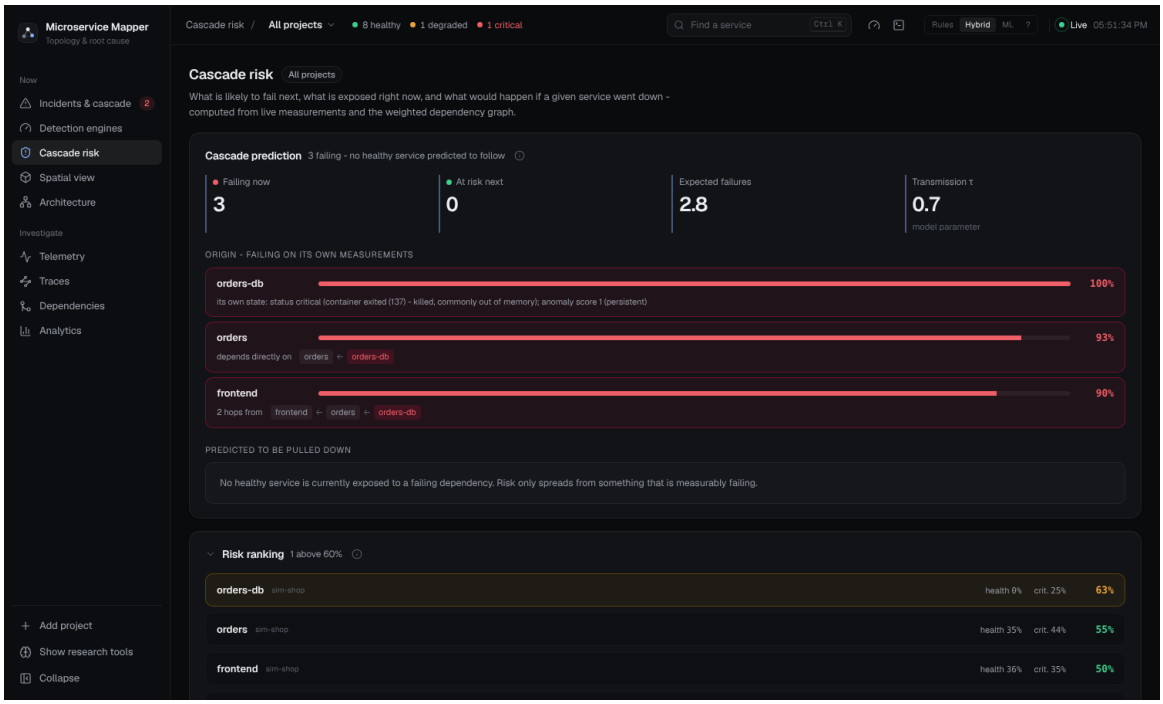


Figure 5.5: Cascade risk during the fault: the origin and the services failing on their own measurements, expected failures and transmission.

5.7.5 Telemetry, traces, dependencies and analytics

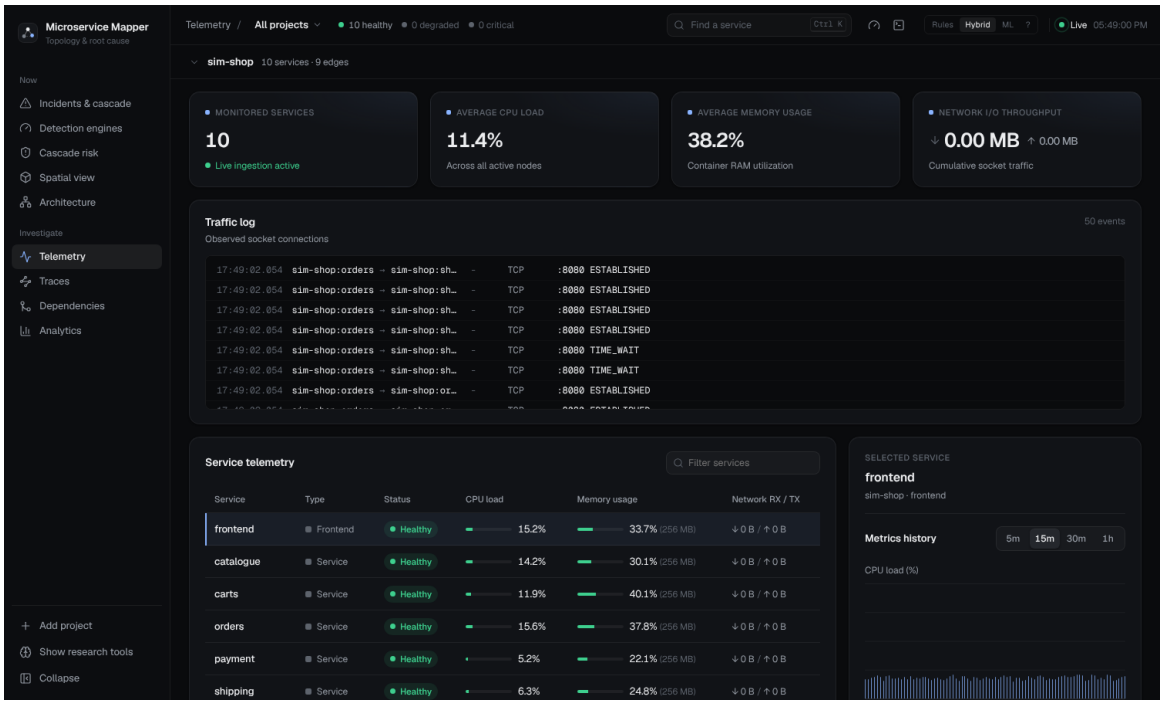


Figure 5.6: Telemetry: fleet totals, the live connection log and a sortable per-service table with a history chart.

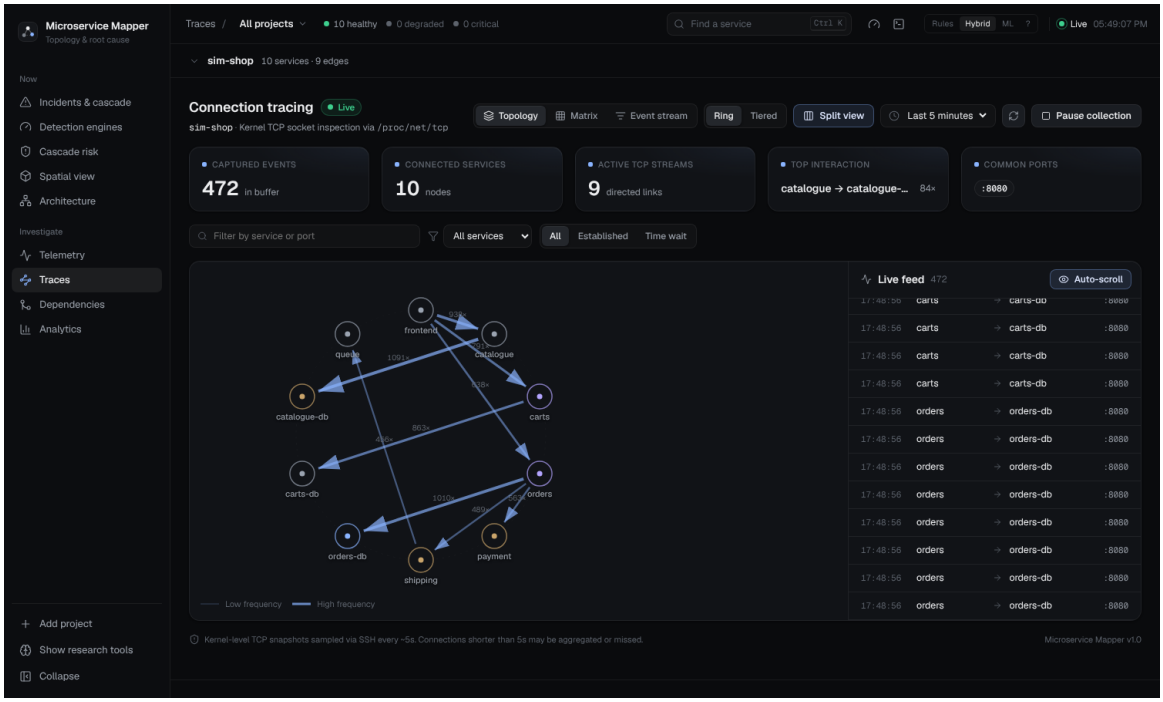


Figure 5.7: Traces: connection tracing with a ring, matrix and event-stream view, summary counters and the live feed of sampled TCP sockets (not requests).

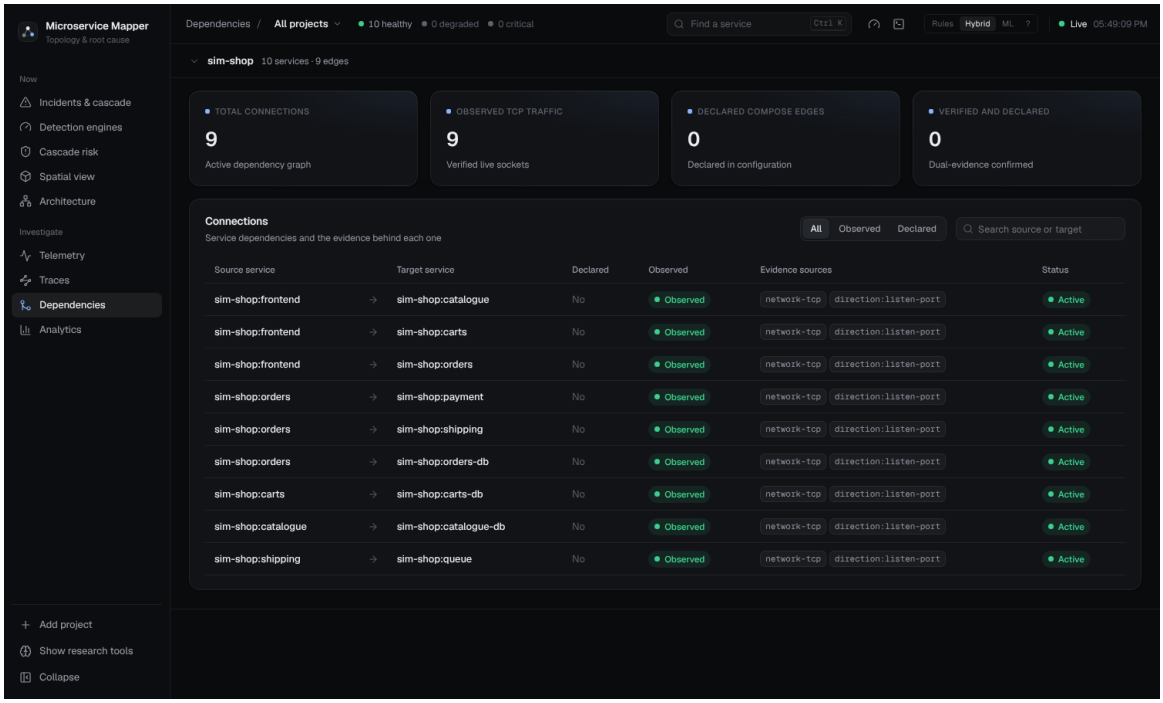


Figure 5.8: Dependencies: every connection with its declared and observed flags and its evidence sources (here *network-tcp* and *direction:listen-port*).

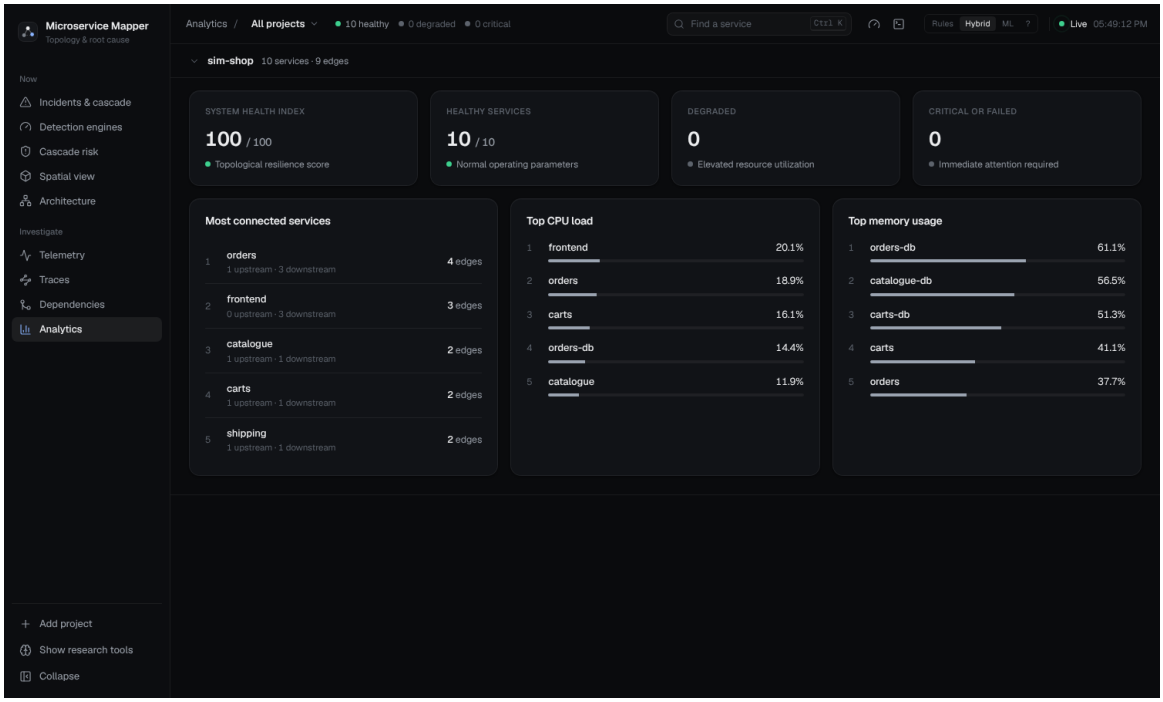


Figure 5.9: Analytics: system health index, counts of healthy, degraded and critical services, the most connected services and the top CPU and memory users.

5.7.6 The 3D spatial view

The 3D scene shows the same graph as a small city. Services are units on platforms; links are cables whose thickness shows the dependency weight and whose colour shows the heat of the link, with pulses that move at a rate driven by the link's request rate. Several arrangements are available (islands, flow, topology, tiers and clusters in the toolbar, more under "More"); Figures 5.10 to 5.13 show four. In Flow the platforms follow the architectural tiers from ingress to data; Topology puts the services on one master deck with a district per role; Tiers stacks the decks in terraces.

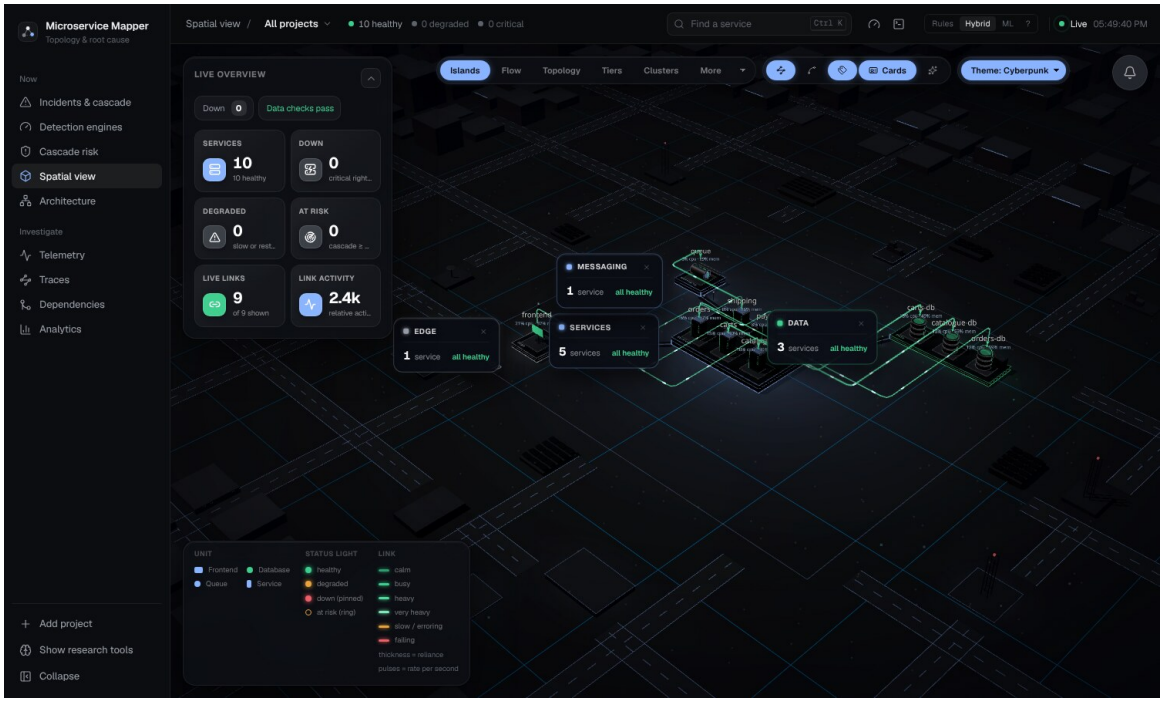


Figure 5.10: Islands: a district with a raised platform per role.

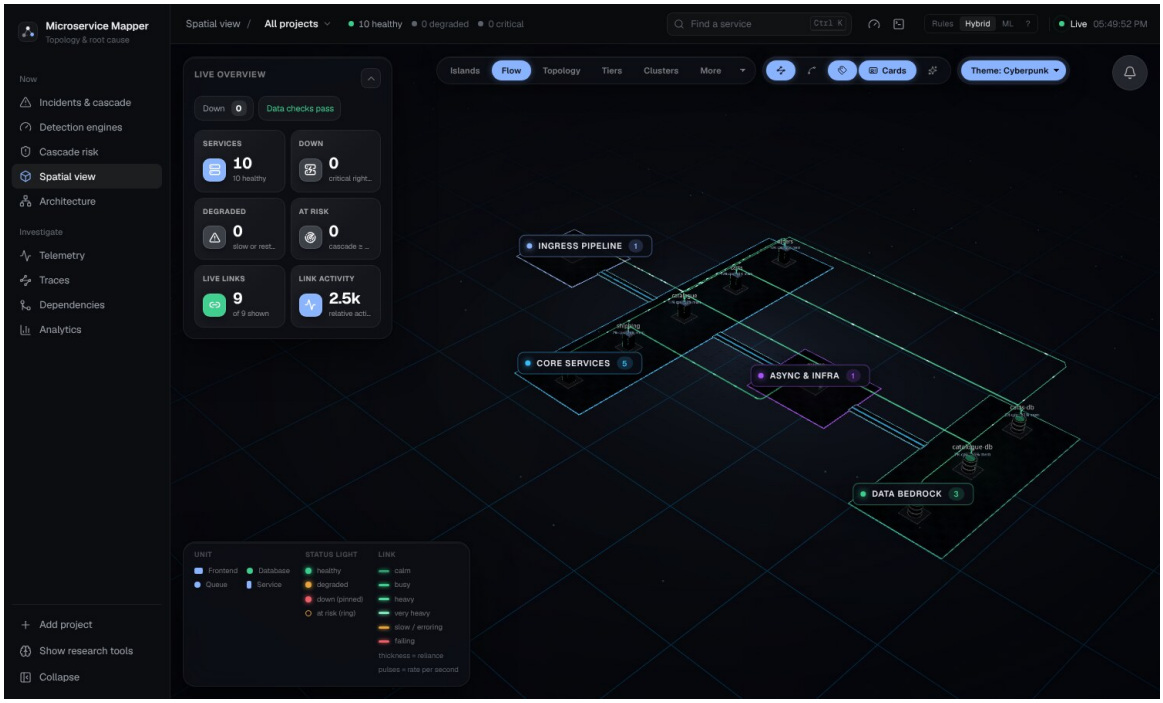


Figure 5.11: Flow: platforms in the order of the request path, from ingress to data.

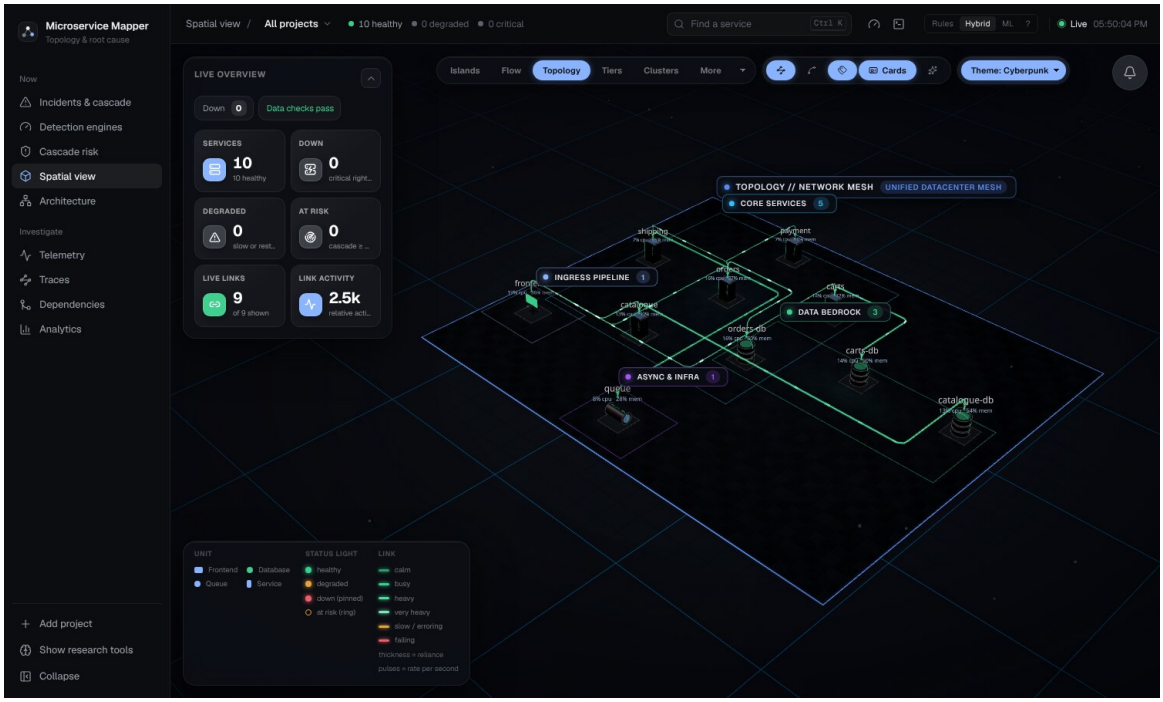


Figure 5.12: Topology: one master deck with a labelled zone per role.

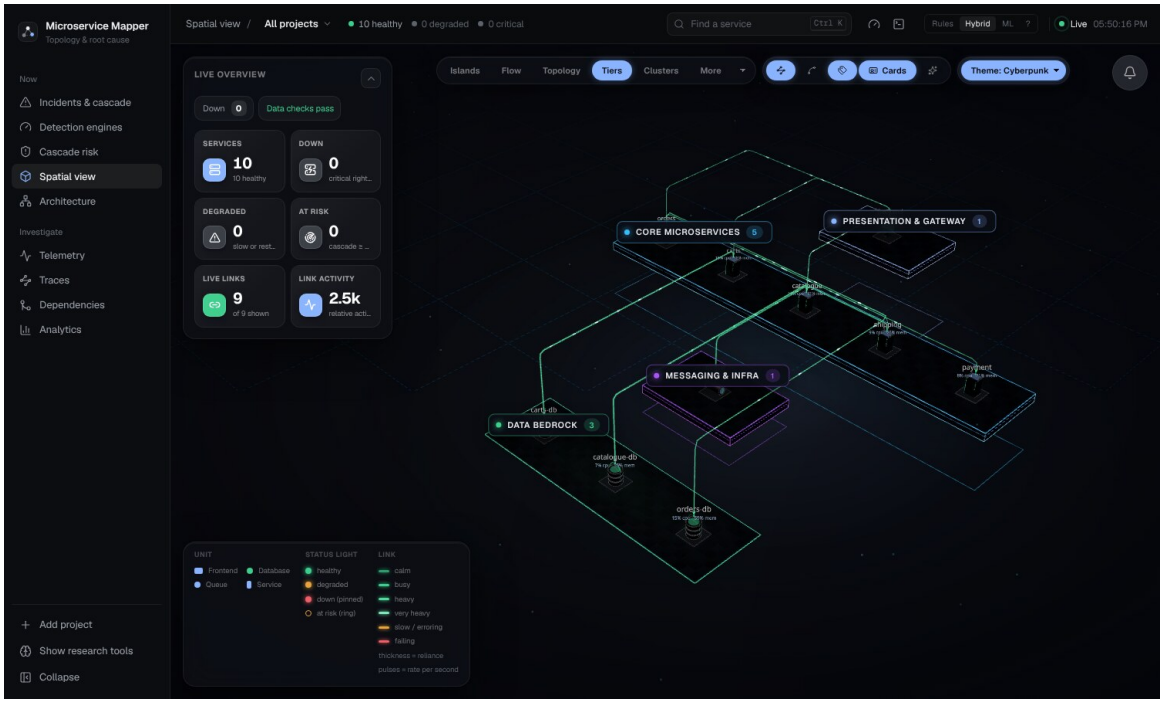


Figure 5.13: Tiers: decks stacked in terraces, one per architectural tier.

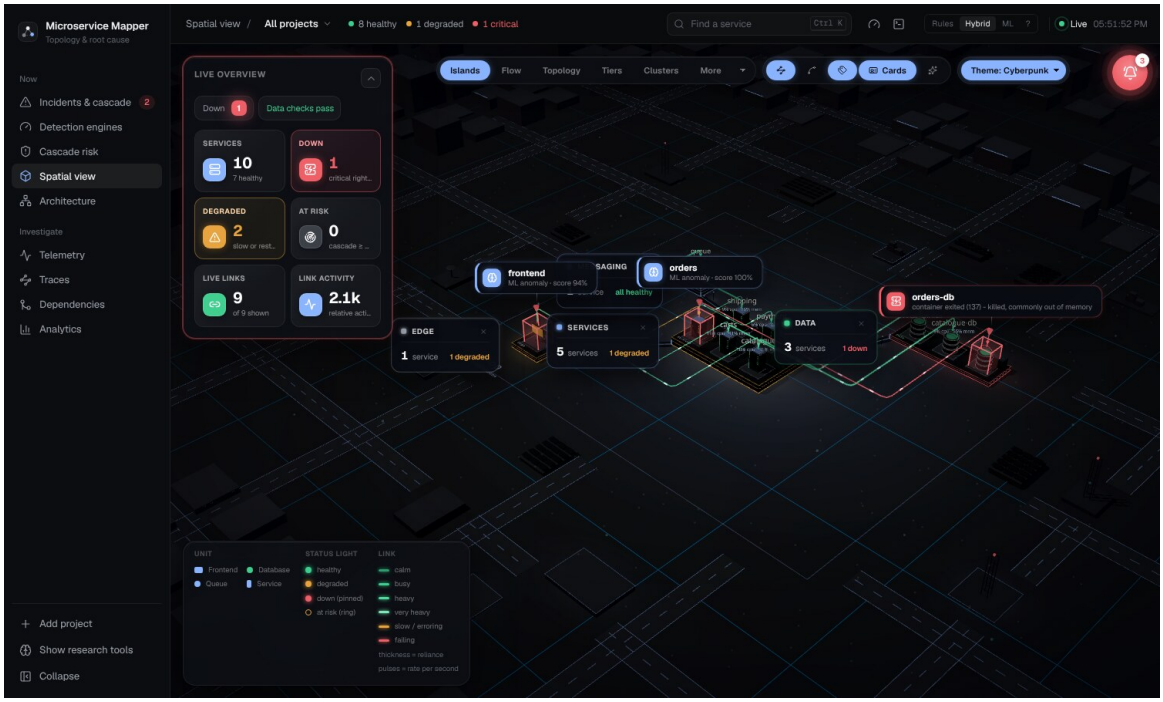


Figure 5.14: Islands during the fault: the failing unit is marked and its callers are lit.

5.7.7 Research pages

Three pages that belong to the study rather than daily operation are hidden until the operator switches them on: the ML pipeline (what the hybrid engine sees per service), the model findings of the optional Python sidecar, and the experiment history.

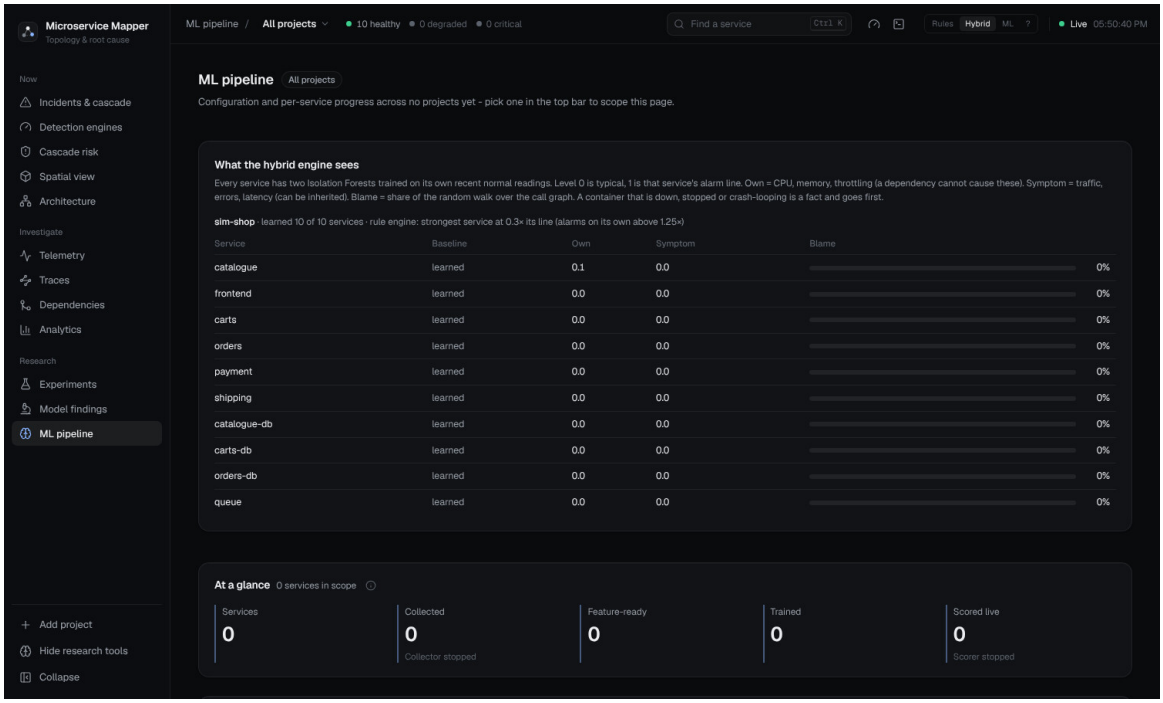


Figure 5.15: ML pipeline page: what the hybrid engine has learned for each service.

5.8 Testing and Validation

The suite has 329 tests in 19 suites and runs in under a minute with `npm test`. Table 5.6 groups them by area. Three kinds of test deserve a note. *Purity tests* check that algorithm modules do not touch the filesystem or the network, so they can be replayed. *Regression tests* hold each bug that was found (for example the ledger-reader bug and the tier layout). *Security tests* check that the command catalogue refuses unsafe identifiers.

Table 5.6: The test suites.

Suite(s)	What is held
`logic`, `riskEngine`, `rca`	Layouts (2D and 3D), edge heat scale, risk, cascade, what-if, root-cause scoring
`ml-engine`, `ml`, `rca-research`	Isolation Forest, walk, propagation, hybrid fusion, learned directions, change points
`pipeline`, `eval`, `purity`	End-to-end replay, scoring against a ledger, purity of algorithm modules
`telemetry-platform.*` (eight files)	SSH connection, command catalogue, Docker parsers, CPU, units, service identity, storage, discovery, metric collector
`sockets`, `requestStats`, `freshness`	Socket direction and recently closed sockets, request statistics, staleness
`security`, `platform`	Command sanitising, portability across operating systems
`edge-cases`, `sparse-traffic`, `target-model`	Missing data, idle services, per-target models

Table 5.7: Representative test cases and the assertions that hold them.

Case	Assertion
3D tiers layout with 19 services	Services wrap into rows instead of one flat line; the row is narrower than 120 units; terraces do not overlap in depth
Socket seen from both ends	The edge points from the end with the ephemeral port to the end with the service port
Recently closed sockets	TIME_WAIT and similar states count as traffic; listening sockets do not
Host-side addresses	Private, loopback and link-local addresses are host-side; public addresses are not
Command catalogue	An identifier with shell metacharacters is refused before any command is built
Ledger with two campaigns	All faults from both campaigns are scored

5.9 The Two Custom Benchmark Hosts: ShopFlow and LedgerLine

The two real systems of the evaluation are not borrowed demos. They were built for this study as *test benches for root-cause-analysis tooling*, each in its own repository and each running on its own EC2 host. The code of both is correct and defensive: every failure comes from configuration, data volume, sizing, topology or a third party, and every failure ships with a machine-readable ground truth (the true root cause, what a monitor should and should not be able to see, and the fix). Each bench has the same four parts: the application, a monitoring stack with Grafana, an always-on traffic generator, and a Fault Lab.

Table 5.8: The two benchmark hosts.

	ShopFlow	**LedgerLine**
Domain	Online retail checkout	Retail-banking transfers
Services	Gateway, catalogue, cart, inventory, payments, orders, an order-event consumer, a third-party payment sandbox; two PostgreSQL, two Redis, NATS (13 services)	Edge, accounts, transfers, fraud screening, FX rates, a double-entry ledger, a statements worker, a third-party FX provider; two PostgreSQL, Redis, NATS (12 services)
Public port	8080 (storefront and API)	8081 (gateway)
Host	EC2 `c7i-flex.large`, Ubuntu 24.04; about 0.6 GiB of containers	the same
Scenarios with ground truth	12 (one baseline control)	7 (one baseline control)
Monitoring	Prometheus, Grafana, Loki, Promtail; 25-panel dashboard	the same stack; 25-panel dashboard

5.9.1 Monitoring: Prometheus, Grafana and Loki

Each bench carries a monitoring stack that is deliberately a *separate Compose project*. The application project stays exactly what a mapper should discover, and monitoring can be started and stopped independently. The stack has Prometheus (24-hour retention), Grafana with a provisioned dashboard, Loki with Promtail shipping every container's logs, and a small exporter that reads per-container CPU, memory, network, restarts, out-of-memory kills and health from the Docker API (the same numbers as `docker stats`). A "dashboard viewer" container queries Grafana every few seconds, as a person watching the dashboard would, so those links are live and visible to a mapper. Everything listens on localhost; Grafana is reached through an SSH tunnel. Figures 5.16 and 5.17 show the provisioned dashboard of each bench: health at a glance (request rate, 5xx ratio, p95 latency, containers running, restarts, memory), request metrics from each service's own endpoint, container metrics from the Docker API, and application internals (database pool waits, unconsumed events, process memory, heap).

Grafana serves two purposes in the study. It is the operator's ordinary view of a healthy and of a faulted bench, and it is a second, independent source against which the mapper's readings can be compared on the same time axis.

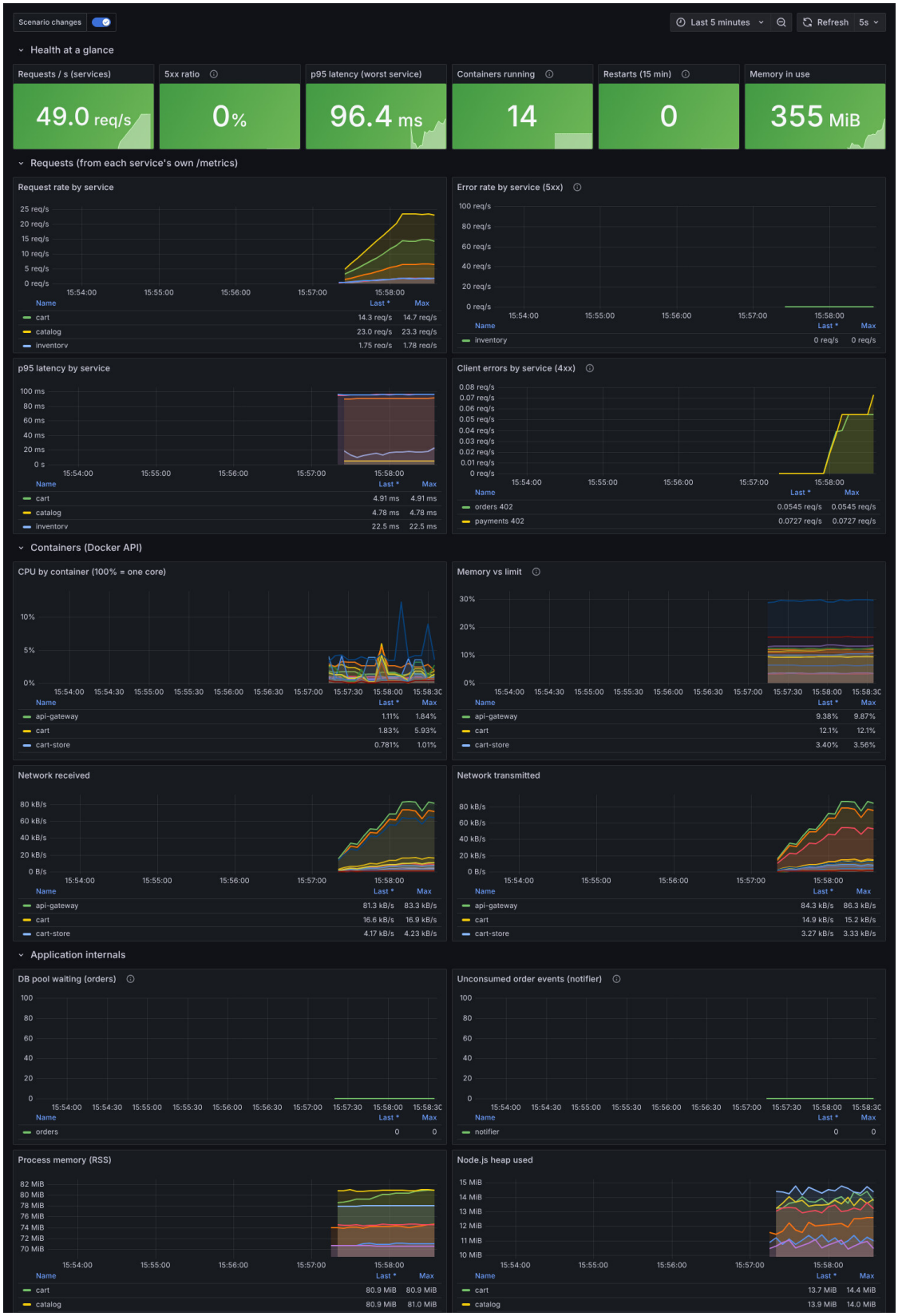


Figure 5.16: The provisioned Grafana dashboard of the ShopFlow bench, taken from the running bench.

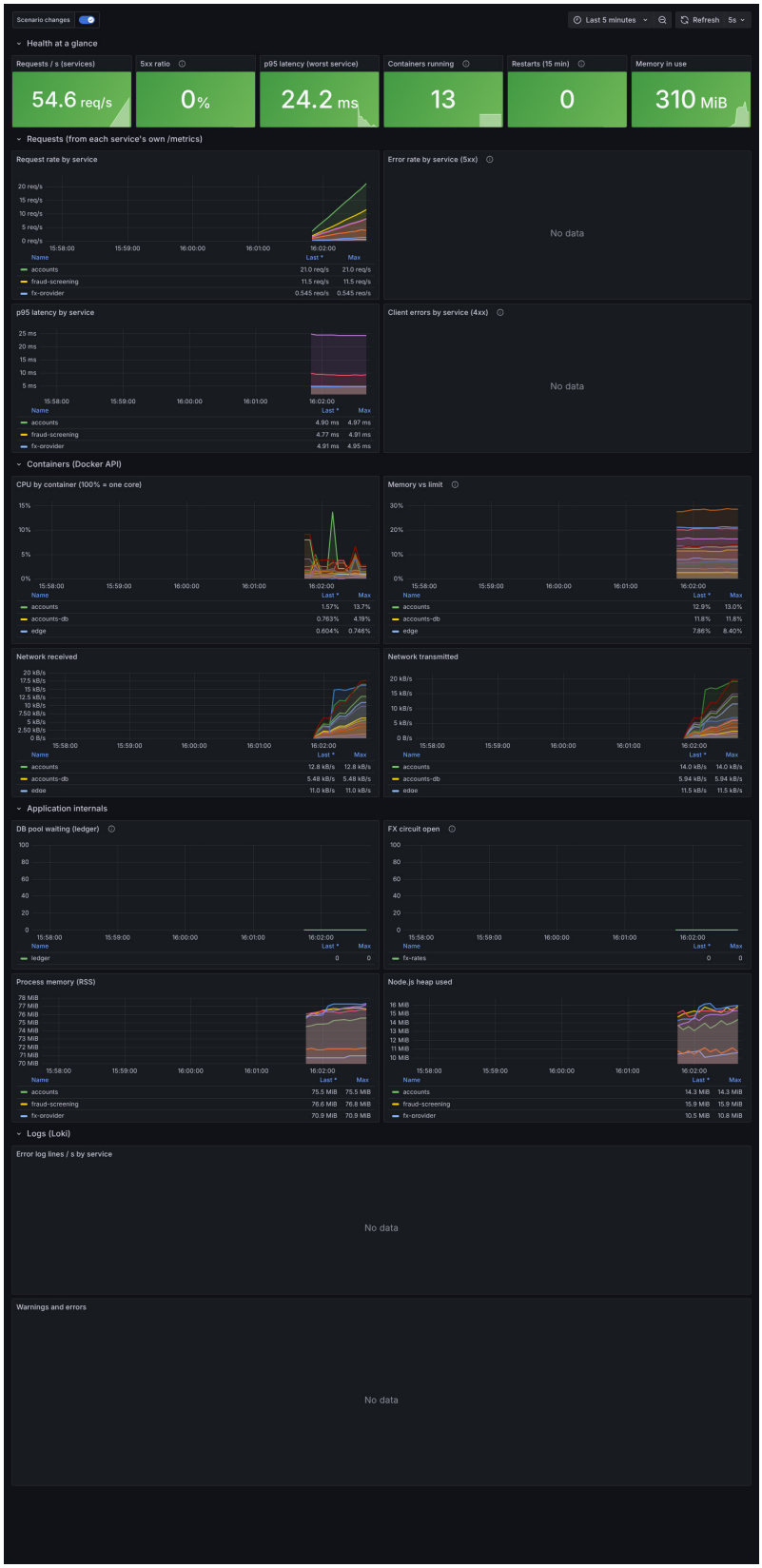


Figure 5.17: The provisioned Grafana dashboard of the LedgerLine bench, taken from the running bench.

5.9.2 Traffic

Each bench has an always-on, open-loop traffic generator. The default mode is *organic*: visitor sessions with think times, popular products, returning users and a day-and-night wave (about twelve workers); a steady mode gives a flat load. A quiet mode stops the generator so that only a person's own clicks reach the gateway. ShopFlow also has a storefront that can be used like a customer, with a live request log and an "autopilot" that browses by itself.

5.9.3 The Fault Lab

The Fault Lab (`lab/faultlab` in each bench repository) is the fault injector of the evaluation. Its properties are those the methodology needs:

- **Blind.** The mapper is never told what is injected. Everything the injector does is written to a private ledger that is read only when scoring.
- **Physical.** Faults act on containers: `SIGKILL` with the restart policy bringing the service back (crash), repeated kills (crash-loop), `docker stop`, pause, resource limits (brownout, memory), network detach (partition) and repeated pause-and-release cycles (flap).
- **Organic and reproducible.** A campaign draws faults from a Poisson process with a profile, a duration, a rate, a maximum number at once and an optional seed; the same seed gives the same campaign.
- **Targeted.** The operator may restrict the fault types and the services that may be hit, or inject one fault by hand with a severity and a hold time, and heal everything at once.

The control page (Figures 5.18 and 5.19) runs on the bench host, listens on localhost only and is reached through an SSH tunnel; it starts the same commands one would type. Its requests need a header the page itself sets, so another site open in the same browser cannot drive it. The same page starts and stops the traffic, sets the average request rate and chooses its shape. The screenshots were taken of the page with no campaign running and away from the bench host, so the service list reads "absent"; on the host each service shows its container state.

5.9.4 Scenarios with ground truth

Besides random campaigns, each bench offers scripted scenarios. A scenario applies a defined cause, drives the load that triggers it, and writes a report beside the expected result stated in its `scenario.yaml`. Tables 5.9 and 5.10 list them. They cover cascades, retry amplification, silent degradation, undocumented dependencies and a circular dependency, which are the cases a diagnosis tool most easily gets wrong.

Table 5.9: ShopFlow scenarios and their true root causes.

Scenario	What happens	Root cause
sf-00-baseline	Healthy control group	none
sf-01-missing-index	Order history is fast in staging, melts the database in production	orders-db
sf-02-cache-thrash	A cache sized for staging evicts itself in production	catalog-cache
sf-03-unbounded-cache	Memory climbs for an hour, then the container is killed for lack of memory	cart
sf-04-retry-storm	A third party throttles; our own retries make it ten times worse	psp-sandbox
sf-05-secret-rotation	A rotated database password; one service still has the old one	orders
sf-06-shadow-dependency	A dependency that exists only in the code	none
sf-07-dead-dependency	A declared dependency nobody calls	none
sf-08-noisy-neighbor	A reporting job on the shared database slows checkout	catalog-db
sf-09-cpu-throttle	Slow but healthy: a container throttled at a quarter of a core	catalog
sf-10-poison-message	One un-processable message stops every confirmation e-mail	notifier
sf-11-wrong-hostname	A renamed service still referenced by its old name	none

Table 5.10: LedgerLine scenarios and their true root causes.

Scenario	What happens	Root cause
ll-00-baseline	Healthy control group	none
ll-01-pool-starvation-cycle	Two healthy services wait on each other and nothing looks busy	accounts ↔ fraud-screening
ll-02-provider-outage	Third-party outage with a circuit breaker: the silence is the symptom	fx-provider
ll-03-hot-row	Every transfer queues on one row	ledger-db
ll-04-connection-budget	Pools that are each reasonable add up to more than the database allows	ledger-db
ll-05-batch-oom	A month-end job that kills itself, restarts and tries again	statements-worker
ll-06-shadow-dependency	fraud-screening quietly depends on fx-rates	none

Not every scenario has been run on EC2: the `verified` field of each `scenario.yaml` records whether it was reproduced in Docker, as plain processes, or only validated statically. The scenario files are the specification against which a running mapper is to be scored; the random campaigns of Chapter 4 are what the reported numbers come from.

5.10 Host and Mapper Diagnostic Tools

Two tools were added after operating the benchmark hosts showed that "nothing appears in the dashboard" has two very different causes: the host cannot answer, or the mapper has a problem.

`npm run check-host` walks one instance step by step and stops at the first failure: whether port 22 answers; whether the key exists with permissions SSH accepts; whether login works; the host's memory, load and disk; whether docker answers within 25 seconds and which containers run; whether the application answers on its port; which peers each front-end container calls and whether they are other containers; and whether the mapper server is up. Each failure says which side to look at.

`npm run mapper-edges` prints what the running mapper holds for a project or service: the target's status and last-seen time, the matching nodes and their status, and every edge in or out with its evidence and last-seen time. Together the two tools separate a host problem from a mapper problem in minutes.

5.11 The Evaluation Harness

Table 5.11: The evaluation harness and the experiments built on it.

Command or folder	Purpose
`npm run eval`	Record a fault-injection campaign and score the live mapper (observe, score, report)
`npm run ml:compare`	Compare detectors and root-cause methods on the recorded runs
`npm run ml:hybrid`	Fusion variants of rules and ML
`npm run ml:predict`	Early warning and cascade prediction
`npm run ml:synth`	Synthetic generator (amplification factor, class filter)
`npm run ml:bench`	Cost of one decision window
`v2/`	Studies: learned directions, change-point onset, fault diagnosis
`paper/exp`	Curves, amplification, cascade, router and scaling data for the figures
`run/recorded-runs`	The two recorded runs

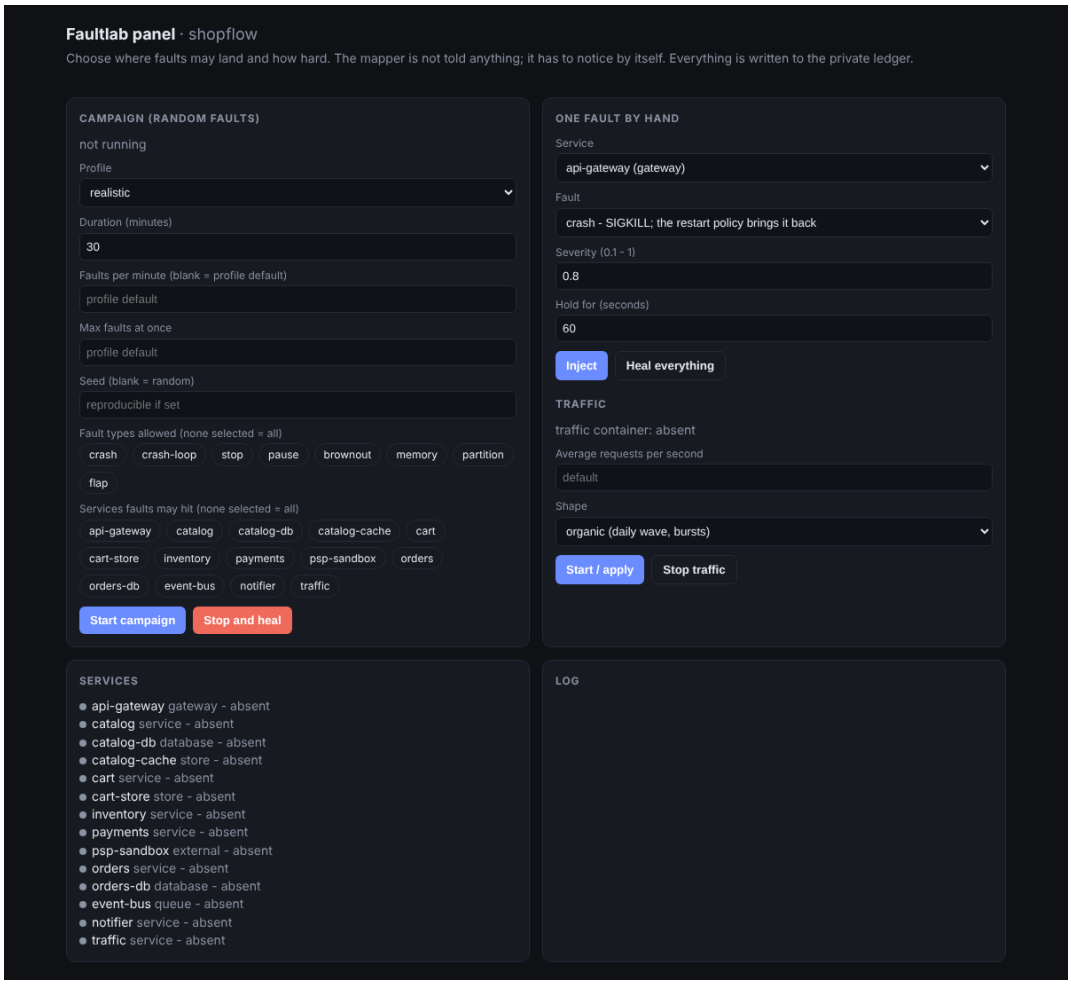


Figure 5.18: The Fault Lab control page of the ShopFlow bench: a random campaign (profile, duration, rate, seed, allowed fault types and services), one fault by hand, traffic control and the service list.

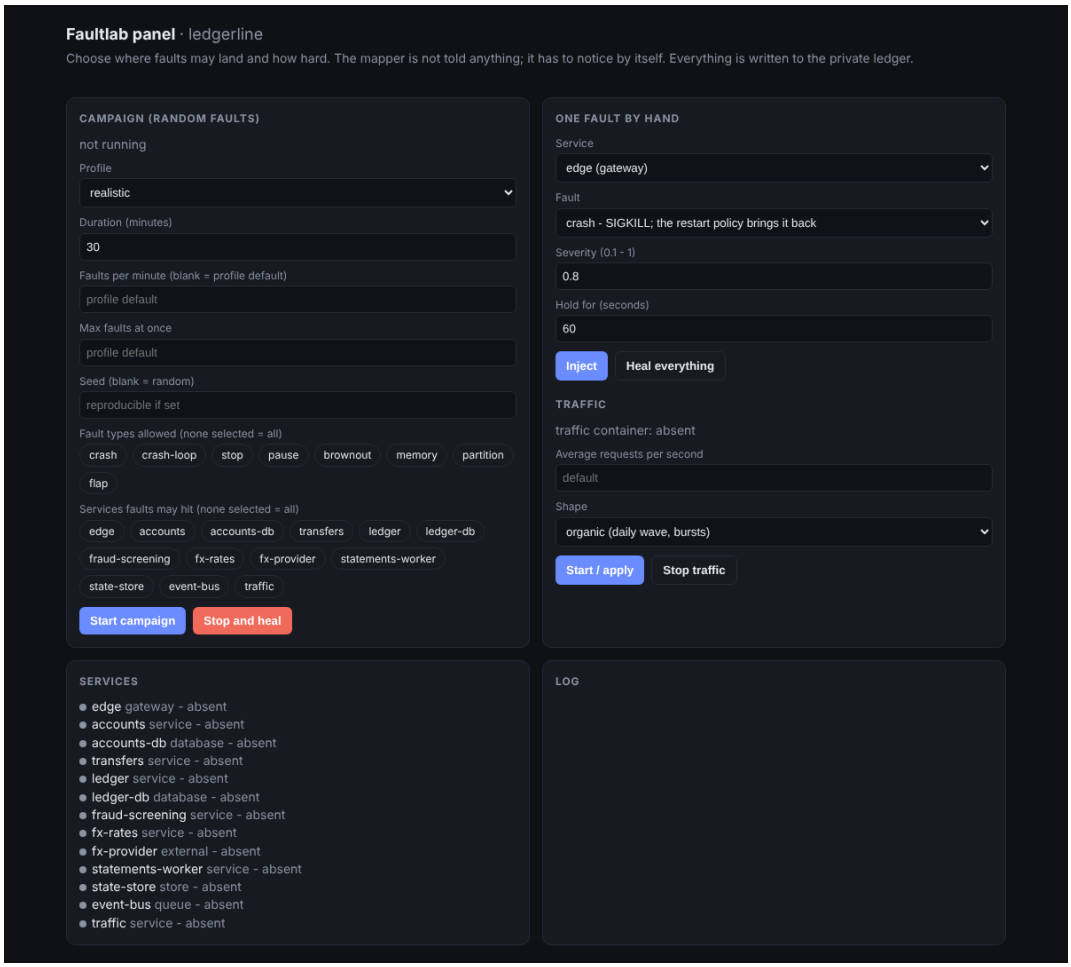


Figure 5.19: The Fault Lab control page of the LedgerLine bench.

CHAPTER 6

RESULTS, ANALYSIS AND DISCUSSION

6.1 Results

6.1.1 Detection and root cause

Table 6.1 compares the engines in one harness. On the real systems the hybrid names the true cause first for 74% of all faults, against 65% for the rule layer, 71% for ranking by anomaly alone and 73% for the ML engine. The paired differences are in Figure 6.2 and Table 6.2. Hybrid over rules is significant on the real systems ($+9.7$ points, 95% CI $[+1.6, +17.7]$, $p = 0.032$) but not on the synthetic set ($+2.2$, $[-3.9, +8.4]$); hybrid over the severity ranking is positive but not significant in either set ($+3.2$ and $+1.1$ points).

Table 6.1: Detection and root cause in one harness (point estimates with 95% bootstrap intervals over faults). Det.: faults detected (%). A@1, A@3: true cause first or in the top three among detected (%). C+N: caught and named first over all faults (%). FA/h: false-alarm episodes per fault-free hour. Alarm: share of fault-free time in alarm (%).

Engine	Det. [95% CI]	A@1	A@3	C+N [95% CI]	FA/h	Alarm
Real benches (62 faults, 217 min)						
Rules	90 [82, 97]	71	91	65 [53, 76]	9.2	9
Severity ranking	90 [82, 97]	79	88	71 [60, 82]	6.2	7
ML engine	87 [77, 95]	83	91	73 [61, 82]	2.3	5
Hybrid	90 [82, 97]	82	89	74 [63, 84]	6.2	7
Hybrid + change-point	95 [89, 100]	78	88	74 [63, 84]	6.2	16
Hybrid + learned dir.	90 [82, 97]	79	89	71 [60, 82]	6.2	7
Synthetic held-out (178 faults, 16 topologies)						
Rules	89 [84, 93]	78	95	70 [62, 76]	2.5	2
Severity ranking	92 [87, 96]	77	91	71 [64, 78]	0.8	6
ML engine	89 [85, 94]	78	93	70 [63, 76]	0.0	5
Hybrid	92 [87, 96]	79	92	72 [65, 79]	0.8	6
Hybrid + change-point	93 [89, 97]	77	91	72 [65, 79]	0.6	7
Hybrid + learned dir.	92 [87, 96]	79	93	72 [65, 79]	0.8	6

The operating curves (Figure 6.1) show why the comparison is delicate: the engines trade detection for false alarms along nearly overlapping curves, and the hybrid sits on the upper envelope in both sets, so its advantage is about operating point rather than a single threshold. Figure 6.3 breaks C+N down by fault class. The gain over the rules is concentrated in *stop* faults (16 of 16 against 9 of 16 on the real systems), which is the effect of the `docker stop`

fact, and, on the synthetic set, in *flap* faults (12 of 12 against 2 of 12).

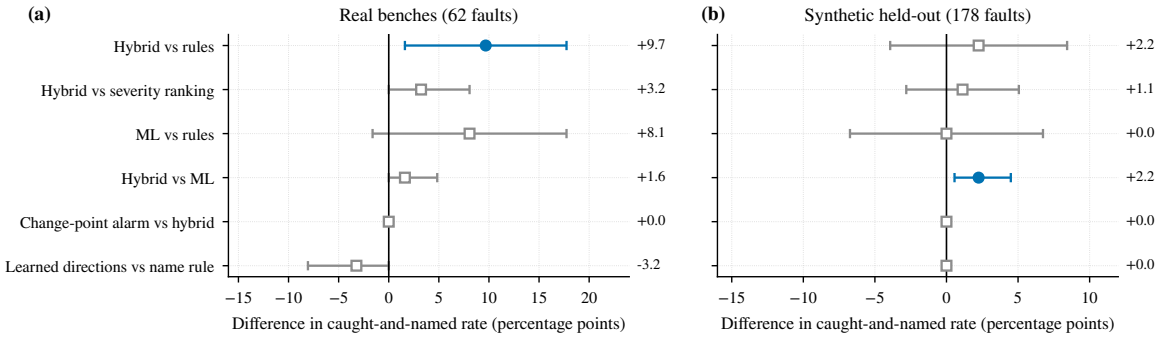


Figure 6.2: Paired differences in caught-and-named rate between engines (percentage points, 95% paired-bootstrap interval); filled markers exclude zero.

Table 6.2: Paired bootstrap tests behind Figure 6.2 (difference in caught-and-named rate).

Set	Contrast	Δ (pp)	95% CI	p
Real	Hybrid vs rules	+9.7	[+1.6, +17.7]	0.032
Real	Hybrid vs severity ranking	+3.2	[+0.0, +8.1]	0.256
Real	ML vs rules	+8.1	[−1.6, +17.7]	0.120
Real	Hybrid vs ML	+1.6	[+0.0, +4.8]	0.707
Real	Change-point alarm vs hybrid	+0.0	[+0.0, +0.0]	1.000
Real	Learned directions vs name rule	−3.2	[−8.1, +0.0]	0.257
Synth.	Hybrid vs rules	+2.2	[−3.9, +8.4]	0.516
Synth.	Hybrid vs severity ranking	+1.1	[−2.8, +5.1]	0.661
Synth.	ML vs rules	+0.0	[−6.7, +6.7]	1.000
Synth.	Hybrid vs ML	+2.2	[+0.6, +4.5]	0.036
Synth.	Change-point alarm vs hybrid	+0.0	[+0.0, +0.0]	1.000
Synth.	Learned directions vs name rule	+0.0	[+0.0, +0.0]	1.000

6.1.2 Which detector

Figure 6.4 and Table 6.3 score each detector as a service-level classifier. The result is not what a learned-detector paper would hope for: the plain rule z-score is as good as or better than every learned detector on discrimination (AUROC 0.696 against 0.635 for the Isolation Forest on the real systems; 0.873 against 0.862 on the synthetic set), and Mahalanobis is the best learned detector on the synthetic set (0.868). Separation on the real systems is weak for every detector (AUROC 0.62 to 0.70), because many real faults hit services that are nearly idle.

The Isolation Forest is therefore kept for what it adds to the hybrid, not for discrimination alone. It is the quietest detector (alarm time of 5% against 9% for the rules on the real

systems, Table 6.1); it separates own from symptom evidence; and its alarms complement those of the rules, so their union detects more faults than either (92% against 89% on the synthetic set). The ensemble of four detectors was not run on the synthetic set to completion and is reported for the real systems only.

Table 6.3: Service-level detection by detector (mean ± s.d. over three seeds; the rule layer is deterministic). Dashes: not run.

	Real		Synthetic	
Detector	AUROC	AP	AUROC	AP
Rule z-score	0.696±0.000	0.340±0.000	0.873±0.000	0.666±0.000
Isolation Forest	0.635±0.009	0.285±0.003	0.862±0.004	0.528±0.020
k-NN	0.657±0.010	0.254±0.024	0.860±0.001	0.517±0.004
Mahalanobis	0.648±0.003	0.309±0.002	0.868±0.001	0.608±0.006
HBOS	0.619±0.008	0.279±0.004	0.853±0.004	0.473±0.030
Ensemble of four	0.648±0.001	0.323±0.003	–	–

6.1.3 Detection delay and root-cause depth

Figure 6.5 shows detection delay as a cumulative distribution. On the real systems the change-point alarm lifts the detected fraction from 90% to 95% and moves the median from 10.5 s to 4.9 s; on the synthetic set, whose faults are abrupt, the curves almost coincide. It does so at a price visible in Table 6.1: alarm time rises from 7% to 16% on the real systems. Figure 6.6 shows the true cause in the top k: the ML-based rankers reach it first more often than the rules on the real systems, and the curves converge by $k = 2$ to 3 near the 90% detection ceiling, so the remaining misses are mostly faults the engine does not see (brownouts on near-idle services) rather than faults it ranks badly.

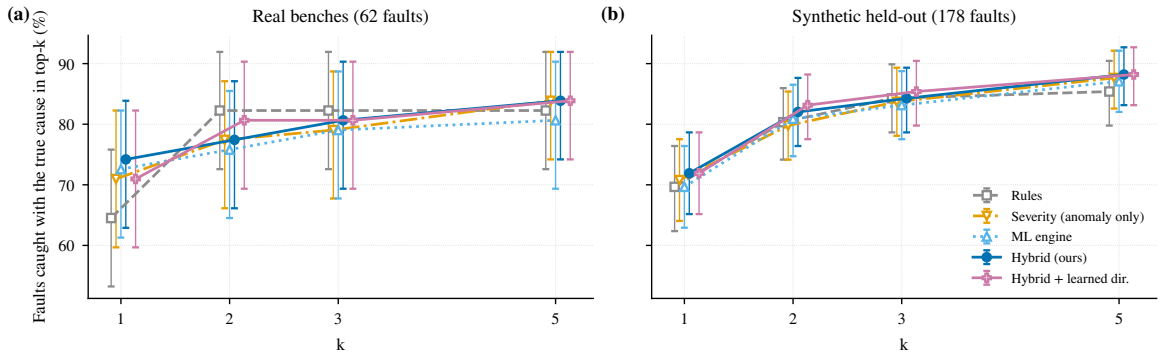


Figure 6.6: Share of faults caught with the true cause in the top k, with 95% bootstrap intervals (offset for legibility), for the real systems (a) and the synthetic held-out set (b).

6.1.4 Ablation

Figure 6.7 removes one mechanism at a time. The most important single component is the `docker stop` fact: without it C+N falls by 19 points on the real systems and 16 on the synthetic set, because every stopped service goes unnamed. The crash-loop fact (3 and 7 points), the range guard (4 and 11 points) and the random walk (3 and 1 points) each contribute. Two mechanisms act on the contract rather than on accuracy. Removing the one-sided features raises alarm time on the real systems from 5% to 30%. Alarming at level one instead of two raises it to 22% and false alarms to 11.3 per hour while changing C+N by only $+2$ points. The one-sided features do not matter on the synthetic set, where restarts do not lower memory: an effect of the real systems that the simulator does not reproduce.

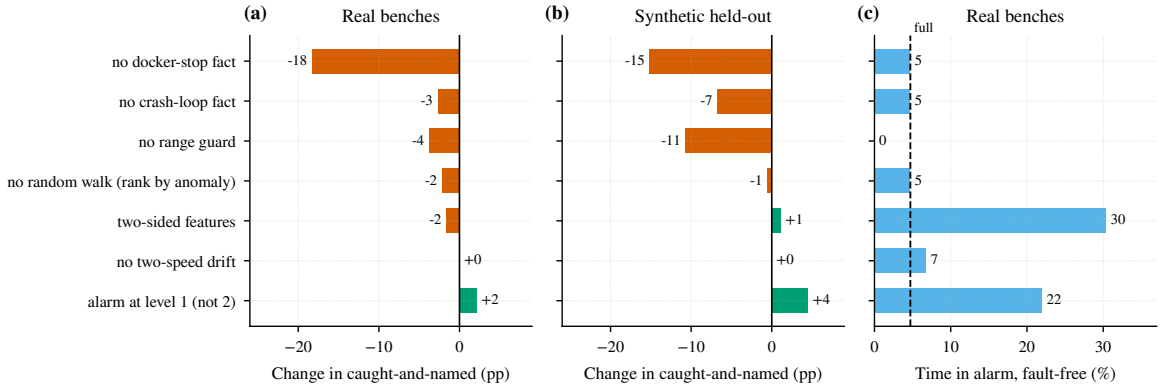


Figure 6.7: Ablation. Each bar removes one mechanism from the full engine. (a), (b): change in caught-and-named rate (percentage points) on the real systems and the synthetic held-out set. (c): share of fault-free time in alarm on the real systems; the dashed line marks the full engine.

6.1.5 When does the graph matter? The amplification study

Standard benchmarks have faults whose own service is also the loudest, which is the case where ranking by severity cannot fail. To probe the regime in which it can, soft faults (brownout, latency, memory) were regenerated with the amplification factor $A \in \{1, 1.5, 2, 3, 4\}$, the hybrid's alarms were kept identical for every ranker, and only the ranking varied (Figure 6.8). As A grows, the share of faults in which some victim is louder than the true cause rises from 15% to 25% (panel c), the severity ranking falls from 67% to 62%, and the walk holds near 67%, so the gain of the walk over severity grows from about one point to about five (panel b). Its intervals include zero at every level ($+4.7$ points, 95% CI $[-0.9, +11.3]$, at $A = 4$). The strongest ranker on soft faults throughout is the own-versus-explained decomposition used by the rule layer (69 to 75%; $+5.7$ to $+9.4$ points over severity, with intervals excluding zero at $A = 1.5$ and $A = 4$). A router that uses it whenever no container fact holds did not change the full benchmarks (within ± 2 points), so the shipped ranking is unchanged. Graph-aware ranking is a modest, regime-dependent gain, and the decomposition matters more than the walk.

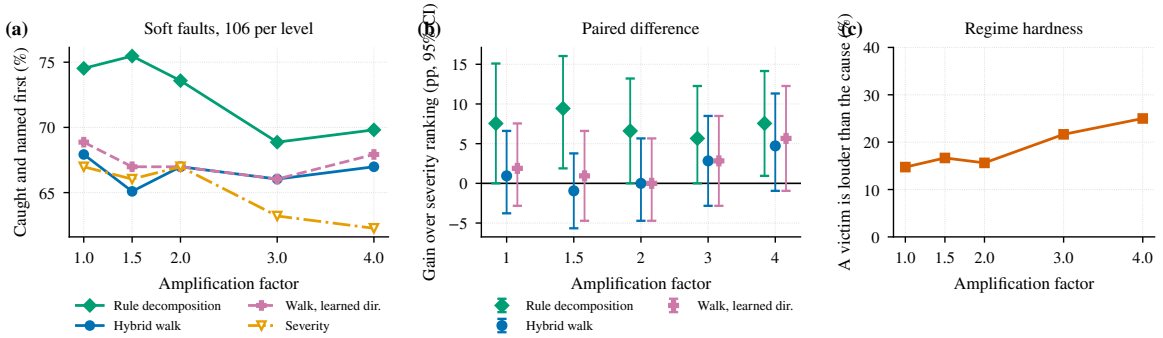


Figure 6.8: Amplification study on soft faults (106 per level). (a) Caught-and-named rate of four rankers on identical alarms. (b) Gain over the severity ranking with 95% paired-bootstrap intervals. (c) Share of faults in which a victim is louder than the true cause.

6.1.6 The cascade model

Figure 6.9 evaluates who-is-next predictions at the first alarm of each fault: for every service still healthy, the probability that it turns unhealthy before the fault ends. On the synthetic set (159 faults, 1 196 predictions, 80 cascades) the learned propagation model reaches AUROC 0.847 against 0.784 for the same noisy-OR with every edge fixed at 0.7, a Brier score of 0.055 against 0.060 and an ECE of 0.039 against 0.058. The reliability curve is noisy in the upper bins, which hold few predictions. *On the real systems the experiment cannot be done well:* only 5 of 420 predictions were followed by a cascade (54 faults), so the real-system comparison (AUROC 0.685 against 0.661; the live mapper's own risk values 0.530) is not informative. The real faults are mostly single-service outages that the systems absorb; whether learned propagation helps on a real cascade is untested.

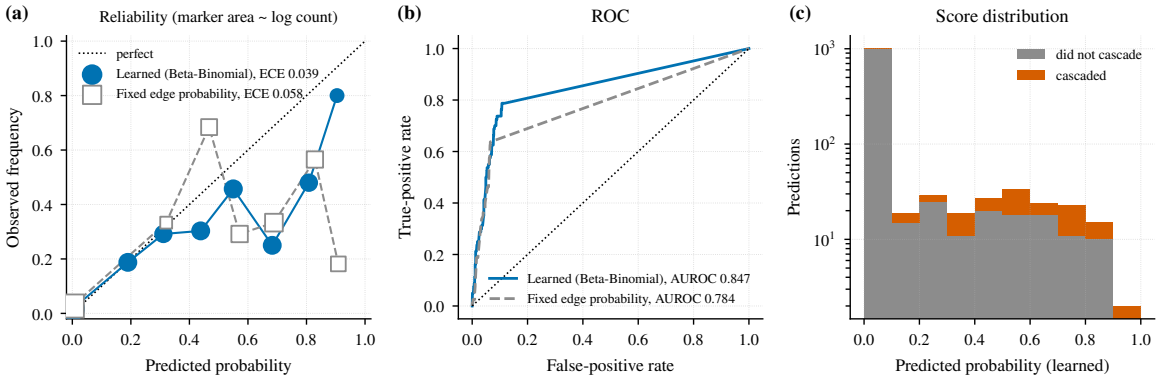


Figure 6.9: Who-is-next prediction on the synthetic held-out set. (a) Reliability diagram with equal-width bins and marker area proportional to the logarithm of the bin count. (b) ROC curves. (c) Distribution of the learned predicted probabilities.

6.1.7 Learned directions

Figure 6.10 reports direction accuracy on observed pairs that are true calls. With names available the name rule is correct for 21 of 31 real pairs (and leaves 10 undecided); the

learned rule is correct for 28 of 31 (3 wrong). With service names hidden the name rule is undecided for every pair while the learned rule is unchanged; the same holds on the synthetic set (95 of 204 against 197 of 204 correct). The effect on ranking is smaller than the effect on directions. With names available the learned directions do *not* beat the name rule (71% against 74% on the real systems, −3.2 points, not significant). With names hidden they reach 71% against 69% for the name rule and 73% for the true graph. Learned directions are a robustness mechanism, not an accuracy gain.

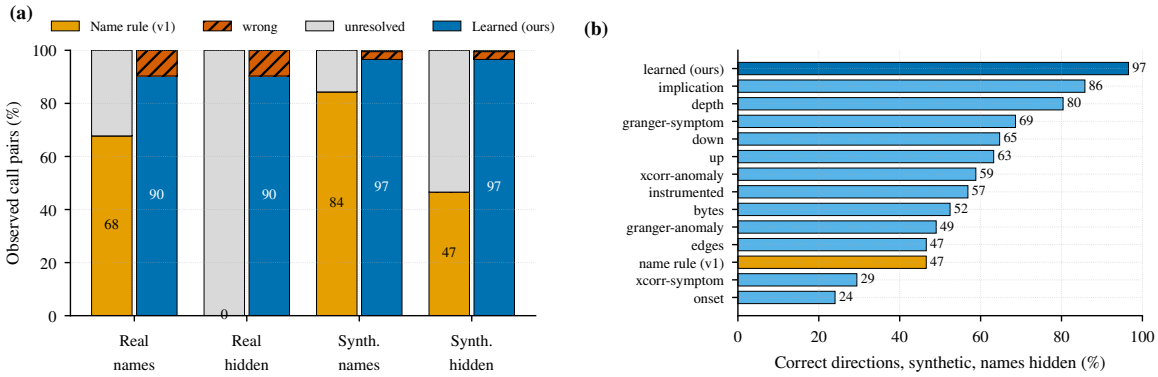


Figure 6.10: Call directions. (a) Share of observed true-call pairs oriented correctly, wrongly, or left unresolved by the name rule and by the learned rule, with service names shown or hidden. (b) Correct share of each individual signal on the synthetic set with names hidden.

6.1.8 Early warning and fault diagnosis

The trend forecast warns before the alarm for 16 of 178 synthetic faults (7 of 18 memory leaks) with a median lead of 30 s and about one false warning per fault-free hour (Figure 6.11); on the real systems it fired for one of 62 faults, since those faults are abrupt. Figure 6.12 shows fault-type diagnosis. With the true service the forest is correct for 99% of synthetic faults 60 s after the alarm; on the engine's own suspect it is 77%, because a wrong suspect is rarely typed correctly. On the real systems a hand-written decision list generalises best (59% on the suspect, 84% on the true service), because real flaps are pause cycles and real brownouts on idle services show no throttling: differences between the simulator and the real systems that the learned models absorb as synthetic-specific patterns.

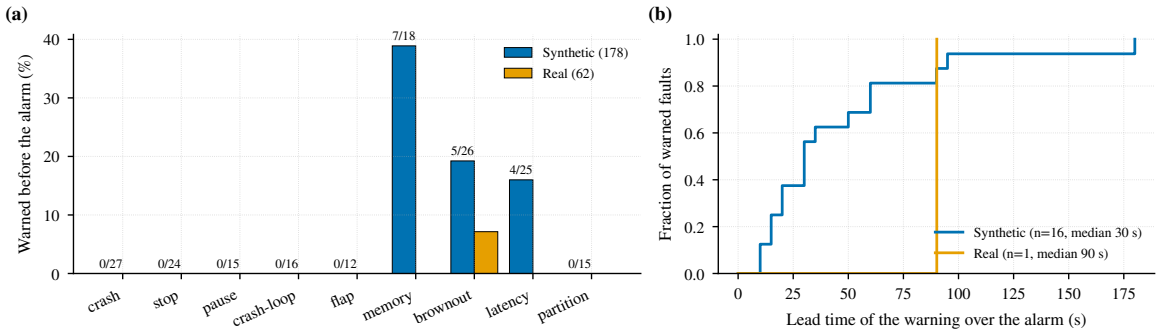


Figure 6.11: Early warning. (a) Share of faults warned before the alarm, by fault class. (b) Lead time of the warnings over the alarm as a cumulative distribution.

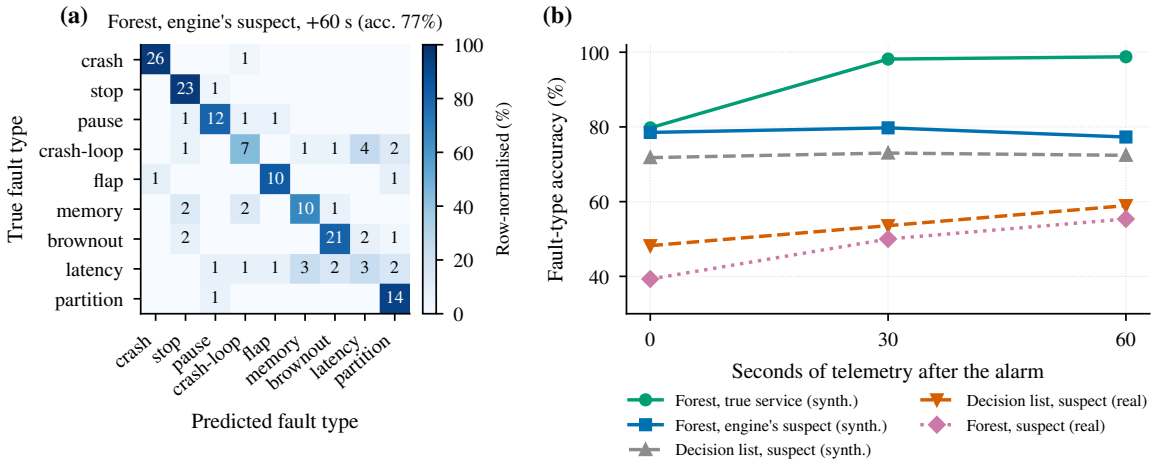


Figure 6.12: Fault-type diagnosis. (a) Confusion matrix (counts; shading is row-normalised) of the random forest on the engine's suspect 60 s after the alarm, synthetic held-out set. (b) Accuracy against seconds of telemetry after the alarm, for the true service and the engine's suspect.

6.1.9 Cost

The pipeline is unsupervised, needs no GPU and runs on one core. In a single replay of the two real systems (12 and 13 services) one decision window, covering the per-service forests, facts, random walk and cascade risk, took about 6 to 7 ms on average, far below the 5 s collection period. How the cost grows with the number of services beyond these sizes was not measured, so no scaling claim is made: the per-service forests are independent and the walk is linear in edges times iterations, but this was not benchmarked.

6.1.10 Case studies

Figure 6.13 shows three real incidents, chosen by class before looking at the outcome. In (a) a memory leak overlaps several outages: the engine names the stopped containers first, as it should, and only a model of concurrent faults could do better. In (b) a crash loop is named correctly about 30 s after its first kill and stays named between kills. In (c) a brownout on an almost idle service never raises an alarm, which is the open problem of Figure 6.3.

6.1.11 The three engines on one live fault

To show what the Detection engines page adds, the simulator was run with a slow `payment` service, the ground-truth root cause being `payment`, while the hybrid was the active engine. All three engines alarmed. Their suspects differed (Figure 5.4 shows the page for a different fault, the loss of the datastore):

Table 6.4: One simulated fault (slow `payment`); each engine's first suspect and peak against its own line.

Engine	First suspect	Second	Peak vs its line
Rules	orders (3.72×)	payment (2.89×)	3.72
ML	payment (54% of the walk)	orders (46%)	1.43
Hybrid	payment	orders	2.98

This is one simulated fault and an illustration, not evidence: it shows the mechanism that the evaluation measures at scale (the rule layer ranks the loudest service first; the ML and hybrid engines let the walk move the blame to the source).

6.1.12 Operational findings from the benchmark hosts

Operating the benchmark hosts produced three findings that changed the tools.

- **A host too small for its workload looks like a mapper fault.** The OpenTelemetry demo ran on a 2-vCPU, 4 GiB instance. The check tool measured a load average of about 25, 3.6 of 3.8 GiB of memory in use, no swap, and docker did not answer within 25 seconds; the dashboard showed no services. A second host with 4 GiB of swap and a load average of 0.2 ran 27 containers of a social-network benchmark normally. The lesson is operational: the mapper depends on the host's Docker daemon answering, and a host that cannot answer shows nothing. The check tool now says so in one step.
- **Idle front ends have no edges.** On the healthy social-network host, the front-end containers showed no outgoing sockets when sampled, because nothing was calling them; an undeclared edge expires ten minutes after its last traffic. A front end that is idle therefore has no edge until traffic arrives, which is correct behaviour but looks like a fault. Declared edges from the Compose file keep an idle link on the screen.
- **Hard-coded project names blocked traffic.** The traffic controller accepted only five named projects, so benchmark hosts with other names could not be driven. Unknown projects now get a generic workflow.

6.2 Analysis

What the evidence supports. Three results are robust. The fact-aware hybrid names the true cause more often than the rules on the real systems (the paired interval excludes zero). The container facts are the single most valuable component (the ablation shows 19 and 16 points). And the learned propagation model discriminates better than fixed edge probabilities on observed cascades (AUROC 0.847 against 0.784).

What did not work. Three claims that were expected to hold did not. The random walk is not statistically better than ranking by anomaly alone on these benchmarks. Learned directions do not beat the name rule when names exist. Onset-precedence re-ranking was neutral. These are reported because a pipeline that is honest about where it stops is more useful to an operator than one that is not. The simple rule z-score is a strong detector on its own, and the Isolation Forest earns its place through quietness and complementarity, not discrimination.

The measurement bug. Before the ledger reader was fixed, every engine was scored on only part of the real faults (14 of 31 per system were hidden) and the outages of the hidden faults were counted as false alarms. That is why every number in this report uses all 62 faults, and why a regression test now guards the reader.

6.3 Discussion and Threats to Validity

Construct validity. Ground truth is the injector's ledger. It records the structural root cause and the services able to be affected; whether an affected service visibly changes depends on protection the ledger does not model, so cascade recall is a lower bound on true propagation. The evidence weighting of Section 4.6.1 is not isolated by the offline benchmarks because the recordings do not retain edge evidence.

Internal validity. Settings were chosen by the authors. On the synthetic track they were chosen on a disjoint tuning split and scored once; on the real track they were not tuned on a held-out system, so the real numbers are optimistic by an unmeasured margin. The two container facts were found by inspecting the real runs.

External validity. Two real applications and one capture. Architecture independence is argued from the cross-system check and the synthetic topologies, not proven; the simulator is cleaner than a real system (for example, restarts do not lower memory in it).

Conclusion validity. With 62 real faults a difference below about eight points cannot be resolved, and overlapping faults make the intervals optimistic.

A caution about auto-restart. The mapper can restart crashed containers on a target (Section 5.10 and Chapter 3). Evaluation runs should be made with it switched off, so that

the mapper does not heal the faults the injector applies; the reader of this report should confirm the setting of any run it intends to repeat.

6.4 Comparison with Existing Approaches

The published methods in Table 2.1 were not re-implemented on our recordings, so there is no head-to-head accuracy table against them. What is compared head to head are the baselines that those methods build on: the rule layer, ranking by anomaly alone (the severity ranking), the walk, and the ML engine. Table 6.5 sets the measured results beside the qualitative properties of Chapter 2.

Table 6.5: This project beside the approaches it builds on. Accuracy figures are caught-and-named on the 62 real faults.

Approach	C+N (real)	Notes
Rule layer (z-score)	65%	Fast, 9 false-alarm episodes per hour, 9% alarm time
Severity ranking (loudest service)	71%	Cannot fail when the cause is loudest; weakens as victims get louder
ML engine (forest + walk)	73%	Fewest false alarms (2.3 per hour), 5% alarm time
Hybrid (this project)	**74%**	Paired +9.7 points over rules; names stops 16 of 16
Learned fusion methods [4, 5, 6]	not run	Need labelled failures and GPU; not auditable
Trace-based rankers [31]	not run	Need tracing in every service

6.5 Significance and Implications

Three practical points follow. An instrumentation-free tool can recover a usable call graph and a useful ranking, and for small teams that is the difference between having a diagnosis tool and not. Container facts are worth more than any statistical refinement: a stopped container is something one knows, and folding it into an anomaly score throws information away. And the right operating contract for an alarm is the one the operator experiences, which is why alarm time is reported beside false-alarm episodes.

6.6 Potential Applications

- On-call support for small teams running Docker Compose stacks on a few hosts.
- Teaching and labs on cascading failure, with a blind fault injector and a recorded-run

format students can replay.

- A first-line triage layer in front of heavier observability systems, giving a ranked suspect and a runbook.
- Capacity and resilience reviews using the what-if and blast-radius views.
- A benchmark harness for new detectors and rankers, with ground truth the detector never sees.

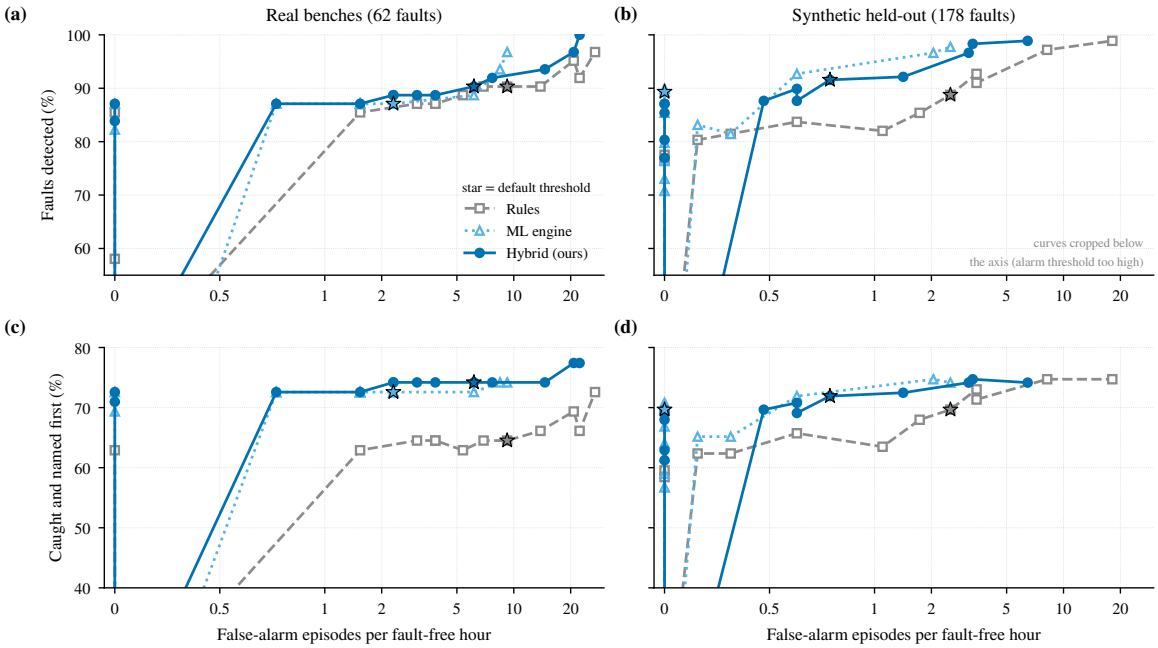


Figure 6.1: Operating curves. Sweeping the alarm threshold of each engine traces detection rate (top) and caught-and-named rate (bottom) against false-alarm episodes per fault-free hour (symmetric-log axis); stars mark the default threshold. The hybrid sits on the upper envelope in both the real systems (a, c) and the synthetic held-out set (b, d).

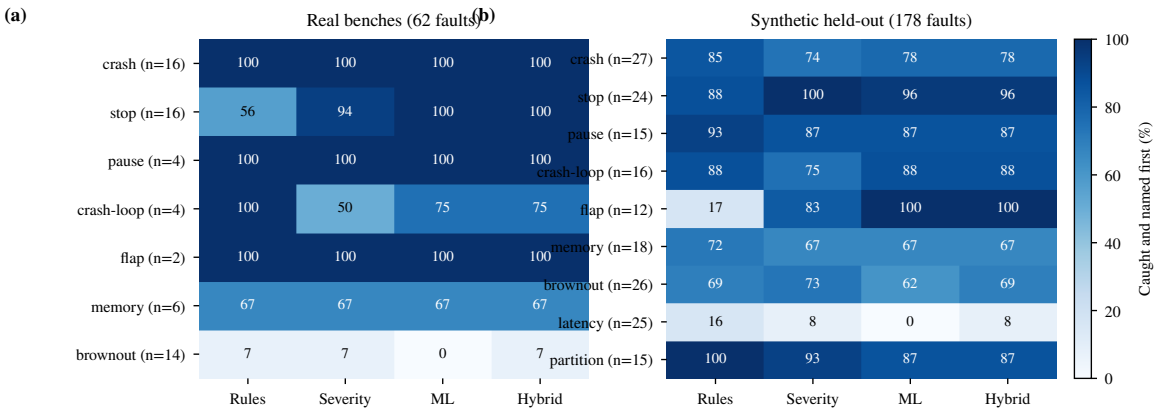


Figure 6.3: Caught-and-named rate by fault class (percent; *n* faults per class), for four rankers on the real systems (a) and the synthetic held-out set (b). Brownouts are the open problem on the real systems.

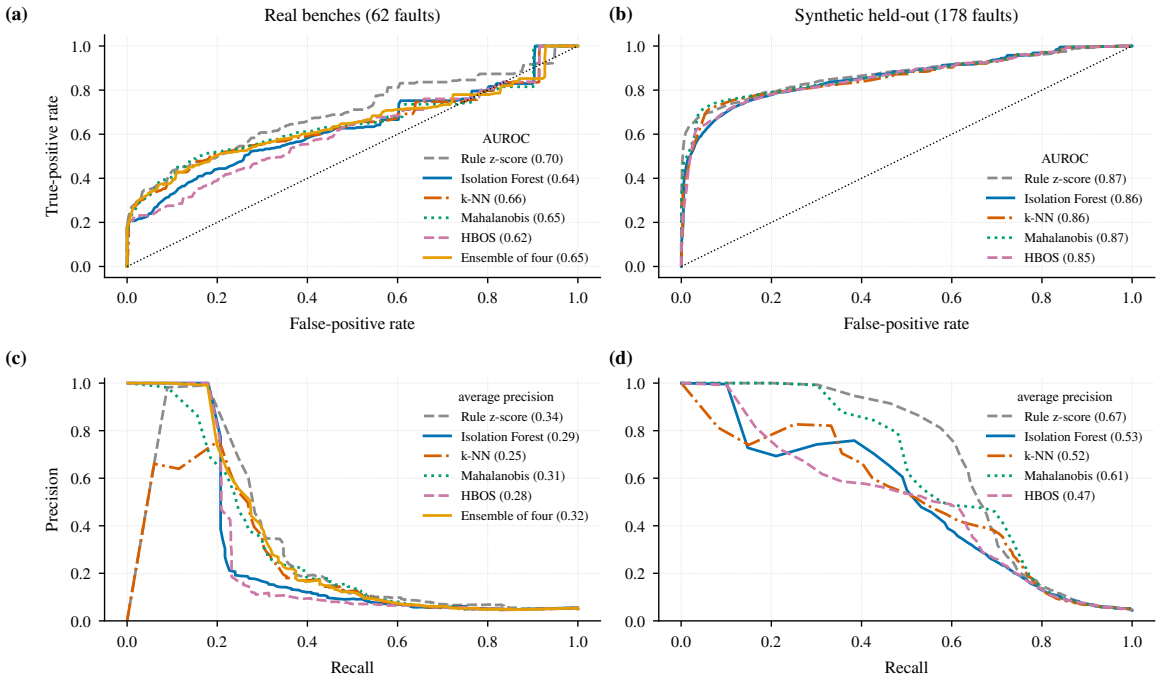


Figure 6.4: Service-level detection by detector: ROC (top) and precision–recall (bottom) for the real systems (left) and the synthetic held-out set (right); legends give AUROC and average precision (means over three seeds for the randomised detectors). Positives are the faulted service during a fault; negatives are any service when no fault is active; container facts are excluded.

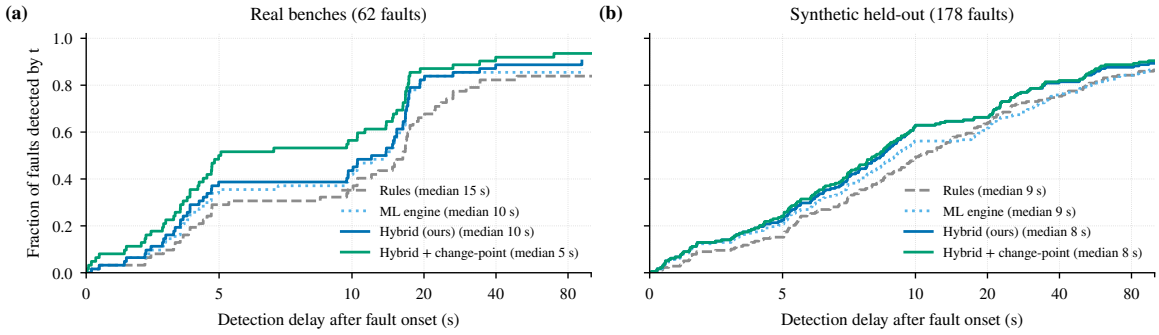


Figure 6.5: Detection delay after fault onset as a cumulative distribution (symmetric-log axis); legends give the median over detected faults. The curves plateau at each engine's detection rate.

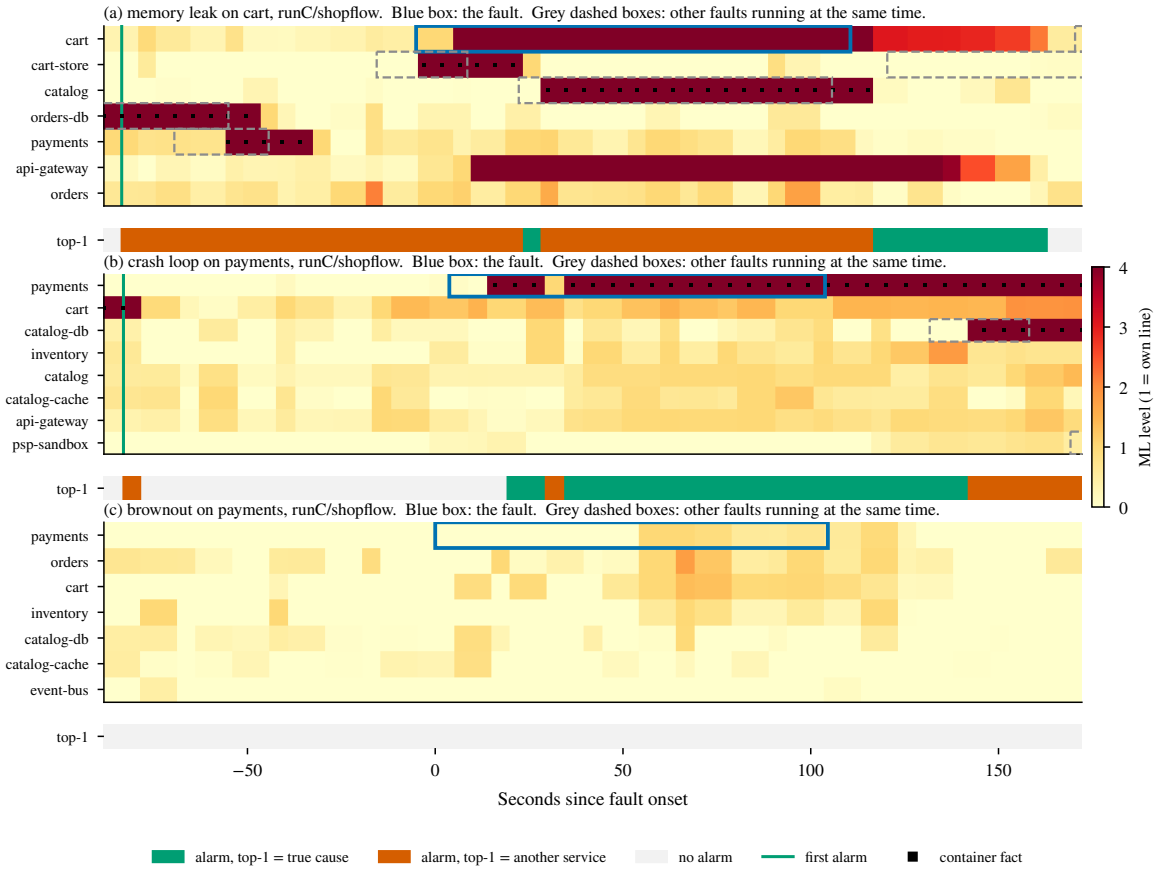


Figure 6.13: Three real incidents. Heat maps show the ML level of the most relevant services; the blue box is the fault, grey dashed boxes are other faults running at the same time, black squares mark container facts, the green line is the first alarm in the window (an alarm already ongoing from an earlier fault starts at the left edge), and the strip below each map shows whether the engine's top-ranked service was the true cause.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Summary of Work

The project built an instrumentation-free platform that watches containerised microservices from outside, recovers their call graph, and runs an unsupervised diagnosis pipeline on it. A collector agent reads Docker state, statistics and socket tables over SSH; an evidence-weighted graph orients each edge by its listening port; a hybrid detector of per-service Isolation Forests and a robust rule layer treats container facts as evidence; a ranker splits each deviation into what neighbours explain and what they do not and runs a random walk over the call graph; a cascade model with learned Beta–Binomial edge probabilities predicts who is next. A dashboard shows the result in 2D and 3D, and a Detection engines page runs Rules, ML and Hybrid in parallel and shows what each detected, computed and predicted. The system was evaluated on 62 faults injected blind into two real containerised systems and on 178 held-out synthetic faults, and operated on further benchmark applications on EC2.

7.2 Conclusions

1. On 62 real faults the hybrid names the true cause first for 74% against 65% for the rule layer (paired $+9.7$ points, 95% CI $[+1.6, +17.7]$, $p = 0.032$); on the synthetic set the difference is not significant ($+2.2$ points).
2. The container facts are the most valuable single component: removing the `docker stop` fact costs 19 points on the real systems and 16 on the synthetic set.
3. Learned propagation discriminates better than fixed edge probabilities on 80 observed cascades (AUROC 0.847 against 0.784), with a lower Brier score and a lower expected calibration error. On the real systems the experiment is uninformative because almost no real fault cascaded.
4. Learned directions stay correct when service names carry no information (28 of 31 real pairs against none for the name rule), but they do not improve ranking when names exist.
5. Change-point alarms roughly halve the median detection delay on the real systems (10.5 s to 4.9 s), at the price of more alarm time (7% to 16%).
6. The simple rule z-score is a strong detector on its own; the Isolation Forest earns its place through quietness and by complementing the rules.

7. The pipeline costs about 6 to 7 ms per five-second window on the systems tested, on one core, with no GPU.
8. A host too small for its workload looks like a mapper fault; the diagnostic tools separate the two.

7.3 Contributions

- An observation layer that recovers topology and direction from the host alone.
- A probabilistic cascading-failure model with learned propagation.
- A hybrid detector with an honest operating contract (alarm time as well as false-alarm episodes).
- A fact-aware, decomposition-based ranker and learned call directions.
- A Detection engines page showing three engines in parallel.
- A blind, reproducible evaluation that includes the negative results and a corrected measurement bug.
- A working platform with 329 automated tests, a traffic generator and host-diagnostic tools.

7.4 Utility and Applications

The platform gives a small team a first-line diagnosis tool for Docker-based systems that nobody wants to instrument, a teaching and benchmarking harness for cascading failure, and a triage layer in front of heavier observability products (Chapter 6, Potential Applications).

7.5 Limitations

- **Coarse signals.** Eight metrics every five seconds. A brownout on an almost idle service never raises an alarm; no engine here catches it.
- **Two real systems.** Architecture independence is argued from a cross-system check and synthetic topologies, not proven. Real cascades were too rare in the recordings to test learned propagation on real data.
- **Edge evidence not isolated offline.** The recordings do not keep edge evidence, so the effect of the evidence weighting is not measured.
- **No scaling measurement.** The cost was measured at 12 and 13 services only.
- **Tuned by the authors.** Real-system settings were not tuned on a held-out system, so those numbers are optimistic by an unmeasured margin.
- **Auto-restart.** The mapper restarts crashed containers by default (switch off with

`MAPPER_AUTO_RESTART=off`); runs meant to measure detection should switch it off.

- **Default ingest token.** The push endpoint has a built-in token; it must be changed when the server is reachable by others.
- **Not tested on Kubernetes**, and the dashboard views were checked on a simulated target, not on every benchmark host.

7.6 Scope for Future Work

- A larger measured campaign on more real systems, and a public labelled data set for comparability.
- Real multi-service cascades, to test learned propagation where it matters; promotion of learned directions and change-point alarms into the live system behind its operator switch.
- Retention of edge evidence in the recordings, so that the evidence weighting can be isolated and measured.
- A language-model agent on top of the structured evidence (cause, evidence, runbook) to answer follow-up questions or draft a change, with the evidence as its only grounding, evaluated with a user study; any action it proposes would go through an approval gate.
- Support for Kubernetes through the same observation idea (node-level access to container sockets), and a scaling benchmark.
- Brownout detection on idle services, perhaps with active probing of the service.

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APPENDIX A

ENGINE CONSTANTS AND THRESHOLDS

Table A.1: Rule-layer thresholds (defaults, adjustable on the Incidents page).

Name	Default	Meaning
`minHistorySamples`	12	Samples a service needs before it is judged
`zScoreAnomaly`	3.5	Deviation counted as anomalous (noise-normalised)
`zScoreHigh`	5.0	Deviation at or above this is HIGH
`zScoreCritical`	8.0	Deviation at or above this is CRITICAL
`minDeltaPercent`	5	Smallest change worth a flag
`absoluteHighPercent`	70	Absolute CPU or memory share counted as high
`absoluteCriticalPercent`	85	Absolute share counted as critical
`flatlineStdDev`, `flatlineDeltaPercent`	0.001, 30	A series that goes flat after moving
`criticalServicesForCritical`	1	Critical services before the incident is CRITICAL
`silenceSamples`	3	Missed readings before a service counts as silent
`monitoringGapShare`	0.8	Share of silent services that means the monitoring itself is down
`explainAwayGamma`	1.0	Weight of what a sick callee explains (γ)
`cascadeEdgeProbability`	0.5	Starting propagation probability of an edge

Table A.2: Edge evidence confidence.

Evidence	Confidence $c(e)$
Seen in request logs, or oriented by a listening port	1.00
Oriented only by the ephemeral-port range	0.85
Observed, no direction evidence	0.70
Observed, direction unresolved	0.50
Declared in Compose, never seen carrying traffic	0.40

APPENDIX B

THE REST INTERFACE

Table B.1: All REST endpoints registered by the server (`server/api/routes.ts`).

Method	Path
GET	/api/graph
GET	/api/nodes/:id
GET	/api/nodes/:id/metrics
GET	/api/traffic
GET	/api/analytics
GET	/api/reliability/blast-radius/:nodeId
GET	/api/reliability/cascade
GET	/api/reliability/whatif/:nodeId
GET	/api/diagnostics
GET	/api/diagnostics/consistency
GET	/api/targets
GET	/api/targets/:targetId/diagnostics
GET	/api/targets/:targetId/discovery
GET	/api/targets/:targetId/services
GET	/api/nodes/:id/logs
GET	/api/targets/:targetId/endpoints
GET	/api/targets/:targetId/routes
POST	/api/discovery/refresh
GET	/api/incidents
GET	/api/engine
GET	/api/engine/info
POST	/api/engine
GET	/api/incidents/thresholds
POST	/api/incidents/thresholds
GET	/api/incidents/:id
GET	/api/incidents/:id/timeline
GET	/api/incidents/:id/evidence
GET	/api/incidents/:id/propagation
GET	/api/services/:id/anomalies
GET	/api/targets/:targetId/resolved-url

Method	Path
GET	/api/targets/:targetId/reachability
POST	/api/ingest
GET	/api/status
GET	/api/traffic/health
GET	/api/traffic/entrypoints
POST	/api/traffic/entrypoint
POST	/api/traffic/entrypoint/probe
POST	/api/traffic/surface/probe
GET	/api/traffic/status
POST	/api/traffic/start
POST	/api/traffic/stop
GET	/api/experiments
POST	/api/experiments/start
POST	/api/experiments/update_stats
GET	/api/ml/status
GET	/api/ml/config
POST	/api/ml/config
GET	/api/ml/processes
POST	/api/ml/processes/:name/start
POST	/api/ml/processes/:name/stop
GET	/api/ml/models
GET	/api/ml/features
GET	/api/ml/normalization
GET	/api/ml/retrain
POST	/api/ml/retrain
GET	/api/ml/source/:name
GET	/api/ml/suggestions
GET	/api/ml/projects
GET	/api/ml/score-history
GET	/api/config/remote
POST	/api/config/remote
GET	/api/models
DELETE	/api/config/remote/:targetId
POST	/api/data/clear

APPENDIX C

FAULT CLASSES

Table C.1: Fault classes, how they are applied, and how many faults of each class the evaluation sets contain.

Class	Applied as	Real (62)	Synthetic (178)
crash	Process killed (`SIGKILL`); the container exits with a non-zero code	16	27
stop	Container stopped with `docker stop` (exit code 0)	16	24
brownout	Resource limit that starves the service without stopping it	14	26
memory	Memory pressure on the service (a leak, or a limit)	6	18
crash-loop	Repeated outages within minutes	4	16
pause	Container paused	4	15
flap	Repeated pause and release cycles	2	12
latency	Slow responses	–	25
partition	Network detached from the service	–	15

APPENDIX D

SIMULATOR SCENARIOS

The ground-truth simulator drives the mapper with a ten-service shop and a known root cause, and was used for the screenshots and for quick checks.

Table D.1: Simulator scenarios.

Scenario	Description	Root cause
baseline	Everything healthy	none
db-crash	`orders-db` is killed; `orders` then the front end degrade as calls to it fail	`orders-db`
slow-payment	`payment` saturates; `orders` degrades waiting on it	`payment`
crash-loop	`carts-db` crash-loops; `carts` degrades	`carts-db`

APPENDIX E

USER GUIDE

Install and start

```
git clone https://github.com/Royson-salis-18/microservice-mapper.
    git
cd microservice-mapper
npm start          # installs what is missing, starts the server
    (:3001) and the dashboard (:5173)
```

Open `http://localhost:5173`. Node.js 20.19 or 22.12 and later is required. `npm run doctor` checks the machine.

Add a host

Click “Add project” and fill in the target identifier, display name, the host’s public IPv4 address, the SSH user and the path to the key file. The key stays on your machine. `npm run hosts` lists the configured hosts and tests port 22; `npm run hosts - set <target> <ip>` changes an address.

Read an incident

Open *Incidents and cascade*. The headline names the most likely root cause with its confidence. The ranking shows what each service’s score is made of; “Why” shows the own-forest, symptom-forest and walk parts and the evidence; the cascade panel shows who is already affected and who is next; the next steps are commands to run on the host.

Compare the three engines

Open *Detection engines*. The three columns show, for the same readings, what each engine detected, its ranked suspects and what it predicts. The switch in the top bar (or the “Use” buttons) chooses which engine decides the incident.

Generate load and reproduce results

```
npm test                    # the 329 tests
```

```
npm run eval              # record and score a campaign against the
    live mapper
npm run ml:compare        # detectors and root-cause methods on the
    recorded runs
npm run ml:hybrid         # fusion variants
npm run ml:predict        # early warning and cascade prediction
report/build.sh           # rebuild this report (needs pdflatex;
   Mermaid optional)
```

When nothing shows up

```
npm run check-host -- <public-ip> ~/path/to/key.pem 8080
npm run mapper-edges -- <project> <service-name-part>
```

The first says whether the host or the mapper is at fault; the second prints what the running mapper holds for a service.